Discrete Mathematics

Lecture notes, Chapters 1–15

Based on Miklós Bóna, *A Walk Through Combinatorics*

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Chapter 1. The Pigeon-Hole Principle

1.1 The basic pigeon-hole principle

Motivation. "Seven is more than six" sounds empty, yet comparing the number of objects with the number of places they can go yields strong existence statements with almost no computation. The art is to decide what the balls and what the boxes are.

Theorem 1.1 (Pigeon-hole principle)

Let $n > k$ be positive integers. If n identical balls are placed into k identical boxes, some box receives at least two balls.

Proof. Suppose no box has two balls. Then every box has 0 or 1 ball; if $m \geq 0$ boxes are empty, the total number of balls is $k - m \leq k < n$, a contradiction. ■

Example 1.2

Some element of $7, 77, 777, \ldots$ is divisible by 2023.
Solution. Suppose none of the first 2023 elements $a_1, \ldots, a_{2023}$ is divisible by 2023. Their remainders lie in $\{1, \ldots, 2022\}$, so by Theorem 1.1 two of them, a_i and a_j with $i < j$, have the same remainder, and $2023 \mid a_j - a_i$. Now $a_j - a_i$ consists of $j - i$ sevens followed by i zeros, i.e. $a_j - a_i = a_{j-i} \cdot 10^i$. Since $\gcd(10^i, 2023) = 1$, $2023 \mid a_{j-i}$, contradicting the assumption. (Remainders are the boxes, the elements the balls.)

Example 1.3

In a round-robin chess tournament with n players, at any moment two players have finished the same number of games.
Solution. The possible numbers of finished games are $0, 1, \ldots, n-1$: n boxes for n players. But 0 and $n-1$ cannot occur simultaneously: a player who has finished all $n-1$ games has played everybody. So at most $n-1$ values occur, and Theorem 1.1 applies.

1.2 The generalized pigeon-hole principle

Motivation. If there are many more balls than boxes, some box must contain many balls. This quantitative version is just as easy to prove and is the form used in geometry, number theory and graph theory.

Theorem 1.4 (Pigeon-hole principle, general version)

Let n, m, r be positive integers with $n > rm$. If n identical balls are distributed into m identical boxes, some box receives at least $r + 1$ balls.

Proof. Otherwise each box holds at most r balls, so all boxes together hold at most $rm < n$ balls, a contradiction. ■

Example 1.5

Ten points lie in a unit square. Then two of them are at distance < 0.48, and three of them can be covered by a disk of radius 0.5.

Solution. Cut the square into nine squares of side $\frac{1}{3}$ (left figure). Some small square contains two points (Theorem 1.1), and their distance is at most the diagonal $\frac{\sqrt{2}}{3} < 0.48$. Cut the square by its diagonals into four triangles (right figure). Since $10 > 2 \cdot 4$, some triangle contains three points (Theorem 1.4). The circumradius of that right isosceles triangle is half its hypotenuse, $\frac{1}{2}$, so a disk of radius 0.5 covers it.

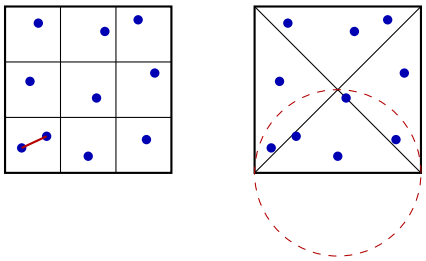


Figure. *Example 1.5: nine small squares force two close points; four triangles force three points under one disk of radius $\frac{1}{2}$ (dashed: the disk around the bottom triangle).*

Example 1.6

Within the last 1000 years, the reader had an ancestor A such that some person P was an ancestor of both parents of A.

Solution. With 25 years per generation, 1000 years are 40 generations, and the family tree of the reader has $1 + 2 + \cdots + 2^{40} = 2^{41} - 1$ nodes. If two nodes are the same person B, then B serves as P (for the nearest common descendant). Otherwise there would be $2^{41} - 1$ distinct people, but fewer than $40 \cdot 10^{10} < 2^{41} - 1$ people lived during those 40 generations (the population never exceeded 10^{10}), a contradiction.

Example 1.7 (The handshake party)

Mr. and Mrs. Smith invite four couples. Acquaintances shake hands (nobody with themself or their spouse). Afterwards Mr. Smith observes that, apart from himself, nobody shook hands the same number of times. How many hands did Mrs. Smith shake?

Solution. Model people as vertices of a **graph**, joining two vertices by an edge if those people shook hands. The nine people other than Mr. Smith have nine distinct handshake numbers in $\{0, \ldots, 8\}$, so every value occurs once; let Y_i be the person with i handshakes.

Y_8 shook hands with everyone except one person, who must be the spouse; since Y_0 shook no hands, Y_8 and Y_0 are married.

Y_7 missed exactly two people, one of them the spouse. These must be Y_0 (no handshakes) and Y_1 (whose only handshake was with Y_8), so the spouse of Y_7 is Y_1.

Continuing in the same way, Y_6–Y_2 and Y_5–Y_3 are couples, so Y_4 is Mrs. Smith: she shook hands **four** times. By Example 1.3, a party in which everyone shakes a different number of hands is impossible; exempting one person leaves exactly one possibility, shown below.

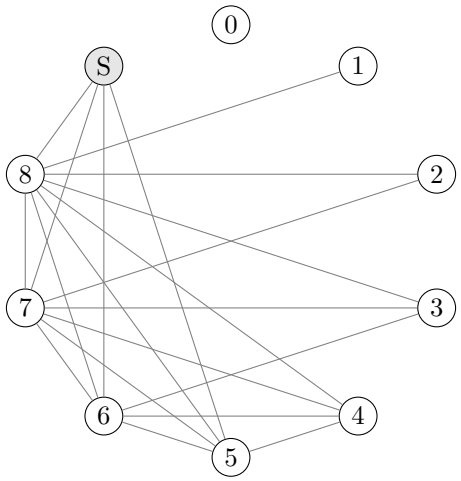


Figure. *The only possible party of Example 1.7: vertex i shook i hands, S is Mr. Smith. Vertex $k \geq 5$ is joined to S and to all $j \geq 9-k$; the couples are $\{8,0\},\{7,1\},\{6,2\},\{5,3\}$ and $\{4,S\}$.*

The notion of a graph will be central in Chapters 9–12.

1.3 Selected exercises

Motivation. The exercises below, taken from the book, train the one real skill of this chapter: choosing the boxes so cleverly that a bare count of balls and boxes gives the conclusion.

Exercise 1.11 (Sum or difference divisible by 1000)

Prove that among any 502 positive integers there are two whose sum or difference is divisible by 1000.

Solution.

Step 1 (boxes). For $r \in \{0,1,\dots,999\}$ let $\bar{r}$ denote the residue class of r modulo 1000. Form the 501 boxes

$$\{\bar{0}\},\quad \{\bar{500}\},\quad \{\bar{1},\overline{999}\},\quad \{\bar{2},\overline{998}\},\quad \dots,\quad \{\overline{499},\overline{501}\},$$

that is, $\{\bar{r},\overline{1000-r}\}$ for $1 \leq r \leq 499$ together with the two singletons. Every residue class lies in exactly one box.

Step 2 (pigeon-hole). Put each of the 502 integers into the box of its residue class. Since $502 > 501$, Theorem 1.1 gives two distinct integers a, b (distinct as members of the list) in the same box.

Step 3 (conclusion). If $a \equiv b \pmod{1000}$, then $1000 \mid a-b$. Otherwise the box is $\{\bar{r},\overline{1000-r}\}$ with $a \equiv r$ and $b \equiv 1000-r$ (or the reverse), so $a+b \equiv 1000 \equiv 0$ and $1000 \mid a+b$. (In the singleton boxes only the first case occurs.) ∎

Exercise 1.15 (Six goals in consecutive games)

A soccer team played 30 games and scored 53 goals, scoring at least one goal in each game.
(a) Prove that there was a sequence of consecutive games in which the team scored exactly six goals.
(b) Prove that the claim of part (a) can fail if the team scored 60 goals, the other conditions unchanged.

Solution.

(a) Let s_k be the number of goals scored in games $1, \dots, k$. Since every game has at least one goal,

$$1 \le s_1 < s_2 < \cdots < s_{30} = 53.$$

Consider the 60 numbers $s_1, \dots, s_{30}, s_1 + 6, \dots, s_{30} + 6$. All of them lie in $\{1, 2, \dots, 59\}$, so by Theorem 1.1 two of them are equal. The s_k are pairwise distinct, and so are the $s_k + 6$; hence $s_j = s_i + 6$ for some i, j. Then $s_j > s_i$, so $j > i$, and games $i+1, \dots, j$ produced exactly $s_j - s_i = 6$ goals.

(b) Let the goals per game be $1, 1, 1, 1, 1, 7$, repeated five times: 30 games, $5 \cdot 12 = 60$ goals, at least one goal per game. A block of consecutive games that contains a game with 7 goals has at least 7 goals. A block that contains no such game lies within one run of five 1's and has at most 5 goals. So no block has exactly 6 goals. ■

Exercise 1.18 (Two 1's in north-east position)

Let $n \ge 2$ and let A be an $n \times n$ matrix with entries 0 and 1 only, at least $2n$ of which are 1. Prove that A contains two entries equal to 1 such that one of them is strictly above and strictly to the right of the other.

Solution. Index the entries by (i, j), row i counted from the top and column j from the left. The entry (i', j') is strictly above and strictly to the right of (i, j) iff $i' < i$ and $j' > j$.

Step 1 (boxes). For $s = 2, 3, \dots, 2n$ let $D_s = \{(i, j) : i + j = s\}$ be the s-th anti-diagonal. These $2n - 1$ sets partition the positions of A.

Step 2 (pigeon-hole). There are at least $2n > 2n - 1$ entries equal to 1, so by Theorem 1.1 two of them, $(i, j) \ne (i', j')$, lie in the same D_s.

Step 3 (conclusion). From $i + j = i' + j'$ and $(i, j) \ne (i', j')$ we get $i \ne i'$; say $i' < i$. Then $j' = s - i' > s - i = j$, so (i', j') is strictly above and strictly to the right of (i, j).

The bound is sharp: the $2n - 1$ ones of the last row and the first column contain no such pair: two of them share a row or a column, or they are $(i, 1)$ and (n, j) with $i < n$ and $j > 1$, and then the upper one is to the left. ■

Exercise 1.35 (One number divides another)

(Part (b) of the book's exercise.) Let $n \ge 1$. Prove that among any $n + 1$ distinct integers selected from $[2n]$ there are two such that one is a multiple of the other.

Solution.

Step 1 (boxes). Every positive integer x can be written uniquely as $x = 2^a m$ with $a \ge 0$ and m odd (divide by 2 as long as possible). Call m the odd part of x. For $x \in [2n]$ the odd part lies in $\{1, 3, \dots, 2n - 1\}$, a set of n elements.

Step 2 (pigeon-hole). The $n + 1$ selected integers have their odd parts among n values, so by Theorem 1.1 two different selected integers $x = 2^a m$ and $y = 2^b m$ have the same odd part m.

Step 3 (conclusion). Since $x \ne y$, $a \ne b$; say $a < b$. Then $y = 2^{b-a} x$ is a multiple of x.

The number $n + 1$ cannot be lowered: the n integers $n + 1, \dots, 2n$ contain no such pair, because twice the smallest of them already exceeds $2n$. ■

Exercise 1.37 (Multiples of an irrational number)

Let r be an irrational real number. Prove that there is a positive integer n such that the distance of nr from the nearest integer is less than 10^{-10}.

Solution. Write $N = 10^{10}$ and $\{x\} = x - \lfloor x \rfloor \in [0,1)$ for the fractional part.

Step 1 (boxes). Split $[0,1)$ into the N half-open intervals $I_a = \left[\frac{a}{N}, \frac{a+1}{N}\right)$, $a = 0, \dots, N-1$.

Step 2 (pigeon-hole). The $N+1$ numbers $\{kr\}$, $k = 0, 1, \dots, N$, lie in $[0,1)$, so by Theorem 1.1 two of them, $\{kr\}$ and $\{\ell r\}$ with $0 \le \ell < k \le N$, lie in the same I_a, hence $\bigl|\{kr\} - \{\ell r\}\bigr| < \frac{1}{N}$.

Step 3 (conclusion). Put $n = k - \ell \ge 1$ and $M = \lfloor kr \rfloor - \lfloor \ell r \rfloor \in \mathbb{Z}$. Then

$$|nr - M| = \bigl|(kr - \lfloor kr \rfloor) - (\ell r - \lfloor \ell r \rfloor)\bigr| = \bigl|\{kr\} - \{\ell r\}\bigr| < 10^{-10},$$

so the distance of nr from the nearest integer is less than 10^{-10}. Irrationality guarantees in addition that this distance is positive, so the multiples of r come arbitrarily close to integers without hitting them. ∎

Chapter 2. The Method of Mathematical Induction

2.1 Weak induction

Motivation. To prove a statement for every natural number it suffices to check the first case and show that each case implies the next. Induction turns one infinite task into two finite ones; its price is that the statement to be proved must already be known or guessed.

The method. To prove a statement $S(m)$ for all natural m:

1. *Initial step:* prove $S(m)$ for the smallest m for which it is defined (usually 0 or 1).
2. *Induction step:* prove that $S(n)$ (the *induction hypothesis*) implies $S(n+1)$.

Validity. If both steps are done but S fails somewhere, let $m+1$ be the smallest failure. By the initial step, $m+1$ is not the first index, so $S(m)$ is defined and true by minimality; the induction step with $n = m$ then gives $S(m+1)$, a contradiction.

Example 2.1

For all positive integers m,

$$1^2 + 2^2 + \cdots + m^2 = \frac{m(m+1)(2m+1)}{6}. \tag{2.1}$$

Solution. For $m = 1$ both sides are 1. Assume (2.1) for n. The statement for $n+1$ differs from it by adding $(n+1)^2$ on the left and

$$\frac{(n+1)(n+2)(2n+3) - n(n+1)(2n+1)}{6} = \frac{(n+1)\big[(n+2)(2n+3) - n(2n+1)\big]}{6} = (n+1)^2$$

on the right. Adding this identity to (2.1) for n gives the statement for $n+1$.

Example 2.2

Let $f(m)$ be the maximum number of regions into which m straight lines divide the plane. Then $f(m) = \frac{m(m+1)}{2} + 1$.
Solution. $f(1) = 2$. Given $n+1$ lines, look at the last line s. It crosses $k \le n$ of the other lines, so it passes through $k+1$ regions formed by the others and cuts each of them in two. Hence $f(n+1) \le f(n)+n+1$, with equality when s meets all other lines in distinct new points (lines in general position). Thus $f(n+1) = f(n)+n+1 = \frac{n(n+1)}{2}+1+n+1 = \frac{(n+1)(n+2)}{2}+1$.

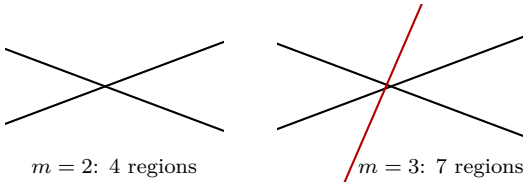


Figure. *Example 2.2: the new (red) line crosses both old lines, passes through 3 old regions and splits each, so* $f(3) = f(2) + 3$.

The formula had to be known in advance; finding it is the weak point of induction. Often it can be guessed from small cases:

Example 2.3

Let $a_0 = 1$ and $a_{n+1} = 3a_n + 1$. Then $a_m = \frac{3^{m+1}-1}{2}$.
Solution. The first values $1, 4, 13, 40, 121$ suggest the formula. It holds for $m = 0$, and if it holds for n, then $a_{n+1} = 3 \cdot \frac{3^{n+1}-1}{2} + 1 = \frac{3^{n+2}-1}{2}$.

A formula like $a_{n+1} = 3a_n + 1$ is a **recurrence relation**; $a_n = \frac{3^{n+1}-1}{2}$ is an **explicit** formula, which is also **closed**: it contains no $\sum$ or $\prod$ over a varying number of terms. For example $f(n) = \sum_{i=1}^{n} i$ is explicit but not closed, while $\frac{n(n+1)}{2}$ is closed. Induction often turns a recurrence into a closed formula.

Sets. A finite **set** is an unordered collection of distinct objects ($\{1, 3, 2\} = \{2, 1, 3\}$); if repetitions are allowed, it is a **multiset**. $B \subseteq A$ means every element of B lies in A. Since enumerative statements usually depend only on the size of a set, we use the canonical n-element set $[n] := \{1, 2, \ldots, n\}$.

Theorem 2.4

For every positive integer n, $[n]$ has 2^n subsets.

Proof. $[1]$ has the two subsets $\emptyset, \{1\}$. Assume the claim for n. Subsets of $[n+1]$ not containing $n+1$ are exactly the subsets of $[n]$: 2^n of them. Those containing $n+1$ are $\{n+1\} \cup T$ with $T \subseteq [n]$ arbitrary: another 2^n. Altogether 2^{n+1}. ■

Pitfalls

The initial step cannot be skipped: "$2n+1$ is even" has a valid induction step ($2(n+1)+1-(2n+1) = 2$) but no true initial case. More subtly, "all horses have the same colour" can be "proved" as follows. Among $n+1$ horses, the first n and the last n are each monochromatic, and they overlap, so all $n+1$ are. But for $n = 1$ the two groups (first horse, last horse) are disjoint, so the step from 1 to 2 fails. The induction step must be valid for *every* n from the initial value on.

2.2 Strong induction

Motivation. Some recurrences express a_{n+1} through all earlier terms, so knowing only a_n is not enough. Strong induction lets the induction step use the statement for *all* smaller values; it is no stronger logically, but it is often the convenient form.

Example 2.5

Let $a_0 = 0$ and $a_{n+1} = a_0 + a_1 + \cdots + a_n + n + 1$ for $n \geq 0$. Then $a_n = 2^n - 1$.
Solution. True for $n = 0$. If $a_k = 2^k - 1$ for all $k \leq n$, then

$$a_{n+1} = \sum_{k=0}^{n} (2^k - 1) + n + 1 = (2^{n+1} - 1) - (n+1) + n + 1 = 2^{n+1} - 1.$$

The method (strong induction).

1. *Initial step:* prove the statement for the smallest n for which it is defined.
2. *Induction step:* prove that the statement for all integers $< n+1$ implies it for $n+1$.

Validity. If $n+1$ were the smallest failure, the statement would hold for all smaller integers (by minimality and the initial step), and the induction step would give it for $n+1$. From now on $\mathbb{N}$ denotes the set of nonnegative integers.

Example 2.6

Let $f(0)=1$, $f(1)=2$ and $f(n+1)=f(n-1)+2f(n)$ for $n \geq 1$. Then $f(n) \leq 3^n$ for all $n \in \mathbb{N}$.
Solution. True for $n=0,1$. If it holds for all values $\leq n$ ($n \geq 1$), then $f(n+1) \leq 3^{n-1}+2\cdot 3^n = 7\cdot 3^{n-1} < 3^{n+1}$. Both $f(n-1) \leq 3^{n-1}$ and $f(n) \leq 3^n$ are used, so the strong hypothesis is needed.

Variations

It is often convenient to shift the parameter (assume the statement for $n-1$, prove it for n), or to induct over even or odd integers only; these variants are equally valid and are used in later chapters.

2.3 Selected exercises

Motivation. In these book exercises the difficulty is not the induction itself but choosing the right statement and the right object to remove in the induction step.

Exercise 2.1 (Iterated differences of a polynomial)

For a polynomial p let $\Delta p(n) = p(n) - p(n-1)$, and let $\Delta^k p$ be the result of applying Δ k times ($\Delta^0 p = p$). Prove that if p has degree d, then $\Delta^d p$ is a constant.

Solution. We prove the sharper claim: if p has degree d and leading coefficient $a \neq 0$, then $\Delta^d p$ is the constant $d!\,a$.

Step 1 (one difference lowers the degree). For $j \geq 1$, the binomial theorem (Theorem 4.1) gives

$$n^j - (n-1)^j = n^j - \sum_{i=0}^{j} \binom{j}{i} n^i (-1)^{j-i} = j\, n^{j-1} + (\text{terms of degree} \leq j-2),$$

and for $j=0$ the difference is 0. Writing $p(n) = \sum_{j=0}^{d} c_j n^j$ with $c_d = a$ and applying this to each term,

$$\Delta p(n) = \sum_{j=1}^{d} c_j (n^j - (n-1)^j) = d\,a\, n^{d-1} + (\text{terms of degree} \leq d-2).$$

So for $d \geq 1$, Δp has degree $d-1$ and leading coefficient $d\,a$.

Step 2 (induction on d). For $d=0$, $\Delta^0 p = p$ is the constant $a = 0!\,a$. Let $d \geq 1$ and assume the claim for degree $d-1$. By Step 1, $q = \Delta p$ has degree $d-1$ and leading coefficient da, so by the induction hypothesis $\Delta^{d-1} q = (d-1)! \cdot da = d!\,a$. Since $\Delta^d p = \Delta^{d-1}(\Delta p) = \Delta^{d-1} q$, the claim holds for d. ∎

Exercise 2.2 (A player who lists everybody)

At a tennis tournament every two players played each other exactly once. Afterwards each player listed the names of those he defeated and of those defeated by someone he defeated.

Prove that some player listed the names of everybody else.

Solution. Say that x *covers* y if x beat y, or x beat some z who beat y. We prove by induction on the number $n \geq 1$ of players that some player covers all others.

Step 1 (initial step). For $n = 1$ there is nobody else to list, and the claim is vacuous.

Step 2 (induction step). Let $n \geq 1$, consider a tournament with $n+1$ players and fix a player X. Among the other n players (with the games between them), the induction hypothesis gives a player K who covers all of them. If K beat X, or K beat someone who beat X, then K also covers X and we are done.

Step 3 (the remaining case). Otherwise X beat K, and X beat every player z beaten by K (else z beat X and K would cover X). Now take any $y \neq X$. If $y = K$, then X beat y. If K beat y, then X beat y. If K beat some z who beat y, then X beat z and z beat y. In every case X covers y, so X lists everybody else. ∎

Exercise 2.15 (Arithmetic and geometric means)

Prove by induction that for all nonnegative reals $a_1, \ldots, a_n$,

$$\sqrt[n]{a_1 a_2 \cdots a_n} \leq \frac{a_1 + a_2 + \cdots + a_n}{n}.$$

Solution. Equivalently, $a_1 \cdots a_n \leq A^n$ where A is the arithmetic mean. We induct on n; for $n = 1$ there is equality.

Step 1 (setting up). Let $n \geq 1$, assume the claim for n numbers, and let $a_1, \ldots, a_{n+1} \geq 0$ have mean A. The claim does not depend on the order, so we may assume $a_n = \min_i a_i$ and $a_{n+1} = \max_i a_i$; then $a_n \leq A \leq a_{n+1}$.

Step 2 (replacing two numbers). Put $b = a_n + a_{n+1} - A$. Then $b \geq a_n \geq 0$, and

$$A\,b - a_n a_{n+1} = (A - a_n)(a_{n+1} - A) \geq 0, \qquad \text{so} \qquad a_n a_{n+1} \leq A\,b.$$

Step 3 (applying the hypothesis). The n numbers $a_1, \ldots, a_{n-1}, b$ are nonnegative with sum $(n+1)A - A = nA$, i.e. with mean A. By the induction hypothesis $a_1 \cdots a_{n-1}\, b \leq A^n$. Hence

$$a_1 \cdots a_{n-1} a_n a_{n+1} \leq a_1 \cdots a_{n-1} \cdot A\,b \leq A^n \cdot A = A^{n+1}.$$

∎

Exercise 2.17 (An L-shape of 1's)

(a) Let $m, n \geq 2$ and let A be an $m \times n$ matrix with entries 0 and 1, at least $m + n - 1$ of which are 1. Prove that some entry 1 has another 1 in its row and another 1 in its column.
(b) Does the statement remain true with only $m + n - 2$ entries 1?

Solution.

(a) Suppose no entry 1 has the required property. Then every 1 is alone in its row or alone in its column. Let R be the set of 1's alone in their row, and C the set of the remaining 1's; each element of C is alone in its column.

Step 1 (counting). Distinct elements of R lie in distinct rows and distinct elements of C in distinct columns, so $|R| \leq m$ and $|C| \leq n$. Since $|R| + |C| \geq m + n - 1$, either $|C| = n$ or $|R| = m$.

Step 2 (case $|C| = n$). Every column contains an element of C, which is the only 1 of that column. So every 1 of A lies in C, and $R = \emptyset$. Then $n = |C| \geq m + n - 1$ gives $m \leq 1$, a contradiction.

Step 3 (case $|R| = m$). Symmetrically, every row contains exactly one 1, all 1's lie in R, $C = \emptyset$, and $m \geq m + n - 1$ gives $n \leq 1$, a contradiction.

(b) No. Put 1's at the positions $(1,1), \ldots, (1, n-1)$ of the first row and $(2, n), \ldots, (m, n)$ of the last column: $m + n - 2$ ones. A 1 in the first row is the only 1 of its column (columns $1, \ldots, n-1$ have no other 1), and a 1 in the last column is the only 1 of its row (rows $2, \ldots, m$). So no 1 has the required property. ■

Chapter 3. Elementary Counting Problems

Pigeon-holes and induction draw conclusions *from* numbers; this chapter develops the basic techniques for *finding* them.

3.1 Permutations

Motivation. Arranging distinct objects in a row is the most basic counting problem: n choices for the first place, $n-1$ for the second, and so on. Once that is settled, repeated objects are handled by labelling them to make them distinct and then dividing out the labellings.

n patients at a dentist can be ordered in n ways for the first chair, then $n-1$ for the second (for *each* first choice), and so on, giving $n(n-1)\cdots 2\cdot 1$ orders.

Definition 3.1 (Permutation, factorial)

An arrangement of distinct objects in a linear order, using each exactly once, is a **permutation** of them. The number $n! := n(n-1)\cdots 2\cdot 1$ of permutations of n objects is n **factorial**.

Theorem 3.2

An n-element set has $n!$ permutations.

Proof. There are n choices for the first position, and for each of them $n-1$ for the second, …, and 1 for the last; the choices multiply. ∎

By convention $0! = 1$: then $n!\,m!$ correctly counts the ways for two rooms of n and m people to form lines, also when $n = 0$. For the size of $n!$ we quote without proof **Stirling's formula**

$$n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n, \qquad \text{i.e.} \quad \lim_{n\to\infty} \frac{n!}{\sqrt{2\pi n}(n/e)^n} = 1. \tag{3.1}$$

Example 3.3

Flags with three horizontal stripes of the distinct colours red, white, green: $3! = 6$ (RWG, RGW, WRG, WGR, GWR, GRW).

Example 3.4

Five red, three yellow and two white flowers can be planted in a row in $A = \frac{10!}{5!\,3!\,2!} = 2520$ ways.
Solution. Label the flowers of each colour to make all ten distinct. Each of the A colour patterns can be labelled in $5!\,3!\,2!$ ways, and the labelled rows are the $10!$ permutations of ten distinct objects. Hence $A \cdot 5!\,3!\,2! = 10!$.

A collection with repetitions is a **multiset**; a linear order of all its elements is a **multiset permutation**.

Theorem 3.5 (Multiset permutations)

Let $a_1 + \cdots + a_k = n$ be nonnegative integers. A multiset with a_i objects of type i ($i \in [k]$)

has

$$\frac{n!}{a_1!\,a_2!\cdots a_k!}$$

linear orders.

Proof [Supplement]. Let A be the number of linear orders. Attach labels $1,\dots,a_i$ to the objects of type i; this makes all n objects distinct. Every permutation of the n labelled objects arises exactly once from a pair (linear order of the multiset, labelling), and for a fixed linear order the labels of type i can be assigned in $a_i!$ ways, independently for different types. Hence $A \cdot a_1!\cdots a_k! = n!$ by Theorem 3.2. ■

3.2 Strings over a finite alphabet

Motivation. Words of length k over an alphabet count functions $[k] \to$ alphabet; forbidding repeated letters counts injections. Bijections let us transfer such counts to other problems, which is the most flexible tool of enumeration.

Theorem 3.6

There are n^k strings of length k over an n-element alphabet.

Proof. Each of the k positions can independently be filled in n ways (letters may repeat). ■

Example 3.7

There are $9 \cdot 10^{k-1}$ positive integers with k digits: of the 10^k strings over $\{0,\dots,9\}$, exactly 10^{k-1} start with 0. (Or: 9 choices for the first digit, 10 for each other.)

The bijective method. To check that a ballroom holds as many men as women, ask them to form couples; if nobody is left over, the numbers agree.

Definition 3.8 (Bijection)

Let X, Y be finite sets and $f : X \to Y$ a function such that (1) $f(a) = f(b) \Rightarrow a = b$, and (2) every $y \in Y$ equals $f(x)$ for some $x \in X$. Then f is a **bijection** from X onto Y; equivalently, each $y \in Y$ has exactly one preimage.

Definition 3.9 (Injection, surjection)

A function satisfying (1) is **one-to-one** (**injective**); one satisfying (2) is **onto** (**surjective**).

Proposition 3.10

If there is a bijection from a finite set X onto a finite set Y, then $|X| = |Y|$.

Proof. A bijection partitions $X \cup Y$ into pairs $\{x, f(x)\}$ with one element from each set, every element lying in exactly one pair. If there are m pairs, then $|X| = |Y| = m$. ■

Example 3.11

An n-element set has 2^n subsets (without induction, cf. Theorem 2.4).
Solution. Map $B \subseteq [n]$ to the binary string whose i-th digit is 1 iff $i \in B$. Every binary string of length n comes from exactly one B, namely the set of positions of its 1's. By Theorem 3.6 there are 2^n such strings.

Example 3.12

Ten new intersections each get a traffic light or not, and lit ones may also get a gas station. Recording each intersection as A (light and gas station), B (light only) or C (neither) is a bijection onto words of length 10 over $\{A, B, C\}$, so there are 3^{10} possibilities.

Theorem 3.13

For $n \geq k$, the number of strings of length k over an n-element alphabet with no repeated letter is

$$(n)_k := n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!}.$$

Proof. n choices for the first letter, $n-1$ for the second, …, $n-k+1$ for the k-th: the argument for $n!$, stopped after k steps. ∎

Example 3.14

Filling five different cabinet posts from 20 candidates: $(20)_5 = 20 \cdot 19 \cdot 18 \cdot 17 \cdot 16$ ways.

3.3 Choice problems

Motivation. When only the set of chosen elements matters and not their order, we count subsets; every ordered selection of k elements corresponds to $k!$ orderings of the same subset. Shifting tricks then handle "no two consecutive" and "repetitions allowed".

How many tickets guarantee the jackpot of a lottery that draws 5 numbers from [90]? The order of the numbers does not matter.

Definition 3.15 (Binomial coefficient)

$\binom{n}{k}$ ("n choose k") denotes the number of k-element subsets of $[n]$; these are the **binomial coefficients** (Chapter 4).

Theorem 3.16

For $0 \leq k \leq n$: $\displaystyle\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{(n)_k}{k!}.$

Proof. By Theorem 3.13 there are $(n)_k$ strings of k distinct elements of $[n]$. Each k-element subset appears among them exactly $k!$ times, once for each ordering of its elements (Theorem 3.2). Hence $\binom{n}{k} \cdot k! = (n)_k$. ∎

So the lottery needs $\binom{90}{5} = 43\,949\,268$ tickets.

Definition 3.17 (Complement)

For $S \subseteq [n]$, the **complement** S^c is the set of $i \in [n]$ with $i \notin S$.

Proposition 3.18

For $0 \le k \le n$:
(1) $\binom{n}{k} = \binom{n}{n-k}$;
(2) $\binom{n}{0} = \binom{n}{n} = 1$.

Proof. (1) $S \mapsto S^c$ maps k-subsets of $[n]$ to $(n-k)$-subsets, and every $(n-k)$-subset T is the image of exactly one S, namely T^c. Apply Proposition 3.10.

(2) The only 0-element subset is $\emptyset$; the other equality follows from (1). ∎

In particular $\binom{0}{0} = 1$, and we set $\binom{n}{k} := 0$ for $k > n$.

Example 3.19

A student must work five days in January, never on two consecutive days. This can be done in $\binom{27}{5}$ ways.
Solution. For chosen days $a_1 < \cdots < a_5$ in $[31]$, "no two consecutive" is equivalent to $1 \le a_1 < a_2 - 1 < a_3 - 2 < a_4 - 3 < a_5 - 4 \le 27$. So $(a_i) \mapsto (a_i - (i-1))$ is a bijection onto the 5-element subsets of $[27]$.

Example 3.20

If the five lottery numbers from $[90]$ are drawn *with replacement* and only the multiset counts, $\binom{94}{5}$ tickets are needed.
Solution. Running the previous trick backwards, $f(b_1, \dots, b_5) = (b_1, b_2+1, b_3+2, b_4+3, b_5+4)$ maps multisets

$$1 \le b_1 \le b_2 \le b_3 \le b_4 \le b_5 \le 90 \tag{3.2}$$

bijectively onto the 5-element subsets of $[94]$: the b_i satisfy (3.2) iff the shifted values are strictly increasing in $[94]$.

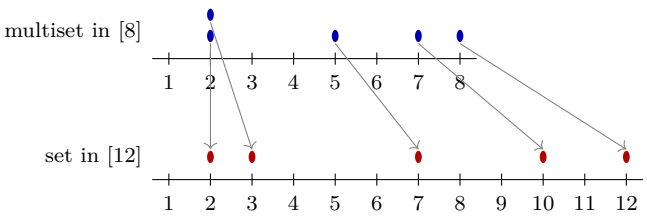


Figure. *Example 3.20 with smaller numbers: the multiset* $\{2,2,5,7,8\}$ *of* $[8]$ *is sent to the set* $\{2,3,7,10,12\}$ *of* $[12]$ *by adding* $0,1,2,3,4$*; equal entries become distinct.*

Theorem 3.21

The number of k-element multisets with elements from $[n]$ is $\binom{n+k-1}{k}$.

Proof [Supplement]. Write a multiset as $1 \le b_1 \le \cdots \le b_k \le n$ and let $f(b_1, \dots, b_k) := (b_1, b_2 + 1, \dots, b_k + k - 1)$. The inequalities $b_i \le b_{i+1}$ are equivalent to $b_i + (i-1) < b_{i+1} + i$, and $b_k \le n$ to $b_k + k - 1 \le n + k - 1$. So f is a bijection onto the k-element subsets of $[n+k-1]$, with inverse

subtracting $0, 1, \dots, k-1$ from the increasingly ordered elements. Conclude with Proposition 3.10. ■

	Parameters	Formula
Permutations	n distinct objects	$n!$
	a_i objects of type i, $\sum a_i = n$	$\frac{n!}{a_1!\cdots a_k!}$
Lists	k of n distinct objects	$(n)_k = \frac{n!}{(n-k)!}$
	words of length k over n letters	n^k
Subsets	k-element subsets of $[n]$	$\binom{n}{k}$
	k-element multisets from $[n]$	$\binom{n+k-1}{k}$

3.4 Selected exercises

Motivation. These book exercises show the two basic tools of the chapter at work: interpreting a quotient of factorials as a count, and building a bijection or recursive construction instead of computing.

Exercise 3.17 (Products of factorials)

Let $k \geq 1$ and let $b_1, \dots, b_k$ be positive integers whose sum is less than the positive integer n. Prove that $b_1!\, b_2! \cdots b_k! < n!$. Can the statement be made stronger?

Solution.

Step 1 (a count). Let $s = b_1 + \cdots + b_k$, so $1 \leq s \leq n-1$. By Theorem 3.5, a multiset with b_i objects of type i has $\frac{s!}{b_1!\cdots b_k!}$ permutations. This number counts a nonempty set (for instance, all objects of type 1 first, then type 2, and so on), so it is a positive integer, and

$$b_1!\, b_2! \cdots b_k! \leq s!.$$

Step 2 (conclusion). Since $s \leq n-1$, $s! \leq (n-1)!$, and $(n-1)! < n!$ because $n \geq 2$. Hence $b_1! \cdots b_k! \leq (n-1)! = \frac{n!}{n} < n!$.

This is the stronger statement: the product is at most $\frac{n!}{n}$, and even at most $(b_1 + \cdots + b_k)!$, with equality iff $k = 1$ (for $k \geq 2$ two different types can be swapped, so there are at least two arrangements). ■

Exercise 3.26 (Permutations without peaks)

How many permutations of $[n]$ have no entry that is larger than both of its neighbours? (The first and the last entry have only one neighbour and never violate the condition.)

Solution. Call such permutations *peakless* and let a_n be their number. We show $a_n = 2^{n-1}$ for $n \geq 1$.

Step 1 (where n is). In a peakless permutation of $[n]$, $n \geq 2$, the entry n is larger than all its neighbours, so it has only one neighbour: n is the first or the last entry.

Step 2 (removing n). Deleting n from a peakless permutation of $[n]$ gives a peakless permutation of $[n-1]$: the entry next to n becomes an end entry, and every other entry keeps its neighbours.

Step 3 (inserting n). Conversely, let q be peakless on $[n-1]$ and put n at the front or at the end. The new entry n is an end entry. The entry next to it now has the neighbour n, which is larger, so it is not a peak; all other entries keep their neighbours. So both results are peakless, and they are different for $n \geq 2$.

Step 4 (counting). Steps 1–3 give a bijection between peakless permutations of $[n]$ and pairs (q, side), so $a_n = 2a_{n-1}$ for $n \geq 2$. With $a_1 = 1$ this gives $a_n = 2^{n-1}$. ■

Exercise 3.57 (Matrices with even lines)

How many $n \times n$ matrices with entries 0 and 1 have an even sum in each row and each column?

Solution. The answer is $2^{(n-1)^2}$. Call such matrices *even*; all congruences below are modulo 2.

Step 1 (the map). Send an even matrix M to its upper left $(n-1) \times (n-1)$ block B. There are $2^{(n-1)^2}$ possible blocks (Theorem 3.6), so it suffices to show that every B extends to exactly one even matrix.

Step 2 (forced entries). Given $B = (b_{ij})$, the row condition forces $m_{in} \equiv \sum_{j<n} b_{ij}$ for $i < n$, and the column condition forces $m_{nj} \equiv \sum_{i<n} b_{ij}$ for $j < n$. Each entry is 0 or 1, so these entries are uniquely determined.

Step 3 (the corner). Row n is even iff $m_{nn} \equiv \sum_{j<n} m_{nj} \equiv \sum_{i,j<n} b_{ij}$, and column n is even iff $m_{nn} \equiv \sum_{i<n} m_{in} \equiv \sum_{i,j<n} b_{ij}$. Both conditions require the same value, so exactly one choice of m_{nn} works.

Step 4 (conclusion). Every block extends to exactly one even matrix, so the map of Step 1 is a bijection and there are $2^{(n-1)^2}$ even matrices (Proposition 3.10). ■

Chapter 4. The Binomial Theorem and Related Identities

This chapter pauses the enumeration of basic situations to study binomial and multinomial coefficients themselves. The proofs matter more than the identities: most show that both sides count the same objects in two different ways (**double counting**).

4.1 The binomial theorem

Motivation. Expanding $(x+y)^n$ means choosing x or y from each factor, so the coefficients count subsets. Specialising x and y produces identities, and double counting produces many more. Together they explain the patterns of the Pascal triangle.

Theorem 4.1 (Binomial theorem)

For every nonnegative integer n,

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}. \tag{4.1}$$

Proof. Multiplying out $(x+y)(x+y)\cdots(x+y)$ means choosing one summand from each of the n factors in all 2^n ways and adding the products. A choice gives $x^k y^{n-k}$ exactly when x is taken from a k-element set of factors, and there are $\binom{n}{k}$ such sets. ■

Theorem 4.2

For every positive integer n: $\sum_{k=0}^{n} (-1)^k \binom{n}{k} = 0$.

Proof. Put $x=-1$, $y=1$ in (4.1): the left side is $0^n = 0$. ■

Theorem 4.3 (Pascal's rule)

For all nonnegative integers n, k:

$$\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}. \tag{4.2}$$

Proof. The right side counts $(k+1)$-subsets S of $[n+1]$. If $n+1 \in S$, the rest of S is a k-subset of $[n]$ (first term); if not, S is a $(k+1)$-subset of $[n]$ (second term). ■

Theorem 4.4

For every nonnegative integer n: $2^n = \sum_{k=0}^{n} \binom{n}{k}$.

Proof. Both sides count the subsets of $[n]$: the left directly (Theorem 2.4), the right grouped by size. (Alternatively, put $x=y=1$ in (4.1).) ■

Writing $\binom{n}{i}$ as the i-th entry of row n (rows starting at 0) gives the **Pascal triangle**. Row n sums to 2^n (Theorem 4.4), each entry is the sum of the two above it (Theorem 4.3), and alternating row sums vanish (Theorem 4.2).

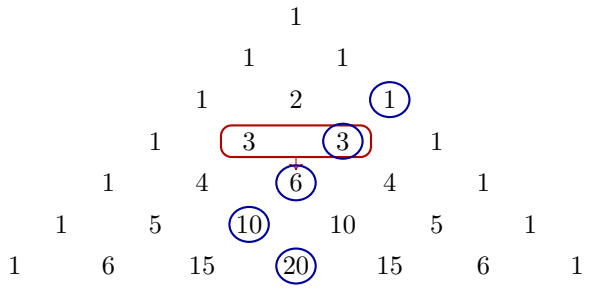


Figure. *The Pascal triangle, rows 0–6. Red: $3+3=6$ (Theorem 4.3). Blue: $1+3+6+10=20$, a "hockey stick" sum down a diagonal (Theorem 4.5).*

Theorem 4.5 (Hockey stick identity)

For all nonnegative integers $k \le n$:

$$\binom{k}{k} + \binom{k+1}{k} + \binom{k+2}{k} + \cdots + \binom{n}{k} = \binom{n+1}{k+1}. \tag{4.3}$$

Proof. The right side counts $(k+1)$-subsets of $[n+1]$. Sort them by their largest element $k+i+1$ $(0 \le i \le n-k)$: the other k elements form a k-subset of $[k+i]$, so there are $\binom{k+i}{k}$ of them. ∎

Theorem 4.6

For every nonnegative integer n:

$$\sum_{k=1}^{n} k\binom{n}{k} = n2^{n-1}. \tag{4.4}$$

Proof.

Step 1 (double counting). Count the ways to choose a committee from n people together with its president. Choosing a k-member committee first gives $k\binom{n}{k}$, summed over k. Choosing the president first (n ways) and then any subset of the other $n-1$ people gives $n2^{n-1}$.

Step 2 (calculus, alternative). Differentiate $(x+1)^n = \sum_{k=0}^{n}\binom{n}{k}x^k$ to get $n(x+1)^{n-1} = \sum_{k\ge 1} k\binom{n}{k}x^{k-1}$, and put $x=1$. ∎

(It is not even obvious a priori that $\sum_k k\binom{n}{k}/2^{n-1}$ is an integer.)

Theorem 4.7 (Vandermonde's identity)

For all positive integers n, m, k:

$$\binom{n+m}{k} = \sum_{i=0}^{k}\binom{n}{i}\binom{m}{k-i}.$$

Proof. Classify the k-subsets of $[n+m]$ by the number i of elements taken from $[n]$: these can be chosen in $\binom{n}{i}$ ways and the other $k-i$ from $\{n+1,\ldots,n+m\}$ in $\binom{m}{k-i}$ ways. ∎

Theorem 4.8

If $0 \le k \le \frac{n-1}{2}$, then

$$\binom{n}{k} \le \binom{n}{k+1}, \tag{4.5}$$

with equality iff $n = 2k+1$.

Proof. Dividing $\frac{n!}{k!(n-k)!} \le \frac{n!}{(k+1)!(n-k-1)!}$ by $n!$ and multiplying by $k!(n-k-1)!$, the inequality becomes $\frac{1}{n-k} \le \frac{1}{k+1}$, i.e. $2k+1 \le n$. The steps are reversible, and equality holds iff $n-k = k+1$. ■

Corollary 4.9

If $k \ge \frac{n-1}{2}$ (and $k+1 \le n$), then

$$\binom{n}{k} \ge \binom{n}{k+1}, \tag{4.6}$$

with equality iff $n = 2k+1$.

Proof. By Proposition 3.18(1), $\binom{n}{k} = \binom{n}{n-k}$ and $\binom{n}{k+1} = \binom{n}{n-k-1}$. With $k' := n-k-1 \le \frac{n-1}{2}$, Theorem 4.8 gives $\binom{n}{k'} \le \binom{n}{k'+1}$, which is the claim, with equality iff $n = 2k'+1 = 2(n-k-1)+1$, i.e. $n = 2k+1$. ■

A sequence $a_0, \dots, a_n$ is **unimodal** if $a_0 \le \cdots \le a_m \ge \cdots \ge a_n$ for some m. By Theorem 4.8 and Corollary 4.9, every row of the Pascal triangle is unimodal, with its peak in the middle.

4.2 The multinomial theorem

Motivation. For $(x_1 + \cdots + x_k)^n$ we choose one of k variables from each factor, so the coefficient of $x_1^{a_1} \cdots x_k^{a_k}$ counts words with prescribed letter multiplicities: the multiset permutations of Chapter 3.

Example 4.10

$$(x+y+z)^3 = x^3 + y^3 + z^3 + 3x^2y + 3x^2z + 3y^2x + 3y^2z + 3z^2x + 3z^2y + 6xyz. \tag{4.7}$$

Solution. Each of the 27 choices of one variable per factor gives a degree-3 monomial. x^2y arises in 3 ways (choose the factor giving y), x^3 in one way, and xyz in $3 \cdot 2 = 6$ ways.

Definition 4.11 (Multinomial coefficient)

For nonnegative integers with $n = \sum_{i=1}^{k} a_i$,

$$\binom{n}{a_1, a_2, \dots, a_k} := \frac{n!}{a_1!\, a_2! \cdots a_k!}. \tag{4.8}$$

For $k = 2$, $\binom{n}{a_1, a_2} = \binom{n}{a_1}$.

Theorem 4.12 (Multinomial theorem)

For all nonnegative integers n, k,

$$(x_1 + \cdots + x_k)^n = \sum_{a_1+\cdots+a_k=n} \binom{n}{a_1, \ldots, a_k} x_1^{a_1} \cdots x_k^{a_k}, \tag{4.9}$$

the sum over all k-tuples of nonnegative integers with sum n.

Proof. A term $x_1^{a_1} \cdots x_k^{a_k}$ arises from choosing x_i in exactly a_i of the n factors. Such choices correspond bijectively to words of length n with a_i letters x_i: the j-th letter is the variable chosen from the j-th factor. By Theorem 3.5 there are $\binom{n}{a_1,\ldots,a_k}$ such words. ■

Theorem 4.13

For nonnegative integers with $n = \sum_i a_i$:

$$\binom{n}{a_1, \ldots, a_k} = \binom{n}{a_1}\binom{n-a_1}{a_2} \cdots \binom{n-a_1-\cdots-a_{k-1}}{a_k} \tag{4.10}$$

(the last factor is $\binom{a_k}{a_k} = 1$).

Proof. The left side counts words with a_i copies of x_i. Build such a word by choosing the a_1 positions of x_1 ($\binom{n}{a_1}$ ways), then the a_2 positions of x_2 among the remaining $n - a_1$, and so on. ■

4.3 When the exponent is not a positive integer

Motivation. Generating functions (Chapter 8) need expansions like $(1-4x)^{-1/2}$. Extending $\binom{m}{k}$ to real m turns the binomial theorem into a power series, Newton's binomial series.

Definition 4.14 (General binomial coefficients)

For real m and integer $k \geq 0$: $\binom{m}{0} := 1$ and $\binom{m}{k} := \frac{m(m-1)\cdots(m-k+1)}{k!}$ for $k > 0$.

Theorem 4.15 (Binomial theorem, general version)

For every real m and $|x| < 1$,

$$(1+x)^m = \sum_{n \geq 0} \binom{m}{n} x^n.$$

Proof [Supplement]. The book derives this from Taylor's theorem, since the n-th derivative of $(1+x)^m$ at 0 is $m(m-1)\cdots(m-n+1)$. Here is a complete argument.

Step 1 (convergence). If $m \in \mathbb{N}$ the series is a finite sum and (4.1) applies. Otherwise all $\binom{m}{n} \neq 0$ and $\left|\binom{m}{n+1}x^{n+1} / \binom{m}{n}x^n\right| = \frac{|m-n|}{n+1}|x| \to |x| < 1$. By the ratio test the series converges on $(-1, 1)$ and defines a differentiable f with $f(0) = 1$, differentiable term by term.

Step 2 (differential equation). Using $(n+1)\binom{m}{n+1} = (m-n)\binom{m}{n}$,

$$(1+x)f'(x) = \sum_n \left[(n+1)\binom{m}{n+1} + n\binom{m}{n}\right]x^n = \sum_n m\binom{m}{n}x^n = mf(x).$$

Step 3. Then $\frac{d}{dx}\left[f(x)(1+x)^{-m}\right] = (1+x)^{-m-1}\left[(1+x)f' - mf\right] = 0$, so $f(x)(1+x)^{-m}$ is constant, equal to its value 1 at $x = 0$. ■

If m is a positive integer, $\binom{m}{n} = 0$ for $n > m$ and we recover (4.1).

Example 4.16

$$\frac{1}{\sqrt{1-4x}} = \sum_{n\geq 0} \binom{2n}{n} x^n.$$

Solution. By Theorem 4.15, $(1-4x)^{-1/2} = \sum_n \binom{-1/2}{n}(-4x)^n$ and

$$\binom{-1/2}{n} = \frac{(-\frac{1}{2})(-\frac{3}{2})\cdots(-\frac{2n-1}{2})}{n!} = (-1)^n \frac{(2n-1)!!}{2^n n!}, \tag{4.11}$$

where $(2n-1)!! = 1\cdot 3\cdots(2n-1)$ is the **semifactorial** and $(-1)!! := 1$. The coefficient of x^n is therefore $\frac{2^n(2n-1)!!}{n!}$. Multiplying numerator and denominator by $n!$ and using $2^n n! = 2\cdot 4\cdots 2n$, this equals $\frac{(2n)!}{n!\,n!} = \binom{2n}{n}$, which also holds for $n = 0$.

4.4 Selected exercises

Motivation. The book exercises below prove identities and inequalities for binomial coefficients in three typical ways: double counting, comparing consecutive ratios, and a bijection on lattice paths.

Exercise 4.4 (Choosing two disjoint subsets)

Let k, m, n be nonnegative integers with $k + m \leq n$. Prove that

$$\binom{n}{m}\binom{n-m}{k} = \binom{n}{k}\binom{n-k}{m}.$$

Solution. Count the ordered pairs (M, K) of disjoint subsets of $[n]$ with $|M| = m$ and $|K| = k$.

Step 1 (M first). Choose M in $\binom{n}{m}$ ways, then K among the $n-m$ remaining elements in $\binom{n-m}{k}$ ways. Every pair arises exactly once, so there are $\binom{n}{m}\binom{n-m}{k}$ pairs.

Step 2 (K first). Choosing K first and then M in $[n]\setminus K$ gives $\binom{n}{k}\binom{n-k}{m}$ for the same number.

Comparing the two counts proves the identity. (Both sides equal $\binom{n}{m,k,n-m-k}$, by Theorem 4.13.) ■

Exercise 4.21 (Log-concavity of binomial coefficients)

(a) Prove that for positive integers $k \leq n-1$,

$$\binom{n}{k-1}\binom{n}{k+1} \leq \binom{n}{k}^2.$$

(b) (Exercise 4.23.) A sequence $a_0, \ldots, a_n$ of positive reals is *log-concave* if $a_{i-1}a_{i+1} \leq a_i^2$ for $1 \leq i \leq n-1$. Prove that a log-concave sequence is unimodal.

Solution.

(a) From $\binom{n}{j} = \frac{n!}{j!(n-j)!}$ we get, for $1 \le k \le n-1$,

$$\frac{\binom{n}{k-1}}{\binom{n}{k}} = \frac{k}{n-k+1}, \qquad \frac{\binom{n}{k+1}}{\binom{n}{k}} = \frac{n-k}{k+1}.$$

Multiplying, $\binom{n}{k-1}\binom{n}{k+1}/\binom{n}{k}^2 = \frac{k(n-k)}{(k+1)(n-k+1)} < 1$, since $k < k+1$ and $0 < n-k < n-k+1$. This is the claim, even with strict inequality.

(b)

Step 1 (ratios). Let $r_i = a_i/a_{i-1}$ for $1 \le i \le n$. Dividing $a_{i-1}a_{i+1} \le a_i^2$ by the positive number $a_{i-1}a_i$ gives $r_{i+1} \le r_i$, so $r_1 \ge r_2 \ge \cdots \ge r_n$.

Step 2 (the peak). Let m be the largest index $i \in [n]$ with $r_i \ge 1$, and $m = 0$ if there is none. For $i \le m$ we have $r_i \ge r_m \ge 1$, i.e. $a_{i-1} \le a_i$; for $i > m$ we have $r_i < 1$, i.e. $a_{i-1} > a_i$. Hence $a_0 \le \cdots \le a_m > a_{m+1} > \cdots > a_n$, and the sequence is unimodal.

Together, (a) and (b) give a second proof that the rows of the Pascal triangle are unimodal. ■

Exercise 4.24 (Catalan numbers by reflection)

Let C_n be the number of north-eastern lattice paths (steps $(1,0)$ and $(0,1)$) from $(0,0)$ to (n,n) that never go above the diagonal $x = y$. Prove that

$$C_n = \binom{2n}{n} - \binom{2n}{n-1} = \frac{1}{n+1}\binom{2n}{n}.$$

Solution. A path from $(0,0)$ to (n,n) has n east and n north steps, so there are $\binom{2n}{n}$ such paths (choose the positions of the east steps). Along a path let $d = y - x$; each step changes d by ± 1. A path is *bad* if it goes above the diagonal, i.e. if it visits a point with $d = 1$. We show that the bad paths are equinumerous with the paths from $(-1,1)$ to (n,n), of which there are $\binom{2n}{n-1}$ ($n+1$ east and $n-1$ north steps).

Step 1 (reflection). Let $\varphi(x,y) = (y-1, x+1)$ be the reflection in the line $y = x+1$. It fixes the points with $d = 1$, maps a point with d to one with $2-d$, and turns an east step into a north step and vice versa. Also $\varphi(0,0) = (-1,1)$.

Step 2 (bad paths to paths from $(-1,1)$). For a bad path P let t be its first point with $d = 1$. Reflect the part of P from $(0,0)$ to t by φ and keep the rest. The result is a lattice path from $(-1,1)$ to (n,n).

Step 3 (the inverse). A lattice path Q from $(-1,1)$ to (n,n) starts with $d = 2$ and ends with $d = 0$, and d moves by ± 1, so Q visits a point with $d = 1$. Reflect Q up to its first such point t. Since φ maps points with $d \ne 1$ to points with $d \ne 1$, the first point with $d = 1$ is the same before and after reflecting. Hence the two operations are inverse to each other, and Step 2 is a bijection.

Step 4 (conclusion). $C_n = \binom{2n}{n} - \binom{2n}{n-1}$. Since $\binom{2n}{n-1} = \binom{2n}{n} \cdot \frac{n}{n+1}$ (by the ratio in Exercise 4.21(a)), $C_n = \binom{2n}{n}\left(1 - \frac{n}{n+1}\right) = \frac{1}{n+1}\binom{2n}{n}$. For $n = 0$ both sides equal 1.

This is the formula found with generating functions in Chapter 8. ■

Chapter 5. Partitions

Chapter 3 counted lists. Here we count *distributions* of n balls into k boxes. Balls may be identical or distinct, and so may boxes, which gives compositions, set partitions and integer partitions.

5.1 Compositions

Motivation. Identical balls into distinct boxes: only the number of balls per box matters, so we count ways to write n as an ordered sum. Lining up balls and separating walls turns this into a multiset permutation.

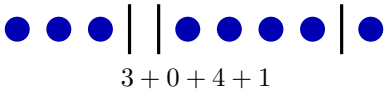
Definition 5.1 (Compositions)

A sequence $(a_1, \dots, a_k)$ of integers with $a_i \geq 0$ and $a_1 + \cdots + a_k = n$ is a **weak composition** of n into k parts; if all $a_i > 0$, it is a **composition**.

Theorem 5.2

For positive integers n, k, the number of weak compositions of n into k parts is $\binom{n+k-1}{k-1} = \binom{n+k-1}{n}$.

Proof. This counts the ways to put n identical balls into k distinct boxes. Line up the boxes and record the contents as a row of n balls and $k-1$ walls. The row determines the distribution and vice versa, so we count permutations of a multiset with n balls and $k-1$ walls: $\frac{(n+k-1)!}{n!(k-1)!}$ by Theorem 3.5. ■



$3 + 0 + 4 + 1$

Figure. *Balls and walls: the weak composition* $(3, 0, 4, 1)$ *of* 8 *into* 4 *parts as an arrangement of* 8 *balls and* 3 *walls.*

Corollary 5.3

The number of compositions of n into k parts is $\binom{n-1}{k-1}$.

Proof [Supplement]. Subtracting 1 from every part is a bijection from compositions of n into k parts onto weak compositions of $n-k$ into k parts (give each box one ball first). By Theorem 5.2 there are $\binom{n-k+k-1}{k-1} = \binom{n-1}{k-1}$ of them. ■

Corollary 5.4

A positive integer n has 2^{n-1} compositions (into any number of parts).

Proof.

Step 1 (summing). A composition has between 1 and n parts, so by Corollary 5.3 and Theorem 4.4 the number is $\sum_{k=1}^{n} \binom{n-1}{k-1} = 2^{n-1}$.

Step 2 (doubling, alternative). By induction: 1 has one composition. From each composition C of n form two compositions of $n+1$: add 1 to the first part, or write a new part 1 in front. Every

composition of $n+1$ arises exactly once (according to whether its first part is ≥ 2 or $= 1$), so $n+1$ has twice as many compositions as n. ■

(Weak compositions into an unspecified number of parts are infinite in number, since zeros can be appended.)

5.2 Set partitions

Motivation. Distinct balls into identical boxes means splitting $[n]$ into blocks. There is no closed formula, but the recurrences come from asking where the largest element goes, and the numbers relate powers to falling factorials.

Definition 5.5 (Set partitions, Stirling numbers of the second kind)

A **partition** of $[n]$ is a collection of nonempty **blocks** such that each element of $[n]$ lies in exactly one block. $S(n,k)$, the **Stirling number of the second kind**, is the number of partitions of $[n]$ into k blocks. $S(n,k) = 0$ for $n < k$; by convention $S(0,0) = 1$ and $S(n,0) = 0$ for $n > 0$.

Example 5.6

$S(n,1) = S(n,n) = 1$ for $n \geq 1$, and $S(n,n-1) = \binom{n}{2}$ for $n \geq 2$: such a partition consists of one doubleton and singletons.

Example 5.7

$S(4,2) = 7$: $\{1,2,3\}\{4\}$, $\{1,2,4\}\{3\}$, $\{1,3,4\}\{2\}$, $\{2,3,4\}\{1\}$, $\{1,2\}\{3,4\}$, $\{1,3\}\{2,4\}$, $\{1,4\}\{2,3\}$.

There is no closed formula for $S(n,k)$; a formula with one summation follows from the sieve (Chapter 7).

Theorem 5.8

For positive integers $k \leq n$:

$$S(n,k) = S(n-1,k-1) + k\,S(n-1,k). \tag{5.1}$$

Proof. Look at the element n. If it forms a singleton block, the others form a partition of $[n-1]$ into $k-1$ blocks: $S(n-1,k-1)$ ways. Otherwise, removing n leaves a partition of $[n-1]$ into k blocks, and n can be put into any of these k blocks: $kS(n-1,k)$ ways. ■

Distinct balls into k distinct boxes, no box empty: partition $[n]$ into k blocks and label the blocks, giving $k!\,S(n,k)$.

Corollary 5.9

The number of surjections $f : [n] \to [k]$ is $k!\,S(n,k)$.

Proof. The fibres $f^{-1}(i)$, $i \in [k]$, form a partition of $[n]$ into k nonempty blocks labelled by $[k]$. Conversely, a partition into k blocks with a bijective labelling by $[k]$ defines a unique surjection. So the count is $S(n,k)$ times the $k!$ labellings. ■

Corollary 5.10

For every real x and nonnegative integer n:

$$x^n = \sum_{k=0}^{n} S(n,k)(x)_k, \tag{5.2}$$

where $(x)_k = x(x-1)\cdots(x-k+1)$.

Proof. For $n = 0$ both sides are 1. For $n > 0$, $S(n,0) = 0$. Both sides are polynomials in x of degree n, so it suffices that they agree for infinitely many x, e.g. all positive integers. For $x \in \mathbb{N}$, x^n counts all functions $[n] \to [x]$ (Theorem 3.6). Group them by image: an image of size k can be chosen in $\binom{x}{k}$ ways and the function onto it in $k!\,S(n,k)$ ways (Corollary 5.9), and $\binom{x}{k}k! = (x)_k$. ■

Definition 5.11 (Bell numbers)

$B(n)$, the n-th **Bell number**, is the number of all partitions of $[n]$; $B(0) = 1$. So $B(n) = \sum_{i=0}^{n} S(n,i)$.

Theorem 5.12

For every nonnegative integer n:

$$B(n+1) = \sum_{i=0}^{n} \binom{n}{i} B(i). \tag{5.3}$$

Proof. In a partition of $[n+1]$, let i be the number of elements *not* in the block of $n+1$. These can be chosen in $\binom{n}{i}$ ways and partitioned in $B(i)$ ways; the rest of $[n]$ joins the block of $n+1$. Sum over i. ■

5.3 Integer partitions

Motivation. When neither balls nor boxes are distinguishable, only the multiset of box sizes remains: a partition of the integer n. There is no usable formula for $p(n)$, but Ferrers diagrams turn many identities into pictures, via conjugation, hook removal and adding a part 1.

Definition 5.13 (Integer partitions)

If $a_1 \geq a_2 \geq \cdots \geq a_k \geq 1$ are integers with $a_1 + \cdots + a_k = n$, then $(a_1, \ldots, a_k)$ is a **partition** of the integer n. $p(n)$ denotes the number of partitions of n, and $p_k(n)$ the number with exactly k parts.

("Set partition" and "integer partition" are different notions; the context, $[n]$ versus n, tells which is meant.)

Example 5.14

$p(5) = 7$: $(5), (4,1), (3,2), (3,1,1), (2,2,1), (2,1,1,1), (1,1,1,1,1)$.

No recurrence computes $p(n)$ simply from smaller values (but see the Notes). Asymptotically

$$p(n) \sim \frac{1}{4n\sqrt{3}} \exp\left(\pi\sqrt{\frac{2n}{3}}\right), \tag{5.4}$$

faster than any polynomial and slower than any c^n, $c > 1$.

The **Ferrers shape** of $(x_1, \ldots, x_k)$ has x_i unit boxes in row i, all rows left-justified; partitions of n and Ferrers shapes of size n correspond bijectively. Reflecting in the main diagonal gives the **conjugate** partition, whose i-th row is the i-th column of the original; the conjugate of $(4, 2, 1)$ is $(3, 2, 1, 1)$.

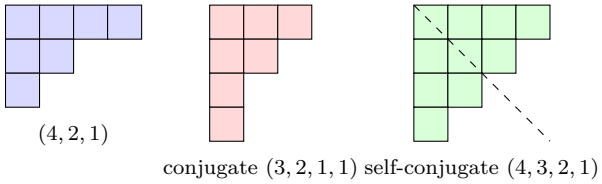


Figure. *Ferrers shapes. Conjugation reflects in the main diagonal (dashed); a self-conjugate shape is symmetric about it.*

Definition 5.15 (Self-conjugate)

A partition is **self-conjugate** if it equals its conjugate.

Example 5.16

$(4, 3, 2, 1)$, $(5, 1, 1, 1, 1)$ and $(4, 2, 1, 1)$ are self-conjugate.

Theorem 5.17

The number of partitions of n into at most k parts equals the number of partitions of n into parts not larger than k.

Proof. The first counts Ferrers shapes of size n with at most k rows, the second those with at most k columns; conjugation is a bijection between them. ∎

(For $n = 5$, $k = 2$: $(5), (4, 1), (3, 2)$ versus $(2, 2, 1), (2, 1, 1, 1), (1^5)$.)

Theorem 5.18

The number of partitions of n into distinct odd parts equals the number of self-conjugate partitions of n.

Proof.

Step 1 (the map). For self-conjugate $\pi = (\pi_1, \ldots, \pi_t)$, remove the first row and first column of its Ferrers shape: by symmetry these are $2\pi_1 - 1$ boxes, the **hook** of the top left box. Repeat on what remains: the i-th hook has $2\pi_i - (2i - 1)$ boxes, as long as boxes remain. Let $f(\pi)$ be the sequence of hook lengths $(2\pi_1 - 1, 2\pi_2 - 3, \ldots)$, a partition of n into odd parts. The parts are distinct, indeed strictly decreasing, since $\pi_1 \geq \pi_2 \geq \cdots$ gives $(2\pi_i - 2i + 1) - (2\pi_{i+1} - 2i - 1) = 2(\pi_i - \pi_{i+1}) + 2 \geq 2$.

Step 2 (inverse). Given distinct odd parts $\alpha = (2a_1 - 1, 2a_2 - 3, \ldots, 2a_u - (2u - 1))$, any π with $f(\pi) = \alpha$ must have a_1 boxes in its first row and in its first column, a_2 further boxes in row and column 2, and so on. This determines a unique self-conjugate shape, and it has image α. So f is a bijection. ∎

Example 5.19

For $\pi = (6,6,4,3,2,2)$ (self-conjugate), $f(\pi) = (11,9,3)$: the three successive hooks have 11, 9 and 3 boxes.

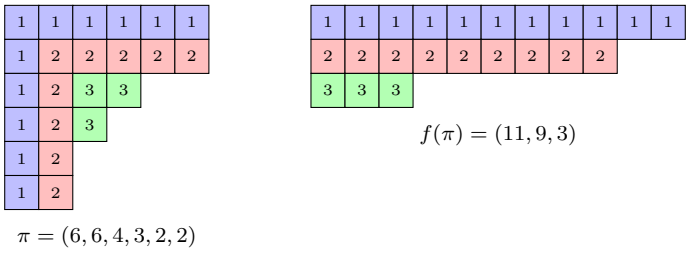


Figure. *Theorem 5.18: hook i of the self-conjugate shape (boxes labelled i) becomes row i of a partition into distinct odd parts.*

Theorem 5.20

Let $q(n)$ be the number of partitions of n with all parts at least 2. Then $q(n) = p(n) - p(n-1)$ for $n \geq 2$.

Proof. Appending a part 1 is a bijection from partitions of $n-1$ onto partitions of n having a part equal to 1. The partitions of n not obtained this way are exactly those counted by $q(n)$. ∎

Type of a set partition. Sorting the block sizes of a set partition π of $[n]$ decreasingly gives a partition of the integer n, the **type** of π.

Example 5.21

$\{1,5,6\},\{2,7\},\{3,9\},\{4,8\},\{10\}$ has type $(3,2,2,2,1)$.

Theorem 5.22

Let $a = (a_1, \ldots, a_k)$ be a partition of n and m_i the multiplicity of i as a part of a. The number of set partitions of $[n]$ of type a is

$$P_a = \frac{\binom{n}{a_1,\ldots,a_k}}{\prod_{i\geq 1} m_i!}.$$

Proof. Take a_j balls of colour j and order all n balls linearly, in $\binom{n}{a_1,\ldots,a_k}$ ways (Theorem 3.5). Put $i,j \in [n]$ in the same block iff positions i and j carry the same colour; this gives a set partition of type a, and every one arises. Two colourings give the same partition iff they differ by permuting colour classes of equal size, and the m_i classes of size i can be permuted in $m_i!$ ways. So each partition of type a arises exactly $\prod_i m_i!$ times. ∎

Example 5.23

There are $\binom{10}{3,2,2,2,1}/(1!\,3!\,1!) = 12600$ set partitions of $[10]$ of type $(3,2,2,2,1)$.

Example 5.24

The partitions of $[2n]$ into blocks of size 2 (perfect matchings) number

$$\frac{\binom{2n}{2,2,\dots,2}}{n!} = \frac{(2n)!}{2\cdot 4\cdots 2n} = \prod_{i=1}^{n}(2i-1) = (2n-1)!!.$$

	Balls	Boxes	Number
Surjections	n distinct	k distinct	$k!\,S(n,k)$
	n distinct	any number, distinct	$\sum_{i=1}^{n} i!\,S(n,i)$
Compositions	n identical	k distinct	$\binom{n-1}{k-1}$
	n identical	any number, distinct	2^{n-1}
Set partitions	n distinct	k identical	$S(n,k)$
	n distinct	any number, identical	$B(n)$
Integer partitions	n identical	k identical	$p_k(n)$
	n identical	any number, identical	$p(n)$
empty boxes allowed (k boxes fixed):			
Functions	n distinct	k distinct	k^n
Weak compositions	n identical	k distinct	$\binom{n+k-1}{k-1}$
Set partitions	n distinct	k identical	$\sum_{i=1}^{k} S(n,i)$
Integer partitions	n identical	k identical	$\sum_{i=1}^{k} p_i(n)$

Notes

With the **pentagonal numbers** $k(3k-1)/2$, $k\in\mathbb{Z}$ (that is $0,1,2,5,7,12,\dots$), MacMahon's recurrence reads

$$p(n) = p(n-1)+p(n-2)-p(n-5)-p(n-7)+\cdots, \tag{5.5}$$

where the i-th term has sign $(-1)^{\lfloor i/2\rfloor}$; e.g. $p(8)=15+11-3-1=22$. Euler's pentagonal number theorem: n has as many partitions into an even number of distinct parts as into an odd number, unless n is pentagonal, when the two counts differ by one.

5.4 Selected exercises

Motivation. The book exercises below compare Bell numbers with factorials, evaluate an alternating Stirling sum by a single substitution, and relate partitions with and without singleton blocks by a bijection.

Exercise 5.2 (Bell numbers versus factorials)

Prove that $B(n) < n!$ for $n \geq 3$.

Solution.

Step 1 (an injection). Given a partition π of $[n]$, define $\sigma_\pi : [n] \to [n]$ as follows: if x is not the largest element of its block, $\sigma_\pi(x)$ is the next larger element of the block; if it is the largest, $\sigma_\pi(x)$

is the smallest element of the block. On a block $\{y_1 < \cdots < y_r\}$ this is $y_1 \mapsto y_2 \mapsto \cdots \mapsto y_r \mapsto y_1$, so σ_π is a permutation of $[n]$ whose cycles are exactly the blocks of π. Hence π can be read off from σ_π, and $\pi \mapsto \sigma_\pi$ is injective. This gives $B(n) \le n!$.

Step 2 (not surjective). Let $n \ge 3$ and let τ be the permutation with $\tau(1) = 3$, $\tau(3) = 2$, $\tau(2) = 1$ and $\tau(x) = x$ for $x \ge 4$. If $\tau = \sigma_\pi$, the blocks of π would be the cycles of τ, so $\{1, 2, 3\}$ would be a block, and then $\sigma_\pi(1) = 2 \ne 3 = \tau(1)$. So τ is not in the image, and $B(n) \le n! - 1 < n!$. ■

Exercise 5.16 (An alternating sum of Stirling numbers)

Prove that for all positive integers n,

$$\sum_{k=1}^{n} (-1)^k k!\, S(n,k) = (-1)^n.$$

Explain the meaning of this identity in words.

Solution. Substitute $x = -1$ in Corollary 5.10, which holds for every real x. Since

$$(-1)_k = (-1)(-2)\cdots(-k) = (-1)^k k!$$

and $S(n, 0) = 0$ for $n \ge 1$, we get $(-1)^n = \sum_{k=1}^{n} S(n,k)(-1)^k k!$, the claim.

Meaning: $k!\, S(n,k)$ is the number of *ordered* partitions of $[n]$ into k blocks (partitions whose blocks are put in a line), equivalently of surjections $[n] \to [k]$ (Corollary 5.9). So among the ordered partitions of $[n]$, those with an even number of blocks and those with an odd number of blocks differ by exactly one, and the even ones are in excess iff n is even. ■

Exercise 5.34 (Partitions without singleton blocks)

Let $F(n)$ be the number of partitions of $[n]$ with no singleton block. Prove that $B(n) = F(n) + F(n+1)$ for $n \ge 1$, by a bijection.

Solution. Split the partitions of $[n]$ into the set $\mathcal{P}_0$ of those without singleton blocks, $|\mathcal{P}_0| = F(n)$, and the set $\mathcal{P}_1$ of those with at least one singleton block. It suffices to give a bijection Φ from $\mathcal{P}_1$ onto the set $\mathcal{F}_{n+1}$ of partitions of $[n+1]$ without singleton blocks.

Step 1 (the map). For $\pi \in \mathcal{P}_1$ let $S \ne \emptyset$ be the union of its singleton blocks. Let $\Phi(\pi)$ be the partition of $[n+1]$ whose blocks are the non-singleton blocks of π together with $S \cup \{n+1\}$. The new block has at least two elements and the other blocks are unchanged, so $\Phi(\pi) \in \mathcal{F}_{n+1}$.

Step 2 (the inverse). For $\rho \in \mathcal{F}_{n+1}$, the block containing $n+1$ is $T \cup \{n+1\}$ with $T \ne \emptyset$, since it is not a singleton. Let $\Psi(\rho)$ be the partition of $[n]$ consisting of the other blocks of ρ and the singletons $\{t\}$, $t \in T$. Then $\Psi(\rho) \in \mathcal{P}_1$.

Step 3 (inverse maps). In $\Psi(\rho)$ the singleton blocks are exactly those coming from T, because the other blocks of ρ have at least two elements. Hence $\Phi(\Psi(\rho)) = \rho$. Conversely $\Psi(\Phi(\pi)) = \pi$ by construction. So Φ is a bijection, $|\mathcal{P}_1| = F(n+1)$, and $B(n) = |\mathcal{P}_0| + |\mathcal{P}_1| = F(n) + F(n+1)$. ■

Chapter 6. Cycles in Permutations

A permutation $p = p_1 p_2 \cdots p_n$ of $[n]$ is also the bijection $p : [n] \to [n]$, $p(i) = p_i$. Viewing it as a function, we can compose permutations and follow the orbits of single elements, i.e. the cycles.

Example 6.1

312 is the function f with $f(1) = 3$, $f(2) = 1$, $f(3) = 2$.

The **product** fg of permutations of $[n]$ is the composition "first f, then g": $(fg)(i) = g(f(i))$. (Some authors use the opposite order; this will not matter here.)

Example 6.2

For $f = 312$, $g = 213$: $(fg)(1) = g(3) = 3$, $(fg)(2) = g(1) = 2$, $(fg)(3) = g(2) = 1$, so $fg = 321$.

Example 6.3

For the same f, g: $gf = 132 \neq fg$. Multiplication of permutations is not commutative.

6.1 Cycles in permutations

Motivation. Iterating a permutation moves each element around a closed orbit, so every permutation splits into disjoint cycles. Counting permutations by their cycle structure leads to the Stirling numbers of the first kind, which turn out to be the matrix inverse of those of the second kind.

In $g = 321564$: $g(2) = 2$ is a fixed point, $1 \leftrightarrow 3$ are swapped (a 2-cycle), and $4 \to 5 \to 6 \to 4$ (a 3-cycle).

Lemma 6.4

For a permutation p of $[n]$ and $x \in [n]$, there is $1 \leq i \leq n$ with $p^i(x) = x$.

Proof. Suppose none of $p(x), \ldots, p^n(x)$ equals x. These are n values in the $(n-1)$-element set $[n] \setminus \{x\}$, so by the pigeon-hole principle $p^j(x) = p^k(x)$ for some $1 \leq j < k \leq n$. Applying p^{-1} j times gives $x = p^{k-j}(x)$ with $1 \leq k - j \leq n - 1$, a contradiction. ∎

Definition 6.5 (Cycle)

Let p be a permutation of $[n]$, $x \in [n]$ and i the smallest positive integer with $p^i(x) = x$. Then $x, p(x), \ldots, p^{i-1}(x)$ form an **i-cycle** of p.

Corollary 6.6

Every permutation is the disjoint union of its cycles.

Proof. By Lemma 6.4 every x lies in a cycle. If the cycles of x and y share an element $z = p^a(x) = p^b(y)$, then both cycles are the orbit $\{p^m(z) \mid m \in \mathbb{Z}\}$ of z, so they coincide. Hence distinct cycles are disjoint. ∎

Example 6.7

321564 has the cycles (31), (2), (564).

In cycle notation, $g(i)$ is the entry following i in its cycle (cyclically). The same permutation has many such notations, e.g. $(241)(35) = (53)(412)$. The **canonical cycle form** writes each cycle starting with its largest element and orders the cycles by increasing first elements: $(412)(53)$. The **one-line notation** lists $f(1)f(2)\cdots f(n)$.

Example 6.8

$(412)(53)$ in one-line notation is 24513.

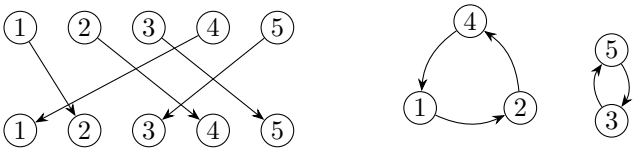


Figure. $24513 = (412)(53)$ *as a function* $[5] \to [5]$ *(left) and as a union of cycles (right).*

Permutations of $[n]$ are called **n-permutations**; their set is the symmetric group S_n.

Theorem 6.9

Let $a_1, \ldots, a_n \geq 0$ with $\sum_i i\, a_i = n$. The number of n-permutations with a_i cycles of length i ($i \in [n]$) is

$$\frac{n!}{a_1!\, a_2! \cdots a_n! \cdot 1^{a_1} 2^{a_2} \cdots n^{a_n}}. \tag{6.1}$$

Proof. Write the elements of $[n]$ in a row ($n!$ ways) and insert parentheses from left to right: a_1 cycles of length 1, then a_2 of length 2, and so on. Every permutation of the given type arises. It arises from exactly $\prod_i i^{a_i} a_i!$ rows: each i-cycle can start at any of its i elements, and the a_i cycles of length i can be listed in any of $a_i!$ orders, independently. ■

$(a_1, \ldots, a_n)$ is the (**cycle**) **type** of the permutation.

Example 6.10

There are $(n-1)!$ n-permutations with a single cycle (type $(0, \ldots, 0, 1)$); equivalently, $(n-1)!$ ways to seat n people around a round table up to rotation.

Definition 6.11 (Stirling numbers of the first kind)

The number of n-permutations with exactly k cycles is the **signless Stirling number of the first kind** $c(n,k)$; $s(n,k) := (-1)^{n-k} c(n,k)$ is the **Stirling number of the first kind**. $c(n,0) = 0$ for $n > 0$, $c(0,0) = 1$, and $c(n,k) = 0$ for $n < k$.

Theorem 6.12

For positive integers $n \geq k$:

$$c(n,k) = c(n-1, k-1) + (n-1)\, c(n-1, k). \tag{6.2}$$

Proof. Either n is a fixed point, and the other $n-1$ entries form $k-1$ cycles: $c(n-1,k-1)$ ways. Or the other entries form k cycles and n is inserted into one of them directly after one of the $n-1$ entries: $(n-1)c(n-1,k)$ ways. Inserting n "in front of" a cycle would give the same cycle as inserting it after the last entry, so these choices are all distinct and exhaustive. ■

Lemma 6.13

For every positive integer n:

$$\sum_{k=0}^{n} c(n,k)x^k = x(x+1)\cdots(x+n-1). \tag{6.3}$$

Proof. Let $G_n(x) := x(x+1)\cdots(x+n-1) = \sum_k a_{n,k}x^k$ (with $G_0 = 1$). From $G_n = (x+n-1)G_{n-1}$,

$$\sum_k a_{n,k}x^k = \sum_k a_{n-1,k-1}x^k + (n-1)\sum_k a_{n-1,k}x^k.$$

Comparing coefficients, $a_{n,k} = a_{n-1,k-1} + (n-1)a_{n-1,k}$: the recurrence (6.2). The initial values also agree ($a_{0,0} = 1$, $a_{n,0} = 0$ for $n > 0$), so $a_{n,k} = c(n,k)$ by induction on n. ■

Replacing x by $-x$ in (6.3) and multiplying by $(-1)^n$:

$$\sum_{k=0}^{n} s(n,k)x^k = (x)_n. \tag{6.4}$$

This explains the sign in $s(n,k)$. Compare with (5.2): $x^n = \sum_k S(n,k)(x)_k$. In the vector space of real polynomials, both $B = \{1, x, x^2, \dots\}$ and $B' = \{1, (x)_1, (x)_2, \dots\}$ are bases ($(x)_k$ has degree k). Let S and s be the infinite lower-triangular matrices with entries $S(n,k)$, $s(n,k)$; then (5.2) and (6.4) say that S and s are the two change-of-basis matrices.

Theorem 6.14

The matrices S and s are inverse to each other: $Ss = sS = I$.

Proof. Both matrices are lower triangular, so all products below are finite sums. Substituting (6.4) into (5.2),

$$x^n = \sum_k S(n,k)(x)_k = \sum_j x^j \sum_k S(n,k)s(k,j).$$

Since the monomials are linearly independent, $\sum_k S(n,k)s(k,j) = \delta_{nj}$, i.e. $Ss = I$. Substituting (5.2) into (6.4) gives $(x)_n = \sum_j (x)_j \sum_k s(n,k)S(k,j)$. The $(x)_j$ are linearly independent, so $sS = I$. ■

6.2 Permutations with restricted cycle structure

Motivation. Erasing the parentheses of the canonical cycle form gives a bijection $S_n \to S_n$ that turns statements about cycles into statements about positions. With it, probabilities about cycles become transparent, and it leads to the surprising square formula for permutations with only odd (or only even) cycles.

Lemma 6.15 (Transition lemma)

For $p \in S_n$ in canonical cycle form, let $g(p)$ be the word obtained by erasing the parentheses, read in one-line notation. Then $g : S_n \to S_n$ is a bijection.

Example 6.16

$g((412)(53)) = 41253$.

Proof of Lemma 6.15. We show that every $q = q_1 \cdots q_n$ admits exactly one way of inserting parentheses that yields a canonical cycle form. In a canonical form, each cycle starts with its maximum and the leading entries increase. So a new cycle starts at q_k iff q_k is larger than all leading entries before it. Since each leading entry is the maximum of its cycle, this happens iff q_k is larger than *all* entries to its left, i.e. q_k is a **left-to-right maximum**. Hence the cycle starts, and so the parentheses, are forced, and they produce a valid canonical form. ■

Example 6.17

$g^{-1}(4356172) = (43)(5)(61)(72)$: the left-to-right maxima $4, 5, 6, 7$ open the cycles.

If q has t left-to-right maxima then $g^{-1}(q)$ has t cycles; q_1 and the entry n are always left-to-right maxima.

Proposition 6.18

For $i \neq j$ in $[n]$, exactly half of all n-permutations contain i and j in the same cycle.

Proof. Relabelling ($i \leftrightarrow n$, $j \leftrightarrow n-1$) preserves cycle structure, so let $i = n$, $j = n-1$. In $q = g(p)$ the entry n is the last left-to-right maximum, so it starts the last cycle of p. That cycle consists of n and the entries to its right in q. Hence $n, n-1$ share a cycle iff $n-1$ is to the right of n in q, which holds for half of all q (swap the two entries). Since g is a bijection, the claim follows. ■

Lemma 6.19

For $i \in [n]$ and every $k \in [n]$, exactly $(n-1)!$ n-permutations have i in a cycle of length k.

Proof. By relabelling let $i = n$. If $q = g(p)$ has $q_j = n$, the cycle of n in p is $q_j \cdots q_n$, of length $n - j + 1$. So its length is k iff n is in position $n + 1 - k$ of q, which holds for $(n-1)!$ words q. ■

So the probability that a given entry lies in a k-cycle is $\frac{1}{n}$, independently of k.

Let ODD(m), resp. EVEN(m), be the m-permutations all of whose cycles have odd, resp. even, length.

Lemma 6.20

$|\mathrm{ODD}(2m)| = |\mathrm{EVEN}(2m)|$ for all positive integers m.

Proof.

Step 1 (the map Φ). Let $\pi \in \mathrm{ODD}(2m)$. Since $2m$ is even, π has an even number $2k$ of odd cycles $C_1, \ldots, C_{2k}$ (canonical order). For each $i \leq k$ move the last element of C_{2i-1} to the end of C_{2i}; a singleton C_{2i-1} disappears. The lengths become $|C_{2i-1}| - 1$ and $|C_{2i}| + 1$, both even. The moved element is smaller than the leading entry of C_{2i-1} (or is that entry, for a singleton), hence smaller than the leading entry of C_{2i}. So canonical form is preserved and $\Phi(\pi) \in \mathrm{EVEN}(2m)$.

Step 2 (the inverse). Let $\sigma \in \mathrm{EVEN}(2m)$ have canonical cycles $c_1, \ldots, c_h$. Process from the right. If the last entry of c_h exceeds the first entry of c_{h-1}, it cannot have come from c_{h-1}, so it was a singleton cycle placed just before c_h: restore it, and continue with c_{h-1} as the new last cycle. Otherwise move it back to the end of c_{h-1} and continue with c_{h-2}. If a single cycle remains, its last entry was a singleton. Each step is forced by Step 1, so the preimage exists and is unique. ■

Example 6.21

$\Phi((4)(513)(726)(8)) = (5134)(72)(86)$.

Example 6.22

$\Phi^{-1}((41)(62)(75)(83)) = (412)(6)(753)(8)$.

Example 6.23

$\Phi^{-1}((21)(53)(64)(87)) = (1)(2)(534)(6)(7)(8)$.

Theorem 6.24

For every positive integer m:

$$|\mathrm{ODD}(2m)| = |\mathrm{EVEN}(2m)| = 1^2 \cdot 3^2 \cdot 5^2 \cdots (2m-1)^2. \tag{6.5}$$

Proof. By Lemma 6.20 it suffices to count $\mathrm{EVEN}(2m)$. Build p by following cycles: start with 1, choose $p(1)$, $p^2(1)$, ..., and whenever a cycle closes, continue with the smallest unused element.

Step 1. $p(1) \neq 1$ (no 1-cycles): $2m-1$ choices. $p^2(1)$: anything except $p(1)$, so again $2m-1$ choices; choosing 1 closes a 2-cycle.

Step 2. In general, when choosing the $(2i-1)$-st image, $2i-2$ elements are used and the current cycle has even length or is just starting. The image may be neither used nor the current start element (an odd cycle, possibly of length 1, would close): $2m-2i+1$ choices. When choosing the $2i$-th image, the current cycle has odd length, so closing it to even length is allowed: the image is any of the $2m-(2i-1)$ elements that are not yet images, again $2m-2i+1$ choices.

Step 3. Multiplying, $|\mathrm{EVEN}(2m)| = \prod_{i=1}^{m}(2m-2i+1)^2$. ■

Of course $\mathrm{EVEN}(n) = \emptyset$ for odd n, but $\mathrm{ODD}(n)$ can still be counted.

Theorem 6.25

For every positive integer m:

$$|\mathrm{ODD}(2m+1)| = (2m+1)\,|\mathrm{ODD}(2m)| = 1^2 \cdot 3^2 \cdots (2m-1)^2\,(2m+1). \tag{6.6}$$

Proof. By Lemma 6.27 below, Ψ is a bijection $\mathrm{ODD}(2m) \times [2m+1] \to \mathrm{ODD}(2m+1)$; combine with Theorem 6.24. ■

An m-permutation in cycle form has $m+1$ **gap positions**: one at the very beginning and one after each entry.

Example 6.26

$(42)(513)$ has six gap positions: $|(4|2|)(5|1|3|)$.

The map Ψ. For $\pi \in \mathrm{ODD}(2m)$ and $k \in [2m+1]$: take $\Phi(\pi) \in \mathrm{EVEN}(2m)$, add 1 to each entry (giving a permutation of $\{2, \dots, 2m+1\}$), and insert the new entry 1 into the k-th gap position. This makes one cycle odd and keeps the canonical form (1 is never a leading entry, and inserting it at the very front puts it in a new fixed point (1) first). Finally apply Φ^{-1} (with relabelling) to the remaining, all-even cycles. The result $\Psi(\pi, k)$ has only odd cycles.

Lemma 6.27

Ψ is a bijection from $\mathrm{ODD}(2m) \times [2m+1]$ onto $\mathrm{ODD}(2m+1)$.

Proof. Given $\pi' \in \mathrm{ODD}(2m+1)$, set aside the cycle containing 1 and apply Φ to the remaining cycles (an even number of elements, all odd cycles) to get even cycles. Put the cycle of 1 back and read off k as the gap position of 1. Delete 1; the cycle of 1 becomes even, so the result lies in EVEN, on $\{2, \dots, 2m+1\}$. Subtract 1 from all entries and apply Φ^{-1} to get π. Every step inverts the corresponding step of Ψ, so Ψ is invertible. ■

6.3 Selected exercises

Motivation. The book exercises below use the cycle decomposition in three ways: to set up a recurrence, to average a statistic with Lemma 6.19, and to decide which permutations are squares.

Exercise 6.5 (Involutions)

Let $r(n)$ be the number of n-permutations p with $p^2 = \mathrm{id}$, and $r(0) = 1$. Prove that $r(n+1) = r(n) + n\, r(n-1)$ for $n \geq 1$.

Solution. $p^2 = \mathrm{id}$ means that every cycle of p has length 1 or 2. Split the involutions p of $[n+1]$ by the cycle containing $n+1$.

Step 1 ($n+1$ is fixed). Then p restricted to $[n]$ is an involution of $[n]$, and every involution of $[n]$ extends in this way by $p(n+1) = n+1$. This gives $r(n)$ involutions.

Step 2 ($n+1$ is in a 2-cycle). Then p contains $(k\ n{+}1)$ for exactly one $k \in [n]$, and the other cycles form an involution of the $(n-1)$-element set $[n] \setminus \{k\}$. Conversely each k and each such involution give one p. This gives $n\, r(n-1)$ involutions (for $n = 1$, $r(0) = 1$ accounts for the empty involution).

Adding the two cases, $r(n+1) = r(n) + n\, r(n-1)$. ■

Exercise 6.7 (The average number of cycles)

Prove that n-permutations have on average $H_n = \sum_{i=1}^{n} \frac{1}{i}$ cycles.

Solution.

Step 1 (cycles as a sum over entries). For $p \in S_n$ and $i \in [n]$ let $\ell_p(i)$ be the length of the cycle of p containing i. A cycle of length ℓ contains ℓ entries, each contributing $\frac{1}{\ell}$, so the number of cycles of p is

$$\mathrm{cyc}(p) = \sum_{i=1}^{n} \frac{1}{\ell_p(i)}.$$

Step 2 (double counting). Summing over all p and exchanging the sums,

$$\sum_{p \in S_n} \mathrm{cyc}(p) = \sum_{i=1}^{n} \sum_{k=1}^{n} \frac{1}{k} \left|\{p : \ell_p(i) = k\}\right| = \sum_{i=1}^{n} \sum_{k=1}^{n} \frac{(n-1)!}{k} = n!\, H_n,$$

where Lemma 6.19 gives $|\{p : \ell_p(i) = k\}| = (n-1)!$.

Step 3 (conclusion). Dividing by $|S_n| = n!$, the average number of cycles is H_n. ■

Exercise 6.21 (Square roots of permutations)

A permutation $p \in S_n$ has a *square root* if $q^2 = p$ for some $q \in S_n$. Prove that p has a square root iff, for every even m, the number of cycles of p of length m is even.

Solution.

Step 1 (square of one cycle). Let $c = (x_0\, x_1 \cdots x_{\ell-1})$, i.e. $c(x_i) = x_{i+1}$ with indices modulo ℓ. Then $c^2(x_i) = x_{i+2}$. If ℓ is odd, the indices $0, 2, 4, \ldots$ run through all residues modulo ℓ before returning to 0 (since 2 is invertible modulo ℓ), so $c^2 = (x_0\, x_2\, x_4 \cdots)$ is a single ℓ-cycle. If ℓ is even, $c^2 = (x_0\, x_2 \cdots x_{\ell-2})(x_1\, x_3 \cdots x_{\ell-1})$ consists of two cycles of length $\ell/2$.

Step 2 (necessity). Let $q^2 = p$. Each cycle C of q is mapped to itself by q, and q^2 restricted to C is the square of that cycle. So the cycles of p arise from the cycles of q as in Step 1. An odd cycle of q yields one odd cycle, so a cycle of p of even length m comes from a cycle of q of length $2m$, and that cycle yields exactly two cycles of length m. Hence the m-cycles of p come in pairs, and their number is even.

Step 3 (sufficiency). Assume the condition. Pair up the cycles of each even length arbitrarily, and treat each odd cycle separately. For an odd cycle c of length $\ell = 2r+1$, let $q = c^{r+1}$ on its elements; then $q^2 = c^{2r+2} = c^\ell c = c$ there. For a pair $(y_0 \cdots y_{m-1})$, $(z_0 \cdots z_{m-1})$ of m-cycles, let q be the $2m$-cycle $(y_0\, z_0\, y_1\, z_1 \cdots y_{m-1}\, z_{m-1})$ on their elements; then $q^2(y_i) = y_{i+1}$ and $q^2(z_i) = z_{i+1}$, so q^2 is the given pair there.

Step 4 (conclusion). The pieces of Step 3 live on disjoint sets covering $[n]$, so together they define $q \in S_n$, and $q^2 = p$ on every piece, hence everywhere. ■

Chapter 7. The Sieve

7.1 Enumerating the elements of intersecting sets

Motivation. Adding the sizes of overlapping sets counts common elements several times. Alternately subtracting and adding back the sizes of intersections corrects every overcount exactly; the reason is that $\sum_j (-1)^j \binom{s}{j} = 0$.

Example 7.1

14 students play soccer, 17 basketball, and 4 both. Then $14 + 17 - 4 = 27$ play at least one of the two.

Example 7.2

Add 18 hockey players; 4 play soccer and basketball, 3 soccer and hockey, 5 basketball and hockey, and 1 all three. Then $14 + 17 + 18 - 4 - 3 - 5 + 1 = 38$ students play at least one game. The student playing all three is counted three times, subtracted three times, and must be added back once.

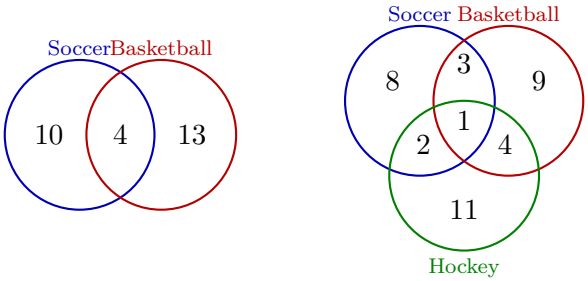


Figure. *Examples 7.1 and 7.2: the counts in the regions of the Venn diagrams sum to* 27 *and* 38.

Theorem 7.3 (Sieve formula, inclusion–exclusion)

For finite sets $A_1, \dots, A_n$,

$$|A_1 \cup \cdots \cup A_n| = \sum_{j=1}^{n} (-1)^{j-1} \sum_{\{i_1,\dots,i_j\}} |A_{i_1} \cap \cdots \cap A_{i_j}|, \tag{7.1}$$

where $\{i_1, \dots, i_j\}$ runs over the j-element subsets of $[n]$.

Proof. An element outside the union is counted by no term. Let x be in the union, $S := \{i \mid x \in A_i\}$ and $s := |S| \geq 1$. An intersection $A_{i_1} \cap \cdots \cap A_{i_j}$ contains x iff $\{i_1, \dots, i_j\} \subseteq S$, so x is counted

$$s - \binom{s}{2} + \binom{s}{3} - \cdots + (-1)^{s-1} \binom{s}{s} = 1 - (1-1)^s = 1 \tag{7.2}$$

times, by the binomial theorem. ∎

7.2 Applications of the sieve formula

Motivation. Counting objects with *none* of several bad properties is the typical use of the sieve: subtract the objects with at least one bad property, which inclusion–exclusion counts through intersections that are easy to count. Derangements and surjections are the classic cases; the second gives the promised formula for $S(n,k)$.

Example 7.4 (Derangements)

n guests pick up their hats at random. In how many ways does nobody get their own hat, i.e. how many permutations of $[n]$ have no fixed point (**derangements**)?
Solution. Let A_i be the set of permutations fixing i. An intersection of j of the A_i fixes j given points and permutes the rest freely, so it has $(n-j)!$ elements, and there are $\binom{n}{j}$ such intersections. By (7.1),

$$|A_1 \cup \cdots \cup A_n| = \sum_{j=1}^{n} (-1)^{j-1} \binom{n}{j} (n-j)! = \sum_{j=1}^{n} (-1)^{j-1} \frac{n!}{j!},$$

so the number of derangements is

$$D(n) = n! - |A_1 \cup \cdots \cup A_n| = \sum_{i=0}^{n} (-1)^i \frac{n!}{i!}. \tag{7.3}$$

Hence $\frac{D(n)}{n!} = \sum_{i=0}^{n} \frac{(-1)^i}{i!} \to e^{-1}$: roughly 37% of all permutations are derangements. [Supplement] In fact $D(n)$ is the integer nearest to $n!/e$ for $n \geq 1$: the alternating tail $\left|\frac{n!}{e} - D(n)\right| = n! \left|\sum_{i>n} \frac{(-1)^i}{i!}\right|$ is at most its first term $\frac{n!}{(n+1)!} = \frac{1}{n+1} \leq \frac{1}{2}$, with strict inequality since the terms decrease strictly.

Theorem 7.5

For positive integers n, k:

$$S(n,k) = \frac{1}{k!} \sum_{i=0}^{k} (-1)^i \binom{k}{i} (k-i)^n.$$

Proof. By Corollary 5.9, $k!\,S(n,k)$ is the number of surjections $[n] \to [k]$. Of the k^n functions, let A_i be those missing i in their image; an intersection of j such sets consists of functions into a $(k-j)$-element set, so it has $(k-j)^n$ elements. By the sieve, the number of non-surjective functions is $\sum_{j=1}^{k} (-1)^{j-1} \binom{k}{j} (k-j)^n$. Subtract this from k^n. ∎

Theorem 7.6 (Möbius inversion on subsets)

Let f, g be real functions on the subsets of $[n]$ with $g(S) = \sum_{T \subseteq S} f(T)$. Then

$$f(S) = \sum_{T \subseteq S} g(T) (-1)^{|S-T|}. \tag{7.4}$$

Proof. Expanding each $g(T)$, the right side of (7.4) is $\sum_{U \subseteq S} f(U) \sum_{U \subseteq T \subseteq S} (-1)^{|S-T|}$. A set T with $U \subseteq T \subseteq S$ is determined by $S \setminus T \subseteq S \setminus U$, and there are $\binom{|S-U|}{i}$ choices with $|S-T| = i$. So the inner sum is $\sum_i (-1)^i \binom{|S-U|}{i} = (1-1)^{|S-U|}$, which is 0 unless $U = S$, when it is 1. Hence the right side is $f(S)$. ∎

(This is the prototype of Möbius inversion on partially ordered sets.)

7.3 Selected exercises

Motivation. The book exercises below apply the sieve to number theory and to set partitions, derive the dual form of Möbius inversion, and give a bijective proof of the derangement recurrence; the first one is a warm-up in elementary counting.

Exercise 7.2 (Three teams)

A class of at least two students has three sports teams, and any two students are both members of at least one team. Prove that some team contains at least $\frac{2}{3}$ of the students.

Solution. Let $n \geq 2$ be the number of students and T_1, T_2, T_3 the teams.

Step 1 (a student in only one team). Suppose some student a belongs to exactly one team, say T_1. For every other student b, some team contains both a and b; it can only be T_1, so $b \in T_1$. Hence T_1 contains all students.

Step 2 (no student in only one team). Every student a is in some team, since a shares a team with another student. If no student is in exactly one team, every student is in at least two teams, so $|T_1| + |T_2| + |T_3| \geq 2n$, and the largest team has at least $\frac{2n}{3}$ students. ∎

Exercise 7.8 (Euler's function)

Let $\varphi(n)$ be the number of integers in $[n]$ relatively prime to n. Prove that if $p_1, \dots, p_k$ are the distinct prime divisors of n, then

$$\varphi(n) = n\Big(1 - \frac{1}{p_1}\Big) \cdots \Big(1 - \frac{1}{p_k}\Big).$$

Solution.

Step 1 (the sets). Let $A_i = \{x \in [n] : p_i \mid x\}$. An integer x is relatively prime to n iff no p_i divides x, so $\varphi(n) = n - |A_1 \cup \cdots \cup A_k|$.

Step 2 (intersections). For $\emptyset \neq S \subseteq [k]$ let $d_S = \prod_{i \in S} p_i$. Since the p_i are distinct primes, x lies in all A_i, $i \in S$, iff $d_S \mid x$. As $d_S \mid n$, exactly n/d_S elements of $[n]$ are multiples of d_S.

Step 3 (sieve). By Theorem 7.3, with $d_\emptyset = 1$,

$$\varphi(n) = n - \sum_{\emptyset \neq S \subseteq [k]} (-1)^{|S|-1} \frac{n}{d_S} = \sum_{S \subseteq [k]} (-1)^{|S|} \frac{n}{d_S} = n \prod_{i=1}^{k} \Big(1 - \frac{1}{p_i}\Big),$$

the last step by expanding the product: choosing $-\frac{1}{p_i}$ from the factors $i \in S$ and 1 from the others gives the term of S. ∎

Exercise 7.12 (The dual Möbius inversion)

Let h, r be real functions on the subsets of $[n]$ with $r(S) = \sum_{T \supseteq S} h(T)$ for all S. Prove that

$$h(S) = \sum_{T \supseteq S} r(T)(-1)^{|T-S|}.$$

Solution. Write $\bar{X} = [n] \setminus X$; complementation is a bijection of the subsets of $[n]$ that reverses inclusion.

Step 1 (transport). Define $f(A) = h(\bar{A})$ and $g(A) = r(\bar{A})$. Then, substituting $T = \bar{U}$,

$$g(A) = r(\bar{A}) = \sum_{T \supseteq \bar{A}} h(T) = \sum_{U \subseteq A} h(\bar{U}) = \sum_{U \subseteq A} f(U).$$

Step 2 (apply Theorem 7.6). Hence $f(A) = \sum_{U \subseteq A} g(U)(-1)^{|A-U|}$ for all A.

Step 3 (transport back). Take $A = \bar{S}$ and substitute $U = \bar{T}$; then $U \subseteq A$ iff $T \supseteq S$, and $A - U = \bar{S} \cap T = T - S$. So

$$h(S) = f(\bar{S}) = \sum_{T \supseteq S} g(\bar{T})(-1)^{|T-S|} = \sum_{T \supseteq S} r(T)(-1)^{|T-S|}.$$

■

Exercise 7.23 (The derangement recurrence)

Give a combinatorial proof of $D(n+1) = n(D(n) + D(n-1))$ for $n \geq 1$, without using (7.3). Here $D(0) = 1$ and $D(1) = 0$.

Solution. Let p be a derangement of $[n+1]$ and $k = p(n+1)$; then $k \in [n]$, which gives n choices. Fix k; we show that exactly $D(n-1) + D(n)$ derangements have $p(n+1) = k$.

Step 1 ($p(k) = n+1$). Then $(k\ n{+}1)$ is a 2-cycle of p, and the other cycles form a derangement of the $(n-1)$-element set $[n] \setminus \{k\}$. Conversely every such derangement, together with the 2-cycle, gives one p. This gives $D(n-1)$ derangements.

Step 2 ($p(k) \neq n+1$). Then the cycle of $n+1$ has length at least 3. Delete $n+1$ from its cycle: with $m = p^{-1}(n+1)$, let $p'(m) = k$ and $p'(x) = p(x)$ for the other $x \in [n]$. Then p' is a permutation of $[n]$, and it has no fixed point: $p'(m) = k \neq m$ because $m \neq k$ (as $p(k) \neq n+1$), and the other values are those of p.

Step 3 (the inverse of Step 2). Given a derangement q of $[n]$, insert $n+1$ before k in its cycle: with $m = q^{-1}(k)$ put $p(m) = n+1$, $p(n+1) = k$, and $p(x) = q(x)$ otherwise. This is a derangement of $[n+1]$ with $p(n+1) = k$ and $p(k) = q(k) \neq n+1$, and the two constructions are inverse to each other. This gives $D(n)$ derangements.

Summing over the n values of k, $D(n+1) = n(D(n)+D(n-1))$. (For $n = 1$: $D(2) = 1 \cdot (0+1) = 1$.)
■

Exercise 7.37 (No singleton blocks, by the sieve)

Let $F(n)$ be the number of partitions of $[n]$ without singleton blocks. Express $F(n)$ through the Bell numbers.

Solution. We show $F(n) = \sum_{j=0}^{n} (-1)^j \binom{n}{j} B(n-j)$, with $B(0) = 1$.

Step 1 (the sets). Let A_i be the set of partitions of $[n]$ in which $\{i\}$ is a block. Then $F(n) = B(n) - |A_1 \cup \cdots \cup A_n|$.

Step 2 (intersections). For a j-element set $J \subseteq [n]$, the partitions in $\bigcap_{i \in J} A_i$ consist of the singletons $\{i\}$, $i \in J$, and an arbitrary partition of the $n-j$ remaining elements. So the intersection has $B(n-j)$ elements, and there are $\binom{n}{j}$ such J.

Step 3 (sieve). By Theorem 7.3,

$$F(n) = B(n) - \sum_{j=1}^{n} (-1)^{j-1} \binom{n}{j} B(n-j) = \sum_{j=0}^{n} (-1)^j \binom{n}{j} B(n-j).$$

For example $F(3) = 5 - 3 \cdot 2 + 3 \cdot 1 - 1 = 1$, the single partition $\{1,2,3\}$. Compare Exercise 5.34, which relates F and B bijectively. ■

Chapter 8. Generating Functions

A generating function is "a clothesline on which we hang all elements of a sequence" (H. Wilf): one function stores infinitely many numbers. Operations on the function then correspond to combinatorial constructions, which makes generating functions the most powerful tool of enumeration.

8.1 Ordinary generating functions

8.1.1 Recurrence relations and generating functions

Motivation. A recurrence $a_{n+1} = 4a_n - 100$ is infinitely many equations. Multiplying the n-th by x^{n+1} and summing packs them into one equation for a single power series. Solving it and reading off coefficients by partial fractions gives an explicit formula.

A lake's frog population quadruples each year and 100 frogs are removed at the start of each year: $a_0 = 50$ and

$$a_{n+1} = 4a_n - 100 \qquad (n \geq 0). \tag{8.1}$$

Computing a_{87} by iterating (8.1) wastes 86 steps; we want an explicit formula.

Definition 8.1 (Ordinary generating function)

For a sequence $(f_n)_{n\geq 0}$ of reals, the **formal power series** $F(x) = \sum_{n\geq 0} f_n x^n$ is its **ordinary generating function**.

A formal power series is determined by its coefficients and need not converge (e.g. $\sum n!x^n$); two series are equal iff all coefficients agree.

Multiply (8.1) by x^{n+1} and sum over $n \geq 0$. With $G(x) = \sum a_n x^n$ and $\sum_{n\geq 0} x^n = \frac{1}{1-x}$:

$$G(x) - a_0 = 4xG(x) - \frac{100x}{1-x}, \qquad G(x) = \frac{50}{1-4x} - \frac{100x}{(1-x)(1-4x)}. \tag{8.2}$$

Partial fractions give $\frac{100x}{(1-x)(1-4x)} = \frac{100}{3}\left(\frac{1}{1-4x} - \frac{1}{1-x}\right)$. Since $\frac{1}{1-4x} = \sum 4^n x^n$,

$$a_n = 50 \cdot 4^n - 100 \cdot \frac{4^n - 1}{3}, \tag{8.3}$$

and one checks directly that (8.3) satisfies $a_0 = 50$ and (8.1).

The method. (1) Define $G(x) = \sum a_n x^n$. (2) Turn the recurrence into an equation for G (multiply by x^n, x^{n+1} or x^{n+k} and sum over $n \geq 0$). (3) Solve for G. (4) Read off the coefficient of x^n.

Example 8.2

\$1000 earns 5% interest per year and \$500 is deposited each year: $a_0 = 1000$, $a_{n+1} = 1.05a_n + 500$.

Solution. As above, $G(x) - 1000 = 1.05xG(x) + \frac{500x}{1-x}$, so

$$G(x) = \frac{1000}{1-1.05x} + \frac{500x}{(1-x)(1-1.05x)} = \frac{1000}{1-1.05x} + \frac{10000}{1-1.05x} - \frac{10000}{1-x},$$

and $a_n = 11000 \cdot 1.05^n - 10000$.

Example 8.3

$a_0 = 0$, $a_1 = 1$, $a_{n+2} = 3a_{n+1} - 2a_n$. Multiplying by x^{n+2} and summing: $G - x = 3xG - 2x^2G$. So

$$G(x) = \frac{x}{1 - 3x + 2x^2} = \frac{1}{1 - 2x} - \frac{1}{1 - x}, \qquad a_n = 2^n - 1.$$

8.1.2 Products of generating functions

Motivation. Multiplying generating functions convolves their coefficients. Combinatorially this means cutting $[n]$ into an initial and a final interval and building independent structures on the two pieces. Many counts that look like awkward sums become simple products.

Lemma 8.4

If $A(x) = \sum a_n x^n$, $B(x) = \sum b_n x^n$ and $c_n = \sum_{i=0}^{n} a_i b_{n-i}$, then $A(x)B(x) = C(x) := \sum c_n x^n$.

Proof. Multiplying out, every product $a_i x^i \cdot b_j x^j$ appears once, and it contributes to x^n iff $j = n - i$. ■

Theorem 8.5 (Product formula)

Let a_n (b_n) be the number of structures of a first (second) kind on an n-element set. Let c_n be the number of ways to cut $[n]$ into the intervals $S = \{1, \dots, i\}$ and $T = \{i + 1, \dots, n\}$ (possibly empty) and build a structure of the first kind on S and of the second kind on T. Then $A(x)B(x) = C(x)$.

Proof. For a fixed cut point i there are $a_i b_{n-i}$ choices, so $c_n = \sum_{i=0}^{n} a_i b_{n-i}$; apply Lemma 8.4. ■

By induction, the same holds for any fixed number of intervals.

Example 8.6

An n-day semester is split into a theoretical part (first k days) and a lab part ($n - k$ days); one holiday is chosen in the first part and two in the second. The number of plans is $f_n = \binom{n+1}{4}$.

Solution. $f_n = \sum_k k\binom{n-k}{2}$. The pieces have generating functions $A(x) = \sum_{k \geq 1} kx^k = \frac{x}{(1-x)^2}$ (differentiate $\sum x^k$) and $B(x) = \sum_{m \geq 2} \binom{m}{2} x^m = \frac{x^2}{(1-x)^3}$. So

$$F(x) = A(x)B(x) = \frac{x^3}{(1 - x)^5} = x^3 \sum_{n \geq 0} \binom{n + 4}{4} x^n = \sum_{n \geq 3} \binom{n + 1}{4} x^n.$$

Here $(1 - x)^{-5} = \sum_n \binom{n+4}{4} x^n$, by Theorem 4.15 or because $\binom{n+4}{4}$ counts the multisets of size n from [5] (Theorem 3.21).

Example 8.7

If instead any set of independent-study days is chosen in each part, each part has generating function $C(x) = \sum 2^k x^k = \frac{1}{1-2x}$. So $F(x) = \frac{1}{(1-2x)^2} = \frac{1}{2}C'(x) = \sum_n (n + 1)2^n x^n$, and $g_n = (n + 1)2^n$.

Example 8.8

Split an n-day semester into three parts and choose any number of holidays in the first, an odd number in the second and an even number in the third. An m-set has 2^{m-1} odd and 2^{m-1} even subsets for $m \geq 1$ (by Theorem 4.2). So the generating functions are $\frac{1}{1-2x}$, $\frac{x}{1-2x}$ and $1 + \frac{x}{1-2x} = \frac{1-x}{1-2x}$, and

$$G(x) = \frac{x(1-x)}{(1-2x)^3} = -\frac{1}{4} \cdot \frac{1}{1-2x} + \frac{1}{4} \cdot \frac{1}{(1-2x)^3}.$$

With $(1-2x)^{-3} = \sum_n \binom{n+2}{2} 2^n x^n$ this gives $g_n = \frac{1}{4}(\binom{n+2}{2} - 1)2^n = 2^{n-3}n(n+3)$.

Example 8.9

With $p_{\leq k}(n)$ the number of partitions of n into parts of size at most k,

$$\sum_{n \geq 0} p_{\leq k}(n) x^n = \prod_{i=1}^{k} \frac{1}{1-x^i} = (1+x+x^2+\cdots)(1+x^2+x^4+\cdots)\cdots(1+x^k+x^{2k}+\cdots). \quad (8.4)$$

Solution. A term x^n of the product picks x^{ij_i} from the i-th factor with $1j_1+2j_2+\cdots+kj_k = n$. This corresponds to the partition of n with j_i parts equal to i, and this correspondence is bijective. By Theorem 5.17 this is also the generating function of partitions into at most k parts.

Example 8.10

For all partitions,

$$\sum_{n \geq 0} p(n) x^n = \prod_{k \geq 1} \frac{1}{1-x^k}, \quad (8.5)$$

by the same argument. The infinite product is well defined: only the first n factors affect the coefficient of x^n.

Example 8.11 (Euler)

The number $p_{\text{odd}}(n)$ of partitions of n into odd parts equals the number $p_d(n)$ of partitions into distinct parts.

Solution. It suffices to compare generating functions:

$$\sum_n p_{\text{odd}}(n) x^n = \prod_{i \text{ odd}} \frac{1}{1-x^i}, \qquad \sum_n p_d(n) x^n = \prod_{i \geq 1} (1+x^i) = \prod_{i \geq 1} \frac{1-x^{2i}}{1-x^i}.$$

In the last product every numerator $1-x^{2i}$ cancels the denominator factor $1-x^{2i}$, leaving $\prod_{i \text{ odd}} \frac{1}{1-x^i}$.

8.1.2.1 The Catalan numbers

A jar starts empty; each day a coin is put in or taken out, never going negative, and after $2n$ days the jar is empty again. Let c_n ($c_0 = 1$) count such histories: sequences $b_1, \ldots, b_{2n} \in \{\pm 1\}$ with $\sum_j b_j = 0$ and all partial sums $\sum_{j \leq k} b_j \geq 0$ (**good sequences**).

Decomposition. For $n > 0$ let day $2i$ be the first day after the start with an empty jar (the jar

can only be empty after an even number of days). Then days $2i+1,\dots,2n$ form a good sequence of length $2(n-i)$. Days $1,\dots,2i$ form a **very good** sequence, one whose partial sums are positive strictly inside. Stripping its first $+1$ and last -1 gives a good sequence of length $2(i-1)$, bijectively. So very good sequences have generating function $\sum_{i\ge 1} c_{i-1}x^i = xC(x)$, and the product formula gives

$$C(x) - 1 = xC(x)\cdot C(x), \qquad \text{i.e.} \quad xC(x)^2 - C(x) + 1 = 0. \tag{8.6}$$

Of the roots $\frac{1\pm\sqrt{1-4x}}{2x}$ only the minus sign gives a power series with constant term 1 (the plus sign has a pole at 0), so

$$C(x) = \frac{1-\sqrt{1-4x}}{2x}. \tag{8.7}$$

By Theorem 4.15, $\sqrt{1-4x} = \sum_n \binom{1/2}{n}(-4x)^n = 1 - 2\sum_{n\ge 1}\frac{1}{n}\binom{2n-2}{n-1}x^n$ (the same computation as in Example 4.16). Hence

$$C(x) = \sum_{n\ge 0}\frac{1}{n+1}\binom{2n}{n}x^n, \qquad c_n = \frac{1}{n+1}\binom{2n}{n}.$$

These are the **Catalan numbers** $1,1,2,5,14,42,\dots$, which count well over 150 kinds of objects (see Chapter 14).



Figure. *A good sequence as a lattice path ($+1$ up, -1 down) that never goes below the axis. Cutting at the first return to the axis gives the decomposition $C - 1 = xC\cdot C$.*

8.1.3 Compositions of generating functions

Motivation. Cutting $[n]$ into an unspecified number of nonempty intervals, each carrying a structure, corresponds to substituting one generating function into another. This is the composition of formal power series, defined when the inner series has no constant term.

For $F(x) = \frac{1}{1-x}$ we would like $F(G(x)) = 1 + G + G^2 + \cdots$. This sum of series is well defined iff each coefficient receives only finitely many contributions, which holds when G has constant term 0: then G^m is divisible by x^m.

Definition 8.12 (Composition)

For a formal power series $F(x) = \sum f_n x^n$ and G with constant term 0, $F(G(x)) := \sum_{n\ge 0} f_n G(x)^n$.

Theorem 8.13

Let a_n count structures of some kind on an n-set, with $a_0 = 0$. Let h_n ($h_0 = 1$) count the ways to cut $[n]$ into any number of nonempty intervals and put such a structure on each. Then $H(x) = \frac{1}{1-A(x)}$.

Proof. By the product formula (for k factors), $A(x)^k$ counts the cuts into exactly k intervals, each with a structure; nonempty intervals are automatic since $a_0 = 0$. As $a_0 = 0$, A^k has no

terms below x^k, so the sum over k is well defined, and with the constant term $h_0 = 1$ we get $H = 1 + \sum_{k\geq 1} A^k = \frac{1}{1-A}$. ■

Empty intervals must be excluded, since otherwise there would be infinitely many cuts.

Example 8.14

n soldiers in a line are split into nonempty units and a commander is named in each. The number of ways is $h_n = \frac{1}{\sqrt{5}}(\alpha^n - \beta^n)$ for $n \geq 1$, with $\alpha, \beta = \frac{3\pm\sqrt{5}}{2}$.
Solution. Here $a_k = k$, $A(x) = \frac{x}{(1-x)^2}$, so $H(x) = \frac{1}{1-A(x)} = 1 + \frac{x}{1-3x+x^2}$. The roots of $x^2 - 3x + 1$ are α, β with $\alpha\beta = 1$, and partial fractions give

$$\frac{1}{1-3x+x^2} = \frac{1}{\sqrt{5}}\Big(\frac{\alpha}{1-\alpha x} - \frac{\beta}{1-\beta x}\Big) = \frac{1}{\sqrt{5}}\sum_n(\alpha^{n+1} - \beta^{n+1})x^n.$$

So the coefficient of x^n in H is $\frac{1}{\sqrt{5}}(\alpha^n - \beta^n)$ for $n \geq 1$: $1, 3, 8, 21, 55, \ldots$.

Theorem 8.15 (Compositional formula)

Let a_n count structures of a first kind on an n-set ($a_0 = 0$) and b_n structures of a second kind ($b_0 = 1$). Let g_n ($g_0 = 1$) count the ways to cut $[n]$ into nonempty intervals, put a first-kind structure on each interval and a second-kind structure on the set of intervals. Then $G(x) = B(A(x))$.

Proof. With exactly k intervals there are b_k structures on the set of intervals, and the generating function of the interval structures is $A(x)^k$ (product formula). So this case contributes $b_k A(x)^k$, and $G = \sum_k b_k A(x)^k = B(A(x))$. ■

Example 8.16

The officer splits the line into nonempty units and chooses any subset of the units for night duty. Here $a_k = 1$ ($k \geq 1$) and $b_m = 2^m$, so $A = \frac{x}{1-x}$, $B = \frac{1}{1-2x}$ and

$$G(x) = \frac{1}{1 - \frac{2x}{1-x}} = \frac{1-x}{1-3x} = 1 + \sum_{n\geq 1} 2\cdot 3^{n-1}x^n.$$

For $n \geq 1$ there are $2 \cdot 3^{n-1}$ ways.

8.2 Exponential generating functions

8.2.1 Recurrence relations and exponential generating functions

Motivation. Sequences that grow like $n!$ have ordinary generating functions with no closed form. Dividing the n-th term by $n!$ tames them, and the factor $(n+1)$ that typically appears in such recurrences then disappears.

Example 8.17

$a_0 = 1$, $a_{n+1} = (n+1)(a_n - n + 1)$ for $n \geq 0$. Then $a_n = n! + n$.

Definition 8.18 (Exponential generating function)

For a sequence $(f_n)_{n\geq 0}$, the formal power series $F(x) = \sum_{n\geq 0} f_n \frac{x^n}{n!}$ is its **exponential generating function** (the constant sequence 1 has e^x).

Solution of Example 8.17. Multiply the recurrence by $\frac{x^{n+1}}{(n+1)!}$ and sum over $n \geq 0$:

$$\sum_{n\geq 0} a_{n+1}\frac{x^{n+1}}{(n+1)!} = \sum_{n\geq 0} a_n \frac{x^{n+1}}{n!} - \sum_{n\geq 0}(n-1)\frac{x^{n+1}}{n!}, \tag{8.8}$$

i.e. $A(x) - 1 = xA(x) - x^2e^x + xe^x$. Hence

$$A(x) = \frac{1}{1-x} + xe^x = \sum_{n\geq 0} n!\frac{x^n}{n!} + \sum_{n\geq 1} n\frac{x^n}{n!},$$

since the equation reads $A(x)(1-x) = 1 + xe^x(1-x)$; hence $a_n = n! + n$.

Example 8.19

$f_0 = 0$, $f_{n+1} = 2(n+1)f_n + (n+1)!$. Multiplying by $\frac{x^{n+1}}{(n+1)!}$ and summing: $F(x) = 2xF(x) + \frac{x}{1-x}$. So $F(x) = \frac{x}{(1-x)(1-2x)} = \sum_n (2^n - 1)x^n$, and $f_n = (2^n - 1)\, n!$.

8.2.2 Products of exponential generating functions

Motivation. Exponential generating functions multiply by a binomial convolution. Combinatorially, this splits $[n]$ into two *arbitrary* subsets rather than two intervals, the natural operation for labelled structures.

Lemma 8.20

For exponential generating functions $A(x) = \sum a_i \frac{x^i}{i!}$, $B(x) = \sum b_k \frac{x^k}{k!}$ and $c_n = \sum_{i=0}^{n} \binom{n}{i} a_i b_{n-i}$, we have $A(x)B(x) = C(x) := \sum c_n \frac{x^n}{n!}$.

Proof. A general term of the product is $a_i \frac{x^i}{i!} \cdot b_j \frac{x^j}{j!} = a_i b_j \binom{i+j}{i} \frac{x^{i+j}}{(i+j)!}$, and it contributes to $\frac{x^n}{n!}$ iff $i + j = n$. ∎

Theorem 8.21 (Product formula, exponential version)

Let a_n, b_n count structures of two kinds on n-sets, and let c_n count the ways to split $[n]$ into disjoint subsets $S \cup T = [n]$ (no other restriction) and build a first-kind structure on S and a second-kind structure on T. Then $A(x)B(x) = C(x)$ for the exponential generating functions.

Proof. If $|S| = i$, there are $\binom{n}{i}$ choices of S, then a_i and b_{n-i} structures; so $c_n = \sum_i \binom{n}{i} a_i b_{n-i}$, and Lemma 8.20 applies. ∎

Theorem 8.5 cuts a *linearly ordered* set into intervals; Theorem 8.21 splits a set into arbitrary subsets.

Example 8.22

A coach splits n players into two groups, lines up each group, and gives everyone in the first group an orange, white or blue shirt. Here $a_k = k!\,3^k$, so $A(x) = \frac{1}{1-3x}$, and $b_m = m!$, so $B(x) = \frac{1}{1-x}$. Then

$$C(x) = \frac{1}{(1-3x)(1-x)} = \frac{3}{2}\cdot\frac{1}{1-3x} - \frac{1}{2}\cdot\frac{1}{1-x} = \sum_n \frac{3^{n+1}-1}{2}x^n,$$

and $c_n = n!\,\frac{3^{n+1}-1}{2}$.

Example 8.23

For fixed k, the exponential generating function of the Stirling numbers of the second kind is

$$S_k(x) := \sum_{n\geq k} S(n,k)\frac{x^n}{n!} = \frac{1}{k!}(e^x-1)^k. \tag{8.9}$$

Solution. A nonempty block carries "no structure" in exactly one way, with exponential generating function $\sum_{m\geq 1}\frac{x^m}{m!} = e^x - 1$. By the product formula, $(e^x-1)^k$ counts partitions into k *ordered* blocks; each unordered partition arises $k!$ times. Expanding $(e^x-1)^k = \sum_i \binom{k}{i}(-1)^i e^{(k-i)x}$ recovers Theorem 7.5.

Differentiation shifts exponential generating functions: $\left(\sum a_n \frac{x^n}{n!}\right)' = \sum a_{n+1}\frac{x^n}{n!}$.

Example 8.24

The exponential generating function of the Bell numbers is $B(x) = e^{e^x-1}$.
Solution. Multiply $B(n+1) = \sum_i \binom{n}{i}B(i)$ (Theorem 5.12) by $\frac{x^n}{n!}$ and sum. The left side is $B'(x)$, and by Lemma 8.20 the right side is $B(x)e^x$. So $(\ln B)' = e^x$, $\ln B(x) = e^x + C$, and $B(0) = 1$ forces $C = -1$.

8.2.3 Compositions of exponential generating functions

Motivation. Partitioning a labelled set into blocks, each carrying a structure, corresponds to exponentiating the generating function of the structure. This exponential formula explains at once why permutations (sets of cycles) are counted by $e^{\ln\frac{1}{1-x}} = \frac{1}{1-x}$.

Theorem 8.25 (Exponential formula)

Let a_n count structures on n-sets ($a_0 = 0$) and let h_n ($h_0 = 1$) count the ways to partition $[n]$ into any number of nonempty blocks and put a structure on each block. Then $H(x) = e^{A(x)}$ for the exponential generating functions.

Proof. By Theorem 8.21 (for k factors), $A(x)^k$ counts the decompositions of $[n]$ into k *ordered* blocks with structures, so $A(x)^k/k!$ counts those into exactly k unordered blocks. As $a_0 = 0$, A^k has no terms below x^k, so $H = 1 + \sum_{k\geq 1}\frac{A^k}{k!} = e^{A(x)}$. ∎

Integrating $\frac{1}{1-x} = \sum x^n$ termwise (the formal integral of $\sum a_n x^n$ is $\sum a_n \frac{x^{n+1}}{n+1}$) gives

$$\sum_{k\geq 1}\frac{x^k}{k} = \ln\frac{1}{1-x}.$$

Example 8.26

n people form groups and each group sits at a circular table. With $a_k = (k-1)!$ (Example 6.10), $A(x) = \sum_k \frac{x^k}{k} = \ln \frac{1}{1-x}$, so $H(x) = \frac{1}{1-x} = \sum n! \frac{x^n}{n!}$ and $h_n = n!$. Combinatorially: a permutation is exactly a set of cycles.

Example 8.27

The partitions of $[n]$ into blocks of sizes 3, 4 and 9 have exponential generating function $e^{x^3/3! + x^4/4! + x^9/9!}$. Blocks of size 3 only give $e^{x^3/3!}$ (exponential formula with $T(x) = x^3/3!$), and similarly for 4 and 9; the three kinds are combined by the product formula.

Theorem 8.28 (Compositional formula, exponential version)

Let a_n count first-kind structures ($a_0 = 0$) and b_n second-kind structures ($b_0 = 1$). Let g_n count the ways to partition $[n]$ into nonempty blocks, put a first-kind structure on each block and a second-kind structure on the set of blocks. Then $G(x) = B(A(x))$.

Proof. With k blocks there are b_k structures on the set of blocks, and by Theorem 8.21 the blocks with their structures contribute $\frac{A(x)^k}{k!}$. So $G = \sum_k b_k \frac{A(x)^k}{k!} = B(A(x))$; the sum is defined since $a_0 = 0$. ■

Example 8.29

n distinct cards are split into nonempty subsets of even size, the cards in each subset are ordered, and the subsets are put in a line. Here $a_n = n!$ for even $n \geq 2$ and 0 otherwise, so $A(x) = \sum_{n \geq 2 \text{ even}} x^n = \frac{x^2}{1-x^2}$; and $b_n = n!$, so $B(x) = \frac{1}{1-x}$. Then

$$G(x) = \frac{1}{1 - \frac{x^2}{1-x^2}} = \frac{1-x^2}{1-2x^2} = 1 + \sum_{m \geq 0} 2^m x^{2m+2},$$

so $g_n = 0$ for odd n and $g_n = 2^{n/2-1}\, n!$ for even $n \geq 2$.

Chapter 9. The Origins of Graph Theory

In 18th-century Königsberg seven bridges connected the land masses around the river Pregel. In 1736 Euler asked whether one can walk through the town, starting and ending at the same place, crossing every bridge exactly once.

9.1 The notion of graphs. Eulerian trails

Motivation. Euler saw that only the pattern of connections matters, not the geography: land masses become dots and bridges become lines. Whether a closed walk can use every line exactly once then depends only on the parity of the number of lines at each dot.

A **graph** consists of finitely many **vertices** (dots) and finitely many **edges** (lines), each joining two vertices. The **degree** of a vertex is the number of edge-ends at it. Graphs may have multiple edges and **loops** (edges from a vertex to itself); a graph with neither is **simple**.

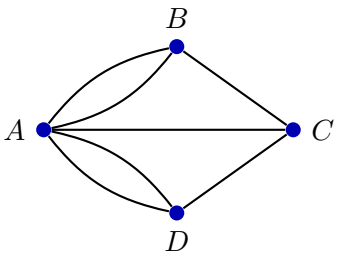


Figure. *The Königsberg bridges as a graph: four land masses, seven edges. All four degrees $(5, 3, 3, 3)$ are odd, so no closed walk uses every bridge once (Theorem 9.2).*

A sequence of distinct edges $e_1 e_2 \cdots e_k$ is a **trail** if the end of e_i is the start of e_{i+1} (it can be drawn without lifting the pencil); a **walk** may repeat edges. A trail is **closed** if it ends where it starts, **Eulerian** if it uses all edges, and a **path** if it visits no vertex twice.

Definition 9.1 (Connected)

G is **connected** if any two vertices are joined by a path.

A **subgraph** H of G has vertices and edges among those of G. It is **induced** if it contains all edges of G between its vertices. The **connected components** are the maximal connected subgraphs; u, v lie in the same component iff a path joins them.

Theorem 9.2 (Euler)

A connected graph has a closed Eulerian trail iff all its degrees are even.

Proof.

Step 1 (only if). Follow a closed Eulerian trail W from its start S. Each visit to a vertex $A \neq S$ enters and leaves along two new edges, and W uses every edge, so $\deg A = 2a$ if A is visited a times. S has the first and last edge plus $2t$ edges for t intermediate visits: $\deg S = 2t + 2$.

Step 2 (a first closed trail). Assume all degrees are even. From any vertex S walk along unused edges. Whenever we enter a vertex other than S, an unused edge is available to leave it (even degree), so we can only get stuck at S; as G is finite we do return to S, giving a closed trail C_1.

Step 3 (a vertex with unused edges). If C_1 is not all of G, some vertex V of C_1 has an edge not in C_1. Otherwise some vertex A is not on C_1, and a path from A to C_1 (connectedness) enters C_1 first at a vertex V through an edge not in C_1, a contradiction.

Step 4 (splicing). Removing the edges of C_1 keeps all degrees even. Starting at V as in Step 2, find a closed trail C_2 among the remaining edges. Walk along C_1 to V, around C_2, and finish C_1: this is a longer closed trail. Repeat; as there are finitely many edges, we end with a closed Eulerian trail. ■

Corollary 9.3

A connected graph G has an Eulerian trail from S to a different vertex T iff S and T have odd degree and all other degrees are even.

Proof. Add a new edge ST to get H. Eulerian trails of G from S to T correspond to closed Eulerian trails of H (close up with the new edge, or cut it out). The degrees of H are those of G with $\deg S$, $\deg T$ increased by 1. Apply Theorem 9.2 to H. ■

Theorem 9.4 (Handshake lemma)

In a graph without loops, the number of vertices of odd degree is even.

Proof. Each of the e edges contributes 1 to the degree of each of its two endpoints, so $d_1 + \cdots + d_n = 2e$ is even. Hence the number of odd summands is even. ■

9.2 Hamiltonian cycles

Motivation. Visiting every *vertex* once is much harder than using every edge once: no efficient test is known, and the problem is NP-complete. Still, enough edges force a Hamiltonian cycle, and Dirac's degree condition is the classic sufficient criterion.

A **cycle** is a closed trail that repeats no vertex except start = end; a cycle with k vertices has k edges. A **Hamiltonian cycle** (**path**) is a cycle (path) through all vertices. Example: seating guests around a round table so that neighbours know each other is a Hamiltonian cycle in the acquaintance graph.

Deciding whether a graph has a Hamiltonian cycle is **NP-complete**. A cycle is easily verified once exhibited, but no algorithm polynomial in n is known to certify that none exists; checking all $(n-1)!$ seatings is far too slow (Chapter 20 of the book).

Theorem 9.5 (Dirac)

Let $n \geq 3$ and let G be a simple graph on n vertices with all degrees at least $n/2$. Then G has a Hamiltonian cycle.

Proof.

Step 1 (connected). A component with fewer than $n/2$ vertices would contain vertices of degree $< n/2$, and if G were disconnected, some component would have fewer than $n/2$ vertices.

Step 2 (a saturated counterexample). Suppose G has no Hamiltonian cycle. Add edges as long as possible without creating one. The resulting G' still has degrees $\geq n/2$ and no Hamiltonian cycle, but adding any missing edge creates one. G' is not complete, since K_n ($n \geq 3$) is Hamiltonian.

Step 3 (a Hamiltonian path). Take non-adjacent x, y in G'. Adding xy creates a Hamiltonian cycle through xy, so G' has a Hamiltonian path $x = z_1, z_2, \ldots, z_n = y$.

Step 4 (pigeon-hole). Let $A = \{i \in \{2, \dots, n-1\} \mid xz_i \in E\}$ and $B = \{i \in \{2, \dots, n-1\} \mid z_{i-1}y \in E\}$. Since $x \not\sim y$, $|A| = \deg x \geq n/2$ and $|B| = \deg y \geq n/2$ (shift the indices of y's neighbours by one). Both lie in an $(n-2)$-element set, so $|A| + |B| \geq n > n-2$ and some $i \in A \cap B$ exists.

Step 5. Then $x\, z_2 \cdots z_{i-1}\, y\, z_{n-1} \cdots z_i\, x$ is a Hamiltonian cycle in G', a contradiction. ■

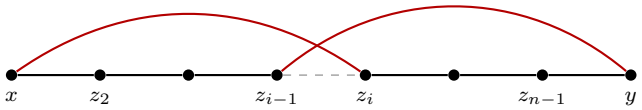


Figure. *Dirac's theorem: if $x \sim z_i$ and $z_{i-1} \sim y$ (red), drop the path edge $z_{i-1}z_i$ (dashed) to obtain the Hamiltonian cycle $x\, z_2 \cdots z_{i-1}\, y\, z_{n-1} \cdots z_i\, x$.*

9.3 Directed graphs

Motivation. One-way streets turn graphs into digraphs. Euler's criterion becomes "in-degree = out-degree". Tournaments (orientations of complete graphs) always have a Hamiltonian path, and a Hamiltonian cycle exactly when every player can be reached from every other.

In a **directed graph** (**digraph**) each edge has a direction, and trails must follow it. G is **strongly connected** if for all vertices a, b there is a directed path from a to b. The **in-degree** (**out-degree**) of a vertex is the number of edges ending (starting) there. G is **balanced** if in-degree = out-degree at every vertex. A vertex is **isolated** if both degrees are 0.

Theorem 9.6

A digraph without isolated vertices has a closed Eulerian trail iff it is balanced and strongly connected.

Proof.

Step 1 (necessity). A closed Eulerian trail leaves each vertex as often as it enters it, so G is balanced. Following the trail from any vertex reaches any other (every vertex has an edge on it), so G is strongly connected.

Step 2 (sufficiency) [Supplement]. We repeat the proof of Theorem 9.2 with directed edges. Starting at S, walk along unused edges in their direction. Balance means that each time we enter a vertex $\neq S$, an unused outgoing edge remains, so we can only stop at S, giving a closed trail C_1. If C_1 misses an edge, some vertex V on C_1 has an unused edge. Otherwise take a vertex A off C_1 (one exists, as there are no isolated vertices) and a directed path from C_1 to A (strong connectivity); its first edge leaving C_1 starts at such a V. Removing C_1 keeps the graph balanced, so a closed trail C_2 from V exists, and splicing C_2 into C_1 at V gives a longer closed trail. Repeat until all edges are used. ■

A simple graph is **complete** if all pairs of distinct vertices are adjacent ($\binom{n}{2}$ edges, K_n). Orienting every edge of K_n gives a **tournament**: the results of a round robin, with ij an edge if i beat j. There are $2^{\binom{n}{2}}$ tournaments on n labelled vertices.

Theorem 9.7

Every tournament has a Hamiltonian path.

Proof. Induction on n; $n \leq 2$ is clear. Remove a vertex V; by induction $T - V$ has a Hamiltonian path $h_1 h_2 \cdots h_{n-1}$. If $h_i V$ and $V h_{i+1}$ are edges for some i, insert V between h_i and h_{i+1}. Otherwise

the edges between V and the path never switch from "into V" to "out of V". So there is k with Vh_j an edge for $j \le k$ and h_jV an edge for $j > k$. Then Vh_1 is an edge (if $k \ge 1$, put V first) or $h_{n-1}V$ is an edge (if $k = 0$, put V last). ■

Theorem 9.8 (Camion)

A tournament has a Hamiltonian cycle iff it is strongly connected.

Proof.

Step 1 (only if). A Hamiltonian cycle gives a directed path between any two vertices.

Step 2 (a cycle exists). Let T be strongly connected. If T had no cycle, then $xy, yz \in E$ would force $xz \in E$ (otherwise zx closes a 3-cycle). So T would be transitive, and its vertices could be listed with every edge pointing to the right; such a tournament is not strongly connected (for $n \ge 2$).

Step 3 (a longest cycle). Let $C = y_1 \cdots y_k$ be a longest cycle and suppose it misses a vertex. By strong connectivity some edge leaves C, say y_1x with $x \notin C$. If $xy_2 \in E$, then $y_1xy_2 \cdots y_k$ is longer, so $y_2x \in E$; inductively $y_ix \in E$ for all i.

Step 4. Let Z be the set of $z \notin C$ with $y_1z \in E$; by Step 3, $y_iz \in E$ for all i and all $z \in Z$. Strong connectivity gives an edge zt with $z \in Z$, $t \notin Z$. Then $t \notin C$, and $t \notin Z$ means $ty_1 \in E$. So $z\,t\,y_1y_2 \cdots y_k$ (preceded by $y_k \to z$) is a cycle longer than C, a contradiction. ■

9.4 The notion of isomorphisms

Motivation. Two drawings may describe the same graph, and two different labellings may hide the same shape. Isomorphism makes "same up to relabelling the vertices" precise.

Two graphs on the same labelled vertex set are **identical** if every pair of vertices is joined by the same number of edges in both, however they are drawn.

Definition 9.9 (Isomorphism)

Graphs G, H are **isomorphic** if there is a bijection f from the vertices of G onto those of H such that X, Y and $f(X), f(Y)$ are joined by the same number of edges, for all X, Y. Such f is an **isomorphism**.

Example 9.10

For the two 4-vertex graphs in the book's Figure 9.8, $f(A) = H$, $f(B) = E$, $f(C) = F$, $f(D) = G$ is an isomorphism. The graphs are not identical (the first contains AB), but they are the same as unlabelled graphs.

Isomorphisms preserve degrees, numbers of cycles, numbers of paths of each length, and so on. Different degree multisets therefore prove non-isomorphism, but equal degree multisets do not prove isomorphism. No efficient general isomorphism test is known (checking all $n!$ bijections is too slow), and it is not known whether the problem is NP-complete; Babai's recent quasi-polynomial algorithm is a major advance.

9.5 Selected exercises

Motivation. The book exercises below combine the basic tools of the chapter (degree counting, Eulerian trails, and longest paths) to force or construct substructures.

Exercise 9.3 (A 4-cycle from many edges)

Let G be a simple graph on 10 vertices with 28 edges. Prove that G contains a cycle of length 4.

Solution.

Step 1 (cherries). Count the pairs $(v, \{x, y\})$ where $x \neq y$ are both neighbours of v. Vertex v contributes $\binom{d_v}{2}$ of them, so there are $\sum_v \binom{d_v}{2}$ pairs.

Step 2 (if there is no 4-cycle). Suppose G has no 4-cycle. If two different vertices $v \neq w$ had two common neighbours $x \neq y$, then $x, y \notin \{v, w\}$ (the graph has no loops) and $v\,x\,w\,y$ would be a 4-cycle. So every pair $\{x, y\}$ occurs in at most one cherry, and $\sum_v \binom{d_v}{2} \leq \binom{10}{2} = 45$.

Step 3 (the degrees force more). We have $\sum_v d_v = 2 \cdot 28 = 56$, and by the Cauchy–Schwarz inequality $\sum_v d_v^2 \geq \frac{1}{10}\left(\sum_v d_v\right)^2 = 313.6$. Hence

$$\sum_v \binom{d_v}{2} = \frac{1}{2}\Big(\sum_v d_v^2 - \sum_v d_v\Big) \geq \frac{313.6 - 56}{2} = 128.8 > 45,$$

contradicting Step 2. So G contains a 4-cycle. ∎

Exercise 9.6 (Two orange and two blue edges at each vertex)

Let G be a simple graph in which every vertex has degree 4. Prove that the edges of G can be coloured orange and blue so that every vertex is incident to two orange and two blue edges.

Solution. It suffices to colour each connected component H separately. All degrees in H equal 4, so H has edges, and by Theorem 9.2 it has a closed Eulerian trail $v_0, e_1, v_1, e_2, \ldots, e_m, v_m = v_0$.

Step 1 (the colouring). Each edge has two ends, so $2m = \sum_{v \in H} \deg v = 4|V(H)|$ and $m = 2|V(H)|$ is even. Colour e_i orange for odd i and blue for even i, and put $e_{m+1} = e_1$.

Step 2 (edges at a vertex). Let $v \in H$. Each edge at v has exactly one end at v (no loops) and occurs exactly once in the trail. So the four edges at v are grouped into the pairs $\{e_i, e_{i+1}\}$ for the indices $i \in [m]$ with $v_i = v$: entering v along e_i and leaving along e_{i+1} (for $i = m$, the trail closes up via e_1). There are two such indices.

Step 3 (conclusion). In each pair the indices i and $i+1$ have different parity; this holds also for the pair $\{e_m, e_1\}$, because m is even. So each pair has one orange and one blue edge, and v is incident to two orange and two blue edges. ∎

Exercise 9.23 (High minimum degree forces connectivity)

(a) Prove that a simple graph on ten vertices in which every vertex has degree at least five is connected.
(b) Generalize to simple graphs on n vertices.

Solution. We prove (b): a simple graph G on n vertices with all degrees at least $\frac{n-1}{2}$ is connected; for $n = 10$ this gives (a), as $5 \geq \frac{9}{2}$.

Step 1. Let $u \neq v$ be vertices that are not adjacent. Their neighbourhoods $N(u), N(v)$ are subsets of the $(n-2)$-element set $V(G) \setminus \{u, v\}$, and

$$|N(u)| + |N(v)| \geq n - 1 > n - 2.$$

So $N(u) \cap N(v) \neq \emptyset$: some w is adjacent to both, and $u\,w\,v$ is a path.

Step 2. Hence any two vertices are joined by a path of length at most 2, and G is connected.

The bound is sharp: for even n, two disjoint copies of $K_{n/2}$ have all degrees $\frac{n}{2} - 1$ and are disconnected. ■

Exercise 9.46 (A long cycle in a regular graph)

Let $k \geq 2$ and let G be a simple graph in which every vertex has degree k. Prove that G contains a cycle of length at least $k + 1$.

Solution.

Step 1 (a longest path). G has finitely many paths and at least one edge, so let $P = v_0 v_1 \cdots v_m$ be a path with the largest number of vertices.

Step 2 (neighbours of an end). Every neighbour of v_0 lies on P: if $u \notin P$ were adjacent to v_0, then $u\,v_0\,v_1 \cdots v_m$ would be a longer path. The k neighbours of v_0 are distinct vertices among $v_1, \ldots, v_m$ (G is simple), so the largest index j with v_j adjacent to v_0 satisfies $j \geq k \geq 2$.

Step 3 (the cycle). Then $v_0 v_1 \cdots v_j v_0$ is a cycle (its vertices are distinct and $j \geq 2$), of length $j + 1 \geq k + 1$.

The proof only uses that all degrees are at least k. The bound is attained by K_{k+1}. ■

Chapter 10. Trees

Networks should be connected with as few links as possible. Graphs that are connected but lose connectivity when any edge is removed are the trees of this chapter.

10.1 Minimally connected graphs

Motivation. "Minimally connected", "connected without cycles", "unique paths" and "connected with $n-1$ edges" all describe the same graphs, the trees. Counting labelled trees gives Cayley's surprisingly clean n^{n-2}, proved here by Joyal's bijection with functions $[n] \to [n]$.

Walks contain paths [Supplement]. If there is a walk from x to y, there is a path: take a shortest walk; if it visited a vertex twice, cutting out the portion between the two visits would give a shorter walk.

Theorem 10.1

For a connected simple graph G the following are equivalent:
(1) G is **minimally connected**: deleting any edge disconnects it;
(2) G contains no cycle.

Definition 10.2 (Tree)

A connected simple graph satisfying (1), equivalently (2), is a **tree**.

Proof of Theorem 10.1.

Step (1)⇒(2). If G had a cycle C containing the edge ab, delete ab. A path using ab can be rerouted along the rest of C, giving a walk and hence a path. So $G-ab$ is still connected, contradicting (1).

Step (2)⇒(1). If deleting an edge AB leaves G connected, a path P from B to A avoids AB, and $AB \cup P$ is a cycle. ∎

Corollary 10.3

A connected graph is a tree iff every two vertices are joined by exactly one path.

Proof.

Step (unique paths ⇒ tree). If deleting an edge rs left the graph connected, r and s would be joined both by the edge rs and by a path avoiding it.

Step (tree ⇒ unique paths) [Supplement]. Suppose two different paths P, Q go from x to y. Let u be the last vertex up to which they agree, so they leave u along different edges. Let w be the first vertex after u on P that also lies on Q after u (it exists, since y is such a vertex). The segments of P and Q from u to w meet only at u and w, so together they form a cycle, contradicting Theorem 10.1. ∎

Deleting one edge of an n-cycle gives a tree with $n-1$ edges; experiments suggest this is always the number of edges.

Theorem 10.4

Every tree on n vertices has $n-1$ edges. Conversely, every connected graph on n vertices with $n-1$ edges is a tree.

Lemma 10.5

A tree on $n \geq 2$ vertices has at least two vertices of degree 1 (**leaves**).

Proof. Take a path p of maximum length; it has two distinct endpoints. If an endpoint a had another neighbour b, then either $b \notin p$ and p could be extended, or $b \in p$ and p together with ab would contain a cycle. Both are impossible, so both endpoints are leaves. ■

Proof of Theorem 10.4.

Step 1 (trees). Induction on n; $n = 1$ is trivial. A tree T on $n+1$ vertices has a leaf l (Lemma 10.5). $T - l$ is connected and acyclic, hence a tree on n vertices with $n - 1$ edges, so T has n edges.

Step 2 (converse) [Supplement]. Let G be connected with n vertices and $n - 1$ edges. While G has a cycle, delete an edge of a cycle; connectivity is preserved (proof of Theorem 10.1). This ends with a tree on the same n vertices, which by Step 1 has $n - 1$ edges. So no edge was deleted, and G was a tree. ■

A **forest** is a graph whose components are trees.

Proposition 10.6

A forest on n vertices with k components has $n - k$ edges.

Proof. Apply Theorem 10.4 to each component: each has one more vertex than edges. ■

Counting labelled trees. Unlabelled trees are counted in Chapter 18 of the book. Trees on the vertex set $[n]$ number $1, 1, 3, 16$ for $n = 1, 2, 3, 4$, suggesting n^{n-2}; this time the "law of small numbers" does not mislead.

Theorem 10.7 (Cayley's formula)

There are $A_n = n^{n-2}$ trees with vertex set $[n]$.

Proof (Joyal).

Step 1 (doubly rooted trees). In each tree mark a vertex Start and a vertex End (possibly equal), in n^2 ways. There are $n^2 A_n$ such *doubly rooted trees*. We give a bijection from the n^n functions $f : [n] \to [n]$ to them, so $n^2 A_n = n^n$.

Step 2 (functions to trees). Let $C = \{c_1 < \cdots < c_k\}$ be the elements lying on cycles of f ($f^i(x) = x$ for some $i \geq 1$), and $d_i := f(c_i)$. Write $d_1, d_2, \ldots, d_k$ along a path, marking d_1 as Start and d_k as End. For $j \notin C$ add the edge $\{j, f(j)\}$. The result is connected, since iterating f from any j reaches C. It has $n - 1$ edges ($k - 1$ on the path, one for each $j \notin C$), so it is a tree by Theorem 10.4.

Step 3 (inverse). Given a doubly rooted tree, for j off the Start–End path let $f(j)$ be the neighbour of j on its unique path to the Start–End path. On the path, list the vertices from Start to End as $d_1, \ldots, d_k$, sort them as $c_1 < \cdots < c_k$, and set $f(c_i) := d_i$. This recovers f uniquely, so the map is a bijection. ■

Example 10.8

$n = 8$: $f(1) = 3$, $f(2) = 4$, $f(3) = 1$, $f(4) = 5$, $f(5) = 5$, $f(6) = 7$, $f(7) = 8$, $f(8) = 6$. The cycles are $(13), (5), (678)$, so $C = \{1, 3, 5, 6, 7, 8\}$ and $(d_i) = (3, 1, 5, 7, 8, 6)$. Hence the Start–End path is $3 - 1 - 5 - 7 - 8 - 6$, and $2 - 4 - 5$ hangs off it.

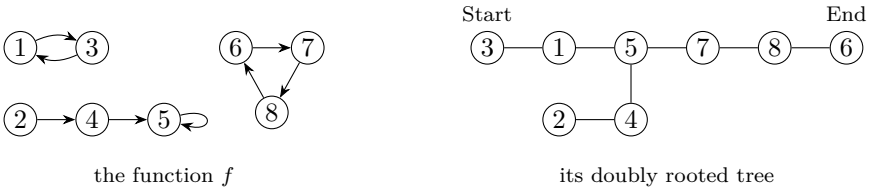


Figure. *Joyal's bijection for Example 10.8: the cyclic part of f becomes the Start–End path (read $f(c_1), \dots, f(c_k)$ with $c_1 < \cdots < c_k$); the other elements hang off it along $j \mapsto f(j)$.*

A **rooted tree** has one marked vertex (the root); there are n^{n-1} rooted trees on $[n]$. A **rooted forest** is a forest whose components are rooted trees.

Corollary 10.9

There are $(n+1)^{n-1}$ rooted forests on $[n]$.

Proof. Joining every root to a new vertex $n+1$ turns a rooted forest on $[n]$ into a tree on $[n+1]$. Conversely, from a tree on $[n+1]$, delete $n+1$ and declare its former neighbours roots. These maps are inverse, and Theorem 10.7 counts $(n+1)^{n-1}$ trees on $[n+1]$. ∎

10.2 Minimum-weight spanning trees. Kruskal's greedy algorithm

Motivation. Among the n^{n-2} spanning trees of a weighted K_n we want the cheapest, and trying them all is hopeless. Greedily taking the cheapest edge that creates no cycle is optimal, thanks to an exchange property of forests (the matroid property).

A **spanning tree** of a connected G is a tree with the same vertex set using edges of G. Every connected graph has one: delete edges of cycles until none remain. K_n has n^{n-2} spanning trees.

Example 10.10 (Minimum-weight spanning tree)

Given a connected simple graph G and weights $w : E(G) \to \mathbb{R}_+$, find a spanning tree T minimising $w(T) = \sum_{e \in T} w(e)$. (For $n = 20$ cities there are $20^{18} > 2.5 \cdot 10^{23}$ spanning trees; at 10^9 per second, checking them all would take more than 91 years.)

Kruskal's greedy algorithm. Start with $T = \emptyset$. At step i, add an edge $e_i \notin T$ of minimum weight among those whose addition creates no cycle. T is always a forest. While it has fewer than $n-1$ edges it is disconnected, and since G is connected some edge of G joins two components of T. So the algorithm runs $n-1$ steps and ends with a spanning tree.

Greedy methods are not always correct: for two disjoint edges of minimum total weight in a path with weights $2, 1, 2$ plus a disjoint edge of weight 10, greedy picks $1 + 10 = 11$ instead of $2 + 2 = 4$ (minimum-weight matching, Chapter 11).

Lemma 10.11 (Exchange property)

If F, F' are forests on the same vertex set and F has fewer edges than F', then some edge e of F' can be added to F so that $F \cup e$ is still a forest.

Proof. Otherwise every edge of F' joins two vertices of the same component of F. Then each component of F' lies inside a component of F, so F' has at least as many components as F. By Proposition 10.6 it has at most as many edges, a contradiction. ∎

Theorem 10.12

Kruskal's greedy algorithm finds a minimum-weight spanning tree.

Proof.

Step 1. Let $T = \{t_1, \dots, t_{n-1}\}$ be the greedy tree and suppose a spanning tree $H = \{h_1, \dots, h_{n-1}\}$ is lighter, both listed by increasing weight. Note $w(t_1) \le \cdots \le w(t_{n-1})$, since the set of eligible edges only shrinks.

Step 2. Let i be the smallest index with $\sum_{j \le i} w(h_j) < \sum_{j \le i} w(t_j)$; it exists because H is lighter overall. Since the inequality fails for $i-1$, we get $w(h_i) < w(t_i)$.

Step 3. Apply Lemma 10.11 to $T_{i-1} = \{t_1, \dots, t_{i-1}\}$ and $H_i = \{h_1, \dots, h_i\}$: some h_j, $j \le i$, can be added to T_{i-1} without a cycle. As $w(h_j) \le w(h_i) < w(t_i)$, the greedy algorithm would not have chosen t_i at step i, a contradiction. ∎

10.3 Adjacency matrices of graphs

Motivation. Encoding a graph by a matrix turns walks into matrix powers, so linear algebra answers counting questions: walk counts, connectivity and, in the next section, spanning trees.

Definition 10.13 (Adjacency matrix)

For a graph G on labelled vertices $1, \dots, n$, the **adjacency matrix** $A = A_G$ has A_{ij} equal to the number of edges between i and j (from i to j for digraphs). It is symmetric for undirected graphs.

Example 10.14

For the undirected graph on A, B, C, D with edges AB, AC, AD, BD:

$$A_G = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}.$$

Example 10.15

For the digraph with edges $A \to B$, $A \to C$, $B \to D$, $D \to A$:

$$A_H = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

Theorem 10.16

$(A^k)_{ij}$ is the number of walks of length k from i to j.

Proof. Induction on k; for $k = 1$ a walk of length 1 is an edge. If b_{iz} is the number of walks of length k from i to z (by induction $B = A^k$), then the walks of length $k+1$ from i to j with next-to-last vertex z number $b_{iz}a_{zj}$. Summing over z gives $(BA)_{ij} = (A^{k+1})_{ij}$. ∎

Theorem 10.17

A simple graph G on n vertices is connected iff all entries of $(I+A)^{n-1}$ are positive.

Proof. Walks contain paths, and paths have length at most $n-1$. So G is connected iff for all $i \neq j$ some $(A^k)_{ij} > 0$ with $1 \le k \le n-1$; the diagonal entries are always positive because of the term I. Since $(I+A)^{n-1} = \sum_{k=0}^{n-1} \binom{n-1}{k} A^k$ with positive coefficients and nonnegative matrices, the claim follows. ∎

10.4 The number of spanning trees of a graph

Motivation. Kirchhoff's matrix-tree theorem counts spanning trees as a determinant. Square submatrices of an incidence matrix have determinant ± 1 exactly on spanning trees, and the Cauchy–Binet formula adds these up.

A **spanning tree** of a digraph is a subgraph whose underlying undirected graph is a spanning tree.

Definition 10.18 (Incidence matrix)

For a loopless digraph with vertices $v_1, \dots, v_n$ and edges $e_1, \dots, e_m$, the **incidence matrix** A is $n \times m$ with $a_{ij} = 1$ if v_i is the start of e_j, -1 if v_i is its end, and 0 otherwise.

Theorem 10.19

Let A be the incidence matrix of a loopless digraph G and A_0 the matrix obtained by deleting any one row. Then the number of spanning trees of G is $\det(A_0 A_0^{\top})$.

Proof. Let the last row be deleted. If $m < n-1$, G is disconnected and both sides are 0.

Step 1 (claim). For every $(n-1) \times (n-1)$ submatrix B of A_0 (a choice of $n-1$ columns, i.e. edges, forming a subgraph G'), $|\det B| = 1$ if G' is a spanning tree and $\det B = 0$ otherwise. We use induction on n; for $n = 2$, B is a 1×1 matrix ± 1 for the single edge.

Step 2 (a leaf other than v_n). If some v_i, $i \neq n$, has degree 1 in G', its row in B has a single nonzero entry ± 1. Expanding along it reduces to the same situation for $G - v_i$ and $G' - v_i$, and G' is a spanning tree iff $G' - v_i$ is one. Induction applies.

Step 3 (no such leaf). Then G' is not a spanning tree (a tree has two leaves, Lemma 10.5). Having $n-1$ edges, it has a vertex of degree 0: otherwise all degrees except possibly that of v_n would be at least 2, giving at least $\frac{2(n-1)+1}{2} > n-1$ edges. If that vertex is v_i, $i \neq n$, then B has a zero row. If it is v_n, each column of B contains one 1 and one -1, so the rows of B sum to 0. Either way $\det B = 0$.

Step 4 (Cauchy–Binet). $\det(A_0 A_0^{\top}) = \sum_B (\det B)^2$, the sum over all $(n-1) \times (n-1)$ submatrices B of A_0. By Step 1 this is the number of spanning trees. ∎

Theorem 10.20 (Matrix-tree theorem)

Let U be a simple undirected graph with vertices $v_1, \dots, v_n$ and L_0 the $(n-1) \times (n-1)$ matrix with $\ell_{ii} = \deg v_i$, $\ell_{ij} = -1$ if $v_i v_j$ is an edge, and 0 otherwise ($1 \le i, j \le n-1$). Then U has exactly $\det L_0$ spanning trees.

Proof.

Step 1. Replace each edge of U by two opposite directed edges to get G, and let A_0 be its incidence matrix without the last row. Entry (i,i) of $A_0 A_0^{\top}$ counts the edges of G at v_i, which is $2 \deg_U v_i$.

Entry (i,j), $i \neq j$, gets -1 from each of the two edges between v_i and v_j if they are adjacent. So $A_0 A_0^{\top} = 2L_0$ and $\det(A_0 A_0^{\top}) = 2^{n-1} \det L_0$.

Step 2. Each spanning tree of U yields 2^{n-1} spanning trees of G (orient each of its $n-1$ edges in either direction), and every spanning tree of G arises this way. By Theorem 10.19, $2^{n-1}\tau(U) = 2^{n-1} \det L_0$. ■

Example 10.21

K_n has n^{n-2} spanning trees. Here L_0 has $n-1$ on the diagonal and -1 elsewhere. Adding all rows to the first gives a first row $(1,\dots,1)$; adding it to every other row gives an upper triangular matrix with diagonal $1, n, \dots, n$. So $\det L_0 = n^{n-2}$.

Example 10.22

The **complete bipartite graph** $K_{m,n}$ (parts of sizes m and n, all mn edges between them) has $n^{m-1}m^{n-1}$ spanning trees. Delete a vertex of the n-part. L_0 has the diagonal blocks nI_m and mI_{n-1}, with -1 entries between the parts. Adding all rows to the first gives $(1,\dots,1,0,\dots,0)$, and adding that row to the last $n-1$ rows makes the matrix block triangular, with determinant $n^{m-1}m^{n-1}$.

Theorem 10.23 (Matrix-tree theorem, eigenvalue version)

Let L be the $n \times n$ **Laplacian** of U (defined like L_0 but over all n vertices), with eigenvalues $\lambda_1, \dots, \lambda_{n-1}, \lambda_n = 0$. Then the number of spanning trees of U is $\frac{1}{n}\lambda_1\lambda_2\cdots\lambda_{n-1}$.

Proof [Supplement]. The book omits the proof. The rows of L sum to 0, so 0 is an eigenvalue. Write $\det(xI - L) = x\prod_{i<n}(x-\lambda_i)$; the coefficient of x is $(-1)^{n-1}\lambda_1\cdots\lambda_{n-1}$. On the other hand, expanding the determinant, the coefficient of x in $\det(xI-L)$ is $(-1)^{n-1}$ times the sum of the n principal $(n-1)\times(n-1)$ minors of L. Deleting row and column k from L gives L_0 for the labelling with v_k last, so by Theorem 10.20 every such minor equals the number $\tau(U)$ of spanning trees. Hence $\lambda_1\cdots\lambda_{n-1} = n\,\tau(U)$. ■

If U is d**-regular** (all degrees d), then $L = dI - A$, so the eigenvalues of L are $d - \alpha_i$ for the eigenvalues α_i of A.

Example 10.24

For K_n, $A + I = J$ (all ones) has rank 1 and trace n, so its eigenvalues are $n, 0, \dots, 0$. Those of A are $n-1, -1, \dots, -1$, those of $L = (n-1)I - A$ are $0, n, \dots, n$, and Theorem 10.23 gives $\frac{1}{n}n^{n-1} = n^{n-2}$ again.

10.5 Selected exercises

Motivation. The book exercises below use the characterizations of trees from Section 10.1: building a tree with prescribed degrees by induction, a connectivity property of complements, and the cut property behind greedy spanning-tree algorithms.

Exercise 10.1 (Degree sequences of trees)

Let $n \geq 2$ and let $a_1 \geq a_2 \geq \cdots \geq a_n$ be positive integers with $a_1 + \cdots + a_n = 2n - 2$. Prove that there is a tree on n vertices whose vertices have degrees $a_1, \ldots, a_n$.

Solution. Induction on n; the order of the a_i plays no role, so we prove the claim for any positive integers $a_1, \ldots, a_n$ with sum $2n - 2$.

Step 1 (initial step). For $n = 2$, $a_1 = a_2 = 1$, and a single edge works.

Step 2 (a 1 and a part ≥ 2). Let $n \geq 3$. If all $a_i \geq 2$, the sum would be at least $2n$; so some $a_i = 1$, say $a_n = 1$. If all $a_i = 1$, the sum would be $n < 2n - 2$; so some $a_i \geq 2$, say $a_1 \geq 2$.

Step 3 (reduction). The $n-1$ positive integers $a_1 - 1, a_2, \ldots, a_{n-1}$ have sum $2n-2-2 = 2(n-1)-2$. By the induction hypothesis there is a tree T' on vertices $v_1, \ldots, v_{n-1}$ with $\deg v_1 = a_1 - 1$ and $\deg v_i = a_i$ for $2 \leq i \leq n - 1$.

Step 4 (adding a leaf). Add a new vertex v_n and the edge $v_1 v_n$. The new graph is connected and has n vertices and $(n - 2) + 1 = n - 1$ edges, so it is a tree by Theorem 10.4. Its degrees are $a_1, \ldots, a_n$. ■

Exercise 10.10 (A graph or its complement is connected)

Prove that for every simple graph G on $[n]$, at least one of G and its complement $\bar{G}$ is connected. Give an example where both are connected.

Solution. Suppose G is not connected, and let $x \neq y$ be vertices; we join them by a path in $\bar{G}$.

Step 1 (different components). If x, y lie in different components of G, then xy is not an edge of G, so it is an edge of $\bar{G}$.

Step 2 (same component). If x, y lie in the same component K, choose a vertex z in another component (there is one, as G is disconnected). Then xz and zy are not edges of G, so $x\,z\,y$ is a path in $\bar{G}$.

Hence $\bar{G}$ is connected. Example: the path $1\,2\,3\,4$ has complement with edges $13, 14, 24$, which is the path $3\,1\,4\,2$; both are connected. ■

Exercise 10.24 (The cut property)

A *cut* in a connected graph G is a minimal set S of edges whose removal splits G into two components. Let w be a weight function on the edges, let S be a cut, and let $e \in S$ satisfy $w(e) < w(f)$ for all other $f \in S$. Prove that every minimum-weight spanning tree of G contains e.

Solution.

Step 1 (the shape of a cut). Let V_1, V_2 be the vertex sets of the two components of $G - S$. Every edge between V_1 and V_2 lies in S, since otherwise it would survive in $G - S$. Conversely, every $f \in S$ joins V_1 to V_2: if both ends of f were in V_1 (or both in V_2), adding f back to $G - S$ would not change the components, so $S \setminus \{f\}$ would also split G into two components, contradicting minimality. So S is exactly the set of edges between V_1 and V_2.

Step 2 (exchange). Let T be a minimum-weight spanning tree and suppose $e = uv \notin T$, with $u \in V_1$, $v \in V_2$. By Corollary 10.3, T contains a unique u–v path P. It starts in V_1 and ends in V_2, so some edge f of P joins V_1 to V_2; then $f \in S$ and $f \neq e$, so $w(f) > w(e)$.

Step 3 (a cheaper tree). The forest $T - f$ has $n-2$ edges, so it has two components (Proposition 10.6), and u, v lie in different ones, since otherwise $T - f$ would contain a second u–v path avoiding f.

Thus $T' = T - f + e$ is connected with $n-1$ edges, a spanning tree by Theorem 10.4, and

$$w(T') = w(T) - w(f) + w(e) < w(T),$$

contradicting the minimality of T. Hence $e \in T$. ■

Chapter 11. Coloring and Matching

11.1 Introduction

Motivation. Assigning frequencies to towers so that nearby towers differ is colouring the vertices of a conflict graph. The least number of colours needed, the chromatic number, measures how constrained the graph is.

A phone company assigns one of four frequencies to each of ten towers; towers within 50 miles must use different frequencies. Join two towers by an edge if they are within 50 miles, and colour each vertex by its frequency.

Proposition 11.1

The four frequencies can be assigned to the ten towers iff the vertices of the graph can be coloured with four colours so that no two adjacent vertices have the same colour.

Proof. A frequency assignment is a colouring with four colours, and the constraint "towers within 50 miles differ" is exactly "adjacent vertices differ". ■

Definition 11.2 (Chromatic number)

The **chromatic number** $\chi(H)$ is the least k such that the vertices of H can be coloured with k colours with adjacent vertices coloured differently (a **proper colouring**). H is **k-colourable** if $\chi(H) \leq k$.

Example 11.3

The pentagon C_5 has $\chi = 3$. Two colours would have to alternate around a cycle of odd length, which is impossible; three colours suffice (red, blue, red, blue, green).

Components can be coloured independently and multiple edges add no constraints, so we may assume graphs are connected and simple.

11.2 Bipartite graphs

Motivation. Two colours suffice exactly when there is no odd cycle. Bipartite graphs can be dense (up to $n^2/4$ edges), but one edge more forces not just an odd cycle but a triangle, and in fact m triangles at once.

Definition 11.4 (Bipartite)

A 2-colourable graph is **bipartite**: its vertex set splits into **colour classes** A, B such that every edge has one end in A and one in B.

Trees are bipartite (colour by parity of the distance from a fixed vertex), as are even cycles. Triangles and all odd cycles are not.

Theorem 11.5 (König)

A graph is bipartite iff it contains no cycle of odd length.

Proof.

Step 1 (only if). Colours must alternate along a cycle $A_1A_2\cdots A_{2m+1}$, so A_1 and A_{2m+1} get the same colour although they are adjacent.

Step 2 (colouring by distance). Let G (wlog connected) have no odd cycle. Fix V, and colour W red if its distance $d(V,W)$ is even, blue if odd. Suppose an edge PQ joins two vertices of the same colour. Shortest paths $V \to P$ and $V \to Q$, followed by the edge PQ, form a closed walk of odd length (two lengths of equal parity plus one).

Step 3 (odd closed walks contain odd cycles) [Supplement]. A closed walk of odd length contains an odd cycle. Induction on its length: if no vertex repeats (other than start = end), it is a cycle. Otherwise split it at a repeated vertex into two shorter closed walks whose lengths add up to an odd number; one of them is odd. This contradicts the assumption, so the colouring is proper. ■

Theorem 11.6

A simple bipartite graph on n vertices has at most $\frac{n^2}{4}$ edges if n is even and at most $\frac{n^2-1}{4}$ if n is odd.

Proof. Take an extremal G with colour classes of sizes a and $n-a$. Every edge between the classes is present (otherwise add it), so G has $a(n-a)$ edges. This quadratic in a is maximised at $a=\lfloor n/2\rfloor$, giving $\frac{n^2}{4}$ or $\frac{n^2-1}{4}$. ■

The extremal graphs are the **complete bipartite graphs** $K_{a,b}$. A graph on $2m$ vertices with m^2 edges can be bipartite ($K_{m,m}$); with more edges it has an odd cycle, and in fact more.

Lemma 11.7 (Mantel)

A simple graph H on $2m$ vertices ($m \geq 2$) with at least m^2+1 edges contains a triangle.

Proof.

Step 1 ($m=2$). $H \subseteq K_4$ has at least 5 edges, so it is not bipartite (Theorem 11.6) and has an odd cycle, which on 4 vertices must be a triangle.

Step 2 (induction). Take an edge FG. If $\deg F + \deg G > 2m$, then F and G have a common neighbour T (they have at least $2m-1$ neighbours in the other $2m-2$ vertices, counting G and F themselves), and FGT is a triangle. Otherwise deleting F and G removes at most $2m-1$ edges (FG is counted twice), leaving $2m-2$ vertices and at least $m^2+1-(2m-1)=(m-1)^2+1$ edges. By induction there is a triangle. ■

Theorem 11.8

A simple graph H on $2m$ vertices ($m \geq 2$) with at least m^2+1 edges contains at least m triangles.

Proof. Assume exactly m^2+1 edges (deleting edges destroys no triangle count in our favour) and use induction on m; for $m=2$, $H=K_4$ minus an edge, which has 2 triangles. By Lemma 11.7 there is a triangle ABC; we need $m-1$ more. Let the number of edges between ABC and the other $2m-3$ vertices be $2m-3+x$. Each outside vertex sending two edges to ABC spans a triangle with them, and by pigeon-holing at least x outside vertices do so: at least x triangles of this kind. Let R be the graph on the outside vertices.

Step 1 ($x \geq m-1$). Done.

Step 2 ($1 \leq x < m-1$). At most $3m-5$ edges meet ABC from outside, so R has at least $m^2+1-(3m-5)-3=(m-1)(m-2)+1$ edges on $2m-3$ vertices. Deleting a vertex of minimum

degree in R removes at most the average degree. So the remaining R' on $2m-4$ vertices has more than $(m-1)(m-2)\frac{2m-4}{2m-3} = (m-2)^2\frac{2m-2}{2m-3} > (m-2)^2$ edges. By induction R' has $m-2$ triangles; with the $x \geq 1$ triangles above we have $m-1$.

Step 3 ($x \leq 0$). At most $2m-3$ edges join ABC to R, so R has at least $m^2+1-(2m-3)-3 = (m-1)^2$ edges. If no such edge exists, R has $m^2-2 \geq (m-1)^2+1$ edges and R plus any vertex of ABC is a graph on $2m-2$ vertices to which induction applies. Otherwise add to R a vertex of ABC adjacent to R, getting $2m-2$ vertices and at least $(m-1)^2+1$ edges, hence $m-1$ further triangles by induction. ■

So as soon as one triangle is guaranteed, m of them are.

11.3 Matchings in bipartite graphs

Motivation. Filling job openings with qualified applicants means choosing disjoint edges in a bipartite graph. Hall's condition (every group of openings has at least as many qualified applicants) is obviously necessary, and remarkably also sufficient. Augmenting paths certify maximality, and deferred acceptance produces stable assignments.

With jobs and applicants as the two classes and edges for qualifications, a set of filled jobs is a set of vertex-disjoint edges.

Proposition 11.9

All m openings can be filled simultaneously iff the job–applicant graph has m vertex-disjoint edges.

Proof. A hiring fills each job with one qualified person and gives each person at most one job. These are exactly the edges "job–person hired for it", and they are pairwise vertex-disjoint; conversely, m such edges are a complete hiring. ■

Definition 11.10 (Matching)

A set S of edges of G, no two sharing a vertex, is a **matching** (independent set of edges). It is **perfect** if it covers every vertex.

Definition 11.11 (Matching X into Y)

For a bipartite $G = (X, Y)$, a matching covering all of X is a **perfect matching of X into Y**; we say X **can be matched into Y**.

11.3.1 Bipartite graphs with perfect matchings

For $T \subseteq X$ let $N(T) = N_G(T)$ be the set of neighbours of T.

Proposition 11.12

If X can be matched into Y, then $|T| \leq |N(T)|$ for all $T \subseteq X$.

Proof. A matching of X into Y matches T injectively into $N(T)$. ■

Theorem 11.13 (Hall's theorem)

X can be matched into Y iff $|T| \le |N(T)|$ for all $T \subseteq X$.

Proof (Halmos–Vaughan). "Only if" is Proposition 11.12. For "if", induct on $|X|$; $|X| \le 1$ is trivial.

Step 1 (strict surplus). Suppose $|T| < |N(T)|$ for all nonempty proper $T \subset X$. Take an edge xy, $x \in X$, and $G' = G - x - y$. For nonempty $A \subseteq X - x$, $|N_{G'}(A)| \ge |N_G(A)| - 1 \ge |A|$. By induction $X - x$ can be matched into $Y - y$ in G'; add xy.

Step 2 (a tight set). Otherwise some nonempty proper $B \subset X$ has $|B| = |N(B)|$. Let G_1 be induced by $B \cup N(B)$ and G_2 the graph obtained by deleting $B \cup N(B)$. In G_1, $N_{G_1}(T) = N_G(T)$ for $T \subseteq B$, so Hall's condition holds. In G_2, for $U \subseteq X - B$, $N_G(U \cup B)$ is the disjoint union of $N_{G_2}(U)$ and $N_G(B)$, so

$$|N_{G_2}(U)| = |N_G(U \cup B)| - |N_G(B)| \ge |U| + |B| - |B| = |U|.$$

By induction B is matched onto $N(B)$ and $X - B$ into $Y - N(B)$; together they match X into Y. ■

A matching is **maximal** if no edge can be added to it, and **maximum** if no matching is larger. Every maximum matching is maximal, but not conversely: a path with three edges has the maximal matching consisting of its middle edge, whereas the two end edges form a larger matching.

Definition 11.14 (Alternating path)

For a matching M, a path $v_1 v_2 \cdots v_r$ is **M-alternating** if its edges alternate between M and not M.

Definition 11.15 (Augmenting path)

An M-alternating path is **M-augmenting** if it starts and ends at vertices not covered by M.

Swapping the edges of an augmenting path P ($M \mapsto M \triangle P$) gives a matching with one more edge.



Figure. *An M-augmenting path (bold: M, dashed: not in M) between two uncovered vertices. Swapping bold and dashed edges increases the matching by one.*

Theorem 11.16 (Berge)

A matching M in a simple graph is maximum iff there is no M-augmenting path.

Proof. "Only if": an augmenting path enlarges M. "If": let M' be a maximum matching and consider $M \triangle M'$. Every vertex meets at most one edge of each, so the components are paths and cycles alternating between M and M', and the cycles are even. A path with more M'-edges than M-edges would be M-augmenting, and one with more M-edges would be M'-augmenting, contradicting maximality of M'. So every component has equally many edges of M and M', and $|M| = |M'|$. ■

11.3.2 Stable matchings in bipartite graphs

n candidates and n jobs; each candidate ranks all jobs and each hiring manager ranks all candidates.

Definition 11.17 (Stable matching)

In a complete bipartite $G(X,Y)$ where every vertex ranks the other side, a perfect matching M is **stable** if there are no $x \in X$, $y' \in Y$ such that (a) $xy \in M$ but x prefers y' to y, and (b) $x'y' \in M$ but y' prefers x to x' (no pair would both rather be together).

Gale–Shapley (deferred acceptance). Each unmatched candidate applies to the best job not yet tried; each manager keeps the best applicant so far (tentatively) and rejects the others. The process ends with every candidate placed: an unplaced candidate C always has a job it has not yet applied to.

The result is stable. Suppose $xy, x'y' \in M$, x prefers y' to y, and y' prefers x to x'. Then x applied to y' before y. If y' accepted x then, it later dropped x for someone it prefers, and its final hire x' is at least as good, hence better than x, a contradiction. If y' rejected x, it already held someone better than x, and its final hire is at least as good, again a contradiction. With unequal sides, add fictitious jobs or candidates ranked last by everyone. In non-bipartite graphs, stable matchings need not exist.

Remark

Remark[Algorithmic note] Gale–Shapley runs in $O(n^2)$ applications, since each candidate applies to each job at most once.

11.4 More than two colours

Motivation. Which k-colourable graphs on n vertices have the most edges? The balanced complete k-partite graph, and any graph with more edges contains K_{k+1} (Turán). Large chromatic number does not, however, require a large clique.

Write $n = tk + r$ with $0 \le r \le k-1$ and split n vertices into k classes, r of size $t+1$ and $k-r$ of size t. Join vertices iff they lie in different classes; this is the **complete k-partite (Turán) graph** H, with

$$T(n,k) = \frac{k-1}{2k}\,n^2 - \frac{r(k-r)}{2k} \tag{11.1}$$

edges. [Supplement] *Reason:* H has $\frac{1}{2}(n^2 - \sum(\text{class sizes})^2) = \frac{1}{2}(n^2 - r(t+1)^2 - (k-r)t^2)$ edges, and substituting $n = tk + r$ gives (11.1). H is k-colourable.

Theorem 11.18 (Turán)

A simple graph on n vertices with more than $T(n,k)$ edges contains K_{k+1}; in particular it is not k-colourable.

Proof.

Step 1. Let G be K_{k+1}-free with n vertices and the maximum number of edges; we show $e(G) \le T(n,k)$ by induction on $t = \lfloor n/k \rfloor$. For $t = 0$, $n < k$ and $T(n,k) = \binom{n}{2}$.

Step 2. For $t \ge 1$, G contains a K_k, call it S. Indeed, if G is complete then $n \ge k$ gives one; otherwise adding a missing edge creates a K_{k+1}, whose other $k-1$ vertices together with one endpoint of that edge form a K_k in G.

Step 3. Edges within S: $\binom{k}{2}$. Each vertex outside S has at most $k-1$ neighbours in S (else K_{k+1}). $G - S$ is K_{k+1}-free on $n-k = (t-1)k + r$ vertices, so by induction it has at most $T(n-k,k)$ edges. Hence

$$e(G) \le \binom{k}{2} + (n-k)(k-1) + T(n-k,k) = T(n,k). \tag{11.2}$$

Step 4 (the identity (11.2)) [Supplement]. Remove one vertex from each class of the Turán graph H on n vertices. What remains is the Turán graph on $n-k$ vertices (class sizes t and $t-1$, same r). The removed edges are the $\binom{k}{2}$ among the removed vertices plus $k-1$ for each of the $n-k$ remaining vertices (one to every removed vertex outside its own class). ■

Odd cycles of length ≥ 5 have $\chi = 3$ without containing K_3. In a sense, odd cycles and complete graphs are the only obstacles:

Theorem 11.19 (Brooks)

If G is connected, neither a complete graph nor an odd cycle, and all degrees are at most d with $d \geq 3$, then $\chi(G) \leq d$.

Proof [Supplement] (Lovász). The book leaves this to a guided exercise.

Step 1 (greedy towards a root). If some vertex w has degree $< d$, order the vertices by non-increasing distance from w and colour greedily with colours $[d]$. Every vertex other than w has a neighbour closer to w, coloured later, so at colouring time at most $d-1$ of its neighbours are coloured; w itself has fewer than d neighbours. So d colours suffice. Hence assume G is d-regular.

Step 2 (a cut vertex). If v is a cut vertex, let $C_1, \dots, C_m$ be the components of $G - v$ and $G_i = G[C_i \cup \{v\}]$. In G_i the vertex v has degree $< d$, so G_i is d-colourable by Step 1 (or is a complete graph on at most d vertices). Permute colours so that v gets the same colour in all G_i and combine.

Step 3 (the key triple). Let G be 2-connected. We find w with non-adjacent neighbours u, v such that $G - u - v$ is connected.

If G is 3-connected: G is connected and not complete, so some w has two non-adjacent neighbours u, v (on a shortest path between non-adjacent vertices), and $G - u - v$ is connected.

If G has a 2-vertex cut $\{w, y\}$: $G - w$ is connected with cut vertex y, so its block structure has at least two leaf blocks. Each leaf block contains a vertex adjacent to w that is not a cut vertex of $G - w$, since otherwise G would have a cut vertex. Choose such u, v in two different leaf blocks; they are non-adjacent. Every vertex has degree $\geq d - 1 \geq 2$ in $G - w$, so leaf blocks are not single edges, and deleting u, v keeps $G - w$ connected. Finally w has $d \geq 3$ neighbours, so it keeps a neighbour in $G - w - u - v$, and $G - u - v$ is connected.

Step 4 (colouring). Colour u, v with colour 1. Order the rest of $G - u - v$ by non-increasing distance from w (in $G - u - v$), ending with w, and colour greedily from $[d]$. Each vertex $\neq w$ has a later neighbour, so at most $d - 1$ coloured neighbours. All d neighbours of w are coloured, but u, v share a colour, so they use at most $d - 1$ colours. Hence d colours suffice. ■

For every n there are triangle-free graphs with chromatic number n, so high chromatic number forces a high degree but not a large clique.

11.5 Matchings in graphs that are not bipartite

Motivation. For general graphs Hall's condition is replaced by Tutte's: deleting any s vertices may leave at most s odd components. Necessity is a parity count; for sufficiency one studies the maximal graphs without a perfect matching, whose structure turns out to be completely rigid.

For a vertex set S, $G - S$ deletes S and its edges, and $c_o(G - S)$ is the number of components of $G - S$ with an odd number of vertices.

Theorem 11.20 (Tutte)

G has a perfect matching iff $c_o(G - S) \le |S|$ for every vertex set S.

Only if. In a perfect matching, every odd component of $G - S$ has a vertex matched outside the component, i.e. into S, and these partners in S are distinct.

A graph is **saturated non-factorizable** if it has no perfect matching but adding any new edge creates one.

Lemma 11.21

If G is saturated non-factorizable and S is the set of vertices adjacent to all other vertices, then the components of $G - S$ are complete graphs.

Proof. Let ab, bc be edges of $G - S$ and suppose $ac \notin E$. Since $b \notin S$, some d is not adjacent to b. $G + ac$ has a perfect matching $F_1 \ni ac$ and $G + bd$ has one, $F_2 \ni bd$. $F_1 \triangle F_2$ is a union of even alternating cycles; let $C_1 \ni ac$ and $C_2 \ni bd$.

Step (a) $C_1 \neq C_2$. $F_1 \triangle C_1$ is a perfect matching avoiding ac and bd, i.e. one of G, a contradiction.

Step (b) $C_1 = C_2$. Traverse C_1 from b through d until reaching a or c, say a, along a path P. Then $P \cup ab$ is F_2-alternating and closed, and $F_2 \triangle (P \cup ab)$ is a perfect matching of G avoiding bd (it uses $ab \in E$), again a contradiction. ■

Theorem 11.22

G is saturated non-factorizable iff either (a) G is complete with an odd number of vertices, or (b) G has an even number of vertices and consists of vertex-disjoint complete graphs $S_0, G_1, \ldots, G_k$ with $k = |S_0| + 2$, each $|G_i|$ odd, and every vertex of every G_i joined to every vertex of S_0.

Proof.

Step 1 (odd order). No graph with an odd number of vertices has a perfect matching, and only K_n admits no new edge.

Step 2 (even order: structure). Let $S_0 = S$ as in Lemma 11.21; then S_0 is complete, the components G_i of $G - S$ are complete, and S_0 is joined to everything. If $c_o(G - S) \le |S|$, a perfect matching would exist: match one vertex of each odd G_i to distinct vertices of S, the rest of each complete G_i internally, and the leftover vertices of S (an even number, by parity) among themselves. Hence $c_o > |S|$, so by parity $c_o \ge |S| + 2$. If $c_o > |S| + 2$, or if there were an even component, adding an edge between two components would leave $c_o(G' - S) > |S|$, so G' would still have no perfect matching, contradicting saturation. So $k = |S_0| + 2$, all odd.

Step 3 (conversely) [Supplement]. A graph as in (b) has $c_o(G - S_0) = |S_0| + 2 > |S_0|$, so it has no perfect matching. Missing edges only join different G_i, G_j. Adding $g_i g_j$ and using it, the remaining $|S_0|$ odd components each match one vertex to a distinct vertex of S_0, and everything else matches inside the complete pieces: a perfect matching. So G is saturated. ■

Proof of Theorem 11.20 ("if", Hetyei–Lovász). Suppose G satisfies Tutte's condition but has no perfect matching. $S = \emptyset$ shows that $|V(G)|$ is even. Add edges as long as no perfect matching arises, reaching a saturated non-factorizable $G' \supseteq G$ on the same vertices, as in Theorem 11.22(b). Let S' be its set of universal vertices. $G' - S'$ has $|S'| + 2$ odd complete components. Removing the added edges may split these components, but every odd component of $G' - S'$ contains an odd component of $G - S'$. Hence $c_o(G - S') \ge |S'| + 2 > |S'|$, contradicting Tutte's condition. ■

11.6 Selected exercises

Motivation. Hall's condition is rarely checked set by set; in the two book exercises below it follows from a single double count of edges, and the second one turns this into a decomposition theorem for matrices.

Exercise 11.4 (A degree condition for Hall)

Let G be a bipartite graph with colour classes X, Y in which every vertex of X has degree at least as large as every vertex of Y, and every vertex of X has at least one neighbour. Prove that X can be matched into Y.

Solution. Let $a = \min_{x \in X} \deg x$ and $b = \max_{y \in Y} \deg y$; by assumption $a \geq b$ and $a \geq 1$. We verify Hall's condition (Theorem 11.13).

Step 1 (double count). Let $T \subseteq X$ and let E_T be the set of edges with an end in T. Each vertex of T has at least a edges, and an edge has only one end in X, so $|E_T| \geq a|T|$. The other ends of these edges lie in $N(T)$, and each vertex of $N(T)$ is the end of at most b of them, so $|E_T| \leq b\,|N(T)|$.

Step 2 (Hall's condition). Hence $a|T| \leq b|N(T)| \leq a|N(T)|$, and dividing by $a \geq 1$ gives $|T| \leq |N(T)|$. By Theorem 11.13, X can be matched into Y.

The extra assumption is needed only to exclude the trivial case without edges, where $a = b = 0$. ■

Exercise 11.9 (Magic squares are sums of permutation matrices)

Let A be an $n \times n$ matrix with nonnegative integer entries in which every row and every column has the same sum r. Prove that A is a sum of r permutation matrices.

Solution. Induction on r. For $r = 0$, $A = 0$ is the empty sum. Let $r \geq 1$.

Step 1 (a bipartite graph). Let G be the bipartite graph with colour classes $X = \{x_1, \dots, x_n\}$ (rows) and $Y = \{y_1, \dots, y_n\}$ (columns), with $x_i y_j$ an edge iff $a_{ij} > 0$.

Step 2 (Hall's condition). Let $T \subseteq X$. The entries in the rows of T add up to $r|T|$. Every nonzero such entry lies in a column of $N(T)$, and the entries of these columns add up to $r|N(T)|$. All entries are nonnegative, so $r|T| \leq r|N(T)|$, i.e. $|T| \leq |N(T)|$.

Step 3 (one permutation matrix). By Theorem 11.13, X can be matched into Y; as $|X| = |Y| = n$, the matching is $x_i y_{\sigma(i)}$ for a permutation σ, and $a_{i\sigma(i)} \geq 1$ for all i. Let P_σ be the permutation matrix with 1's at the positions $(i, \sigma(i))$.

Step 4 (induction). $A - P_\sigma$ has nonnegative integer entries, and all its line sums are $r - 1$. By the induction hypothesis it is a sum of $r - 1$ permutation matrices, so A is a sum of r of them. ■

Chapter 12. Planar Graphs

12.1 Euler's theorem for planar graphs

Motivation. Can three houses be joined to three wells by non-crossing roads? Drawing a graph without crossings creates a new parameter, the number of faces, and Euler's formula $V+F=E+2$ ties it to V and E. Counting edges around faces then rules out $K_{3,3}$ and K_5.

Definition 12.1 (Planar graph)

A graph is **planar** if it can be drawn in the plane with no two edges crossing. The edges of such a drawing cut the plane into regions, the **faces**; one of them, the outer face, is unbounded.

Theorem 12.2 (Euler's theorem on planar graphs)

A connected planar graph with V vertices, E edges and F faces satisfies $V+F=E+2$.

Proof. Induction on E. For $E=1$: either one edge ($V=2$, $F=1$) or one loop ($V=1$, $F=2$). Let $E \geq 2$.

Step 1 (an edge on a cycle). If removing some edge e leaves G' connected, then e lies on a cycle, so the two sides of e belong to different faces, which merge in G'. G' has V vertices, $E-1$ edges and $F-1$ faces, so $V+F-1=E-1+2$ by induction.

Step 2 (a tree). Otherwise G is a tree, so $V=E+1$ (Theorem 10.4) and $F=1$. ■

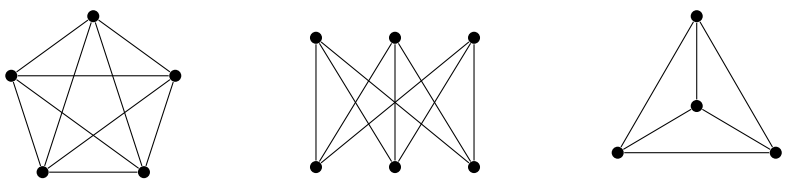
Example 12.3

$K_{3,3}$ is not planar, so the three houses, three wells problem has no solution.
Solution. With $V=6$, $E=9$, a planar drawing would have $F=5$ faces. $K_{3,3}$ is bipartite and simple, so it has no odd cycles and no 2-cycles, and every face is bounded by at least 4 edges. Each edge borders at most two faces, so $4F \leq 2E$, i.e. $20 \leq 18$, a contradiction.

$K_{3,3} \subseteq K_6$, so K_6 is not planar either. K_4 is planar (draw a square with one diagonal inside and the other outside).

Example 12.4

K_5 is not planar.
Solution. With $V=5$, $E=10$, a drawing would have $F=7$ faces, each bounded by at least 3 edges (the graph is simple). Then $3F \leq 2E$, i.e. $21 \leq 20$, a contradiction.



K_5 $K_{3,3}$ K_4, drawn without crossings

Figure. *The two minimal non-planar graphs and a crossing-free drawing of K_4.*

Deleting a vertex of degree 2 and merging its two edges, or subdividing an edge by a new vertex, does not change planarity. Graphs related by a sequence of these operations are **edge-equivalent** (homeomorphic).

Theorem 12.5 (Kuratowski)

A graph is not planar iff it contains a subgraph that is edge-equivalent to K_5 or $K_{3,3}$.

Proof. A subgraph edge-equivalent to K_5 or $K_{3,3}$ is non-planar by Examples 12.3 and 12.4, and a graph with a non-planar subgraph is non-planar. This proves the "if" direction. The book proves only this direction; the "only if" direction is stated without proof. ■

12.2 Polyhedra

Motivation. Regular polygons exist with any number of sides, but there are only five regular polyhedra. The reason is purely combinatorial: Euler's formula, together with "every face has ≥ 3 edges and every vertex has degree ≥ 3", leaves room for only five pairs (d, ℓ).

For a convex polyhedron (identified with its 1-skeleton) every face has at least 3 edges and every vertex lies on at least 3 edges. A polyhedron is **regular** (combinatorially) if every face is an ℓ-cycle and every vertex has degree d.

Theorem 12.6

A convex polyhedron with V vertices, F faces and E edges satisfies $V + F = E + 2$.

Proof (angle counting). Project P onto a plane p perpendicular to no face. Every face projects to a convex polygon, and the outer boundary B of the image is a convex v_1-gon; let v_2 vertices project inside it, so $v_1 + v_2 = V$. Each point inside B is covered by exactly one "front" and one "back" face. Let S be the sum of all angles of all projected faces.

Step 1 (by vertices). At a vertex projecting inside B, the image faces meeting there fill a full neighbourhood, so their angles sum to $360°$; this gives $360v_2$. At a vertex on B, the front faces and the back faces each fill the interior angle of B, so the boundary vertices contribute twice the angle sum $(v_1 - 2)180°$ of B. Hence

$$S = (v_1 - 2)360 + 360v_2 = (V - 2)360. \tag{12.1}$$

Step 2 (by faces). Let $f_1, \ldots, f_F$ be the numbers of edges of the faces. Each edge lies on two faces, so

$$\sum_{i=1}^{F} f_i = 2E, \tag{12.2}$$

and a convex k-gon has angle sum $(k - 2)180°$, so

$$S = \sum_{i=1}^{F} (f_i - 2)180 = 180 \sum_{i=1}^{F} f_i - 360F = 360(E - F). \tag{12.3}$$

Comparing (12.1) and (12.3) gives $V - 2 = E - F$. ■

Corollary 12.7

In a convex polyhedron, $3F \leq 2E$.

Proof. $f_i \geq 3$ in (12.2). ■

Proposition 12.8

In a convex polyhedron, $3V \le 2E$.

Proof. Let $c_1, \dots, c_V$ be the vertex degrees. Each edge has two ends, so

$$\sum_{i=1}^{V} c_i = 2E, \tag{12.4}$$

and $c_i \ge 3$. ■

Lemma 12.9

In a convex polyhedron, $E \le 3V - 6$ and $E \le 3F - 6$.

Proof. By Corollary 12.7, $E + 2 = V + F \le V + \frac{2E}{3}$, so $\frac{E}{3} \le V - 2$. By Proposition 12.8, $E + 2 \le F + \frac{2E}{3}$, so $\frac{E}{3} \le F - 2$. ■

Remark

Remark[$E \le 3V - 6$ for every simple planar graph [Supplement]] Let G be simple and planar with $V \ge 3$; we may assume it connected (adding edges between components keeps it planar). If G is a tree, $E = V - 1 \le 3V - 6$. Otherwise every face is bounded by a closed walk of length at least 3 (no loops or multiple edges) and each edge borders at most two faces (counted twice if it borders one face on both sides), so $3F \le 2E$. Euler's formula then gives $E \le 3V - 6$ as above. Consequently every simple planar graph has a vertex of degree at most 5: otherwise $2E \ge 6V$.

Lemma 12.10

Every convex polyhedron has a face with at most five edges.

Proof. By (12.2) and Lemma 12.9,

$$\sum_{i=1}^{F} f_i = 2E \le 6F - 12 < 6F, \tag{12.5}$$

so not all $f_i \ge 6$. ■

Lemma 12.11

Every convex polyhedron has a vertex of degree at most five.

Proof. By (12.4) and Lemma 12.9,

$$\sum_{i=1}^{V} c_i = 2E \le 6V - 12 < 6V. \tag{12.6}$$

■

The symmetry between V and F has a structural reason. Given a planar G, put a vertex inside each face and, for every edge e of G, join the vertices of the two faces on either side of e by an edge crossing e. The result G^* is planar with F vertices, E edges and V faces.

Definition 12.12 (Dual graph)

G^* is the **dual graph** of the plane graph G.

A statement about (V, E) of a polyhedron P is a statement about (F, E) of P^*. The dual of a regular polyhedron of type (d, ℓ) is regular of type (ℓ, d).

Classification. In a regular polyhedron, Lemmas 12.10 and 12.11 give $d, \ell \in \{3, 4, 5\}$, and by (12.2)

$$E = \frac{F\ell}{2}. \tag{12.7}$$

(A) $d = 3$: $3V = 2E$, so Euler gives $3F = E + 6$, and with (12.7), $(6 - \ell)F = 12$. Then $\ell = 3, 4, 5$ give $(V, E, F) = (4, 6, 4)$ (tetrahedron), $(8, 12, 6)$ (cube), $(20, 30, 12)$ (dodecahedron).

(B) $d = 4$: $4V = 2E$, so $E = 2F - 4$ and $(4 - \ell)F = 8$; only $\ell = 3$ works: $(6, 12, 8)$, the octahedron (dual of the cube).

(C) $d = 5$: $5V = 2E$, so $3E = 5F - 10$ and $(10 - 3\ell)F = 20$; only $\ell = 3$ works: $(12, 30, 20)$, the icosahedron (dual of the dodecahedron).

Theorem 12.13

There are exactly five regular polyhedra: the tetrahedron, the cube, the dodecahedron, the octahedron and the icosahedron.

Proof. The case analysis (A)–(C) leaves exactly these five parameter sets, and each is realised by the named solid. ■

	d	ℓ	V	E	F
Tetrahedron	3	3	4	6	4
Cube	3	4	8	12	6
Octahedron	4	3	6	12	8
Dodecahedron	3	5	20	30	12
Icosahedron	5	3	12	30	20

12.3 Colouring maps

Motivation. Colouring the countries of a map so that neighbours differ is, by duality, properly colouring the vertices of a planar graph. A vertex of degree ≤ 5 gives six colours at once; Kempe's chain-swapping trick brings this down to five. Four colours suffice too, but every known proof uses a computer.

Faces sharing an edge must get different colours; this is the same as properly colouring the vertices of G^*. K_4 is planar, so at least four colours are needed.

Proposition 12.14

Every planar graph can be properly vertex-coloured with six colours.

Proof. Induction on V; loops are excluded and multiple edges do not matter, so let G be simple. By the Remark after Lemma 12.9, G has a vertex A of degree at most 5. Colour $G - A$ with six colours by induction, and give A a colour not used by its at most five neighbours. ■

Theorem 12.15 (Five colour theorem)

Every planar graph can be properly vertex-coloured with five colours.

Proof. Induction on V for simple G. Take A of degree ≤ 5 and colour $G' = G - A$ properly with five colours. If the neighbours of A use at most four colours, colour A with the fifth. Otherwise A has neighbours $y_1, \dots, y_5$ in clockwise order, coloured $1, \dots, 5$.

Step 1 (Kempe chains). Let H_{13} be the subgraph of G' induced by the vertices coloured 1 or 3, and K the component of H_{13} containing y_1. Swapping colours 1 and 3 on K keeps the colouring proper. If $y_3 \notin K$, after the swap y_1 and y_3 both have colour 3, colour 1 is free at A, and we are done.

Step 2 (both chains cannot exist). So assume $y_3 \in K$: there is a path $y_1 \to y_3$ coloured alternately $1, 3$. Together with A it forms a cycle C through A that separates y_2 from y_4 in the plane (the neighbours occur around A in the order y_1, y_2, y_3, y_4). By the same argument with colours $2, 4$, we are done unless there is a path $y_2 \to y_4$ coloured $2, 4$. This path would have to cross C at a vertex, but its vertices are coloured $2, 4$ and those of $C - A$ are coloured $1, 3$. So this case cannot occur. ■

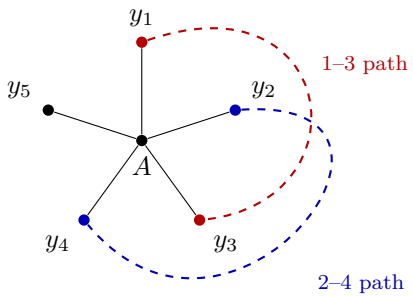


Figure. *Kempe chains in the five colour theorem: a 1–3 path from y_1 to y_3 and a 2–4 path from y_2 to y_4 would have to cross at a vertex.*

Remark

Remark[The four colour theorem] Four colours always suffice for maps with contiguous countries. Appel and Haken proved this in 1976 by a computer check of 1936 configurations, later reduced to 633 by Robertson, Sanders, Seymour and Thomas (1996). A machine-verified proof was given by Gonthier and Werner (2005).

12.4 Selected exercises

Motivation. In the book exercises below, Euler's formula is combined with two double counts (edges around faces, edges at vertices); the face lengths and degrees then satisfy a single linear relation that decides everything.

Throughout, the *length* of a face is the number of edges on its boundary, an edge counted twice if the face lies on both of its sides. Every edge has two sides, so the face lengths add up to $2E$; the degrees add up to $2E$ as well.

Exercise 12.15 (Polyhedra without triangles)

Let P be a convex polyhedron with no triangular faces. Prove that $E \le 2V - 4$.

Solution.

Step 1 (faces). Every face has at least 4 edges, so $4F \le 2E$, i.e. $F \le \frac{E}{2}$.

Step 2 (Euler). By Theorem 12.6, $E + 2 = V + F \le V + \frac{E}{2}$, so $\frac{E}{2} \le V - 2$ and $E \le 2V - 4$.

Equality holds iff all faces are quadrilaterals, e.g. for the cube ($V = 8$, $E = 12$). ■

Exercise 12.19 (Trivalent (a, b)-faced polyhedra)

Let P be a convex polyhedron whose faces are all a-gons or b-gons and in which every vertex lies on exactly three edges. Let p_a, p_b be the numbers of a-gonal and b-gonal faces and n the number of vertices.
(a) Express the number E of edges in two ways.
(b) Prove that $p_a(6 - a) + p_b(6 - b) = 12$.

Solution.

(a) Counting edges around faces, $2E = a\,p_a + b\,p_b$. Counting edges at vertices, $2E = 3n$.

(b) By Theorem 12.6, $n + p_a + p_b = E + 2$. Multiply by 6 and use $6n = 4E$ from (a):

$$4E + 6p_a + 6p_b = 6E + 12, \qquad \text{so} \qquad 6p_a + 6p_b - 2E = 12.$$

Substituting $2E = a\,p_a + b\,p_b$ gives $p_a(6 - a) + p_b(6 - b) = 12$.

For example, a trivalent polyhedron with pentagonal and hexagonal faces only (a fullerene, such as the soccer ball) has $p_5 \cdot 1 + p_6 \cdot 0 = 12$: exactly 12 pentagons, whatever the number of hexagons. ■

Exercise 12.22 (Octahedrites)

Let G be a connected planar graph in which each face has length 2, 3 or 4, with p_2, p_3, p_4 faces of these lengths. Assume that every vertex has degree 4 and that $p_2 + p_3 = 8$. Prove that $p_2 = 0$ and $p_3 = 8$.

Solution.

Step 1 (two counts). Counting at the vertices, $4V = 2E$, so $V = \frac{E}{2}$. Counting around the faces, $2E = 2p_2 + 3p_3 + 4p_4$, and $F = p_2 + p_3 + p_4$.

Step 2 (Euler). By Theorem 12.2, $V + F = E + 2$, i.e. $F = \frac{E}{2} + 2$. Multiplying by 4,

$$4p_2 + 4p_3 + 4p_4 = 2E + 8 = 2p_2 + 3p_3 + 4p_4 + 8, \qquad \text{so} \qquad 2p_2 + p_3 = 8.$$

Step 3 (conclusion). Subtracting $p_2 + p_3 = 8$ gives $p_2 = 0$, and then $p_3 = 8$. ■

Chapter 13. Ramsey Theory

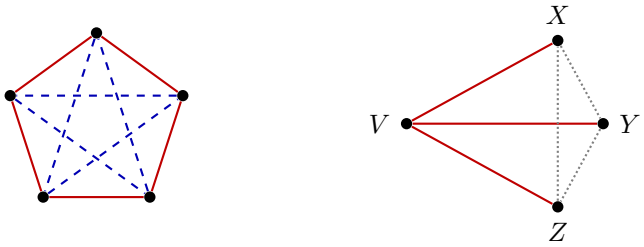
13.1 Ramsey theory for finite graphs

Motivation. Complete disorder is impossible: colour the edges of a large enough complete graph with two colours and a large monochromatic clique must appear. The proof is a double induction built on the recursion $R(k,\ell) \le R(k,\ell-1)+R(k-1,\ell)$; the exact values, however, are known only for very small cases.

Example 13.1

Among any six people there are three who all know each other or three who are mutual strangers.
Solution. Colour the edges of K_6 red (acquainted) or blue (strangers). A vertex V has degree 5, so by pigeon-hole at least three of its edges have the same colour, say red, going to X, Y, Z. If an edge of XYZ is red, it forms a red triangle with V; otherwise XYZ is a blue triangle.

Five people are not enough: colour the sides of a regular pentagon red and its diagonals blue. Every triangle contains both a side and a diagonal.



K_5: no monochromatic triangle Example 13.1

Figure. *Left: a 2-colouring of K_5 (solid red, dashed blue) without monochromatic triangles. Right: three red edges at V; any red edge of XYZ closes a red triangle.*

Equivalently: for every simple graph H on six vertices, H or its complement contains a **clique** (complete subgraph) of size 3.

Theorem 13.2 (Ramsey's theorem for graphs)

For integers $k,\ell \ge 2$ there is a least positive integer $R(k,\ell)$ such that every red/blue colouring of the edges of $K_{R(k,\ell)}$ contains a red K_k or a blue K_ℓ.

It suffices to find one such integer; the least one then exists.

Example 13.3

$R(3,3)=6$ by Example 13.1 and the pentagon. Trivially $R(2,2)=2$.

Proof of Theorem 13.2. Induction on $k+\ell$.

Step 1 (boundary). $R(k,2)=k$: either all edges of K_k are red, or one blue edge is a blue K_2. Likewise $R(2,\ell)=\ell$.

Step 2 (induction step). Assume $R(k,\ell-1)$ and $R(k-1,\ell)$ exist. We show

$$R(k,\ell) \le R(k,\ell-1)+R(k-1,\ell). \tag{13.1}$$

In K_N with $N = R(k, \ell-1) + R(k-1, \ell)$, a vertex V has $N-1$ edges, so it has at least $R(k, \ell-1)$ blue edges or at least $R(k-1, \ell)$ red edges.

(a) In the first case, the set b of blue neighbours contains a red K_k (done) or a blue $K_{\ell-1}$, which together with V is a blue K_ℓ.

(b) In the second case, the set r of red neighbours contains a blue K_ℓ or a red K_{k-1}, which together with V is a red K_k. ■

Corollary 13.4

For $k, \ell \geq 2$,

$$R(k, \ell) \leq R(k, \ell-1) + R(k-1, \ell). \tag{13.2}$$

Proof. This is (13.1), established in the proof of Theorem 13.2. ■

For instance $R(4,3) \leq R(4,2) + R(3,3) = 10$, which is not sharp:

Example 13.5

$R(4,3) = 9$.
Solution.
(1) K_9 *always works.* Some vertex V has (i) at least six red edges or (ii) at least four blue edges. Otherwise every vertex has exactly five red edges, and the red degree sum would be 45, which is odd.
(a) Six red neighbours span a red or blue triangle (Example 13.1); a blue triangle is done, and a red one plus V is a red K_4.
(b) Four blue neighbours: if all edges among them are red, they form a red K_4; a blue edge among them forms a blue triangle with V.
(2) K_8 *need not work.* On vertices $1, \ldots, 8$, colour $\{i, j\}$ $(i < j)$ blue if $j - i \in \{1, 4, 7\}$, red if $j - i \in \{2, 3, 5, 6\}$. A blue triangle with least vertex i needs two of $i+1, i+4, i+7$, but their differences $3, 6, 3$ are red. A red K_4 with least vertex i needs three of $i+2, i+3, i+5, i+6$, which forces $i+2, i+3$ or $i+5, i+6$ together, whose difference 1 is blue.

Example 13.6

$R(4,4) = 18$.
Solution. The upper bound is $R(4,4) \leq R(4,3) + R(3,4) = 18$. For the lower bound, colour K_{17} on $\mathbb{Z}_{17}$: $\{i, j\}$ is red iff $i - j$ is a nonzero square mod 17, i.e. $i - j \in Q = \{\pm 1, \pm 2, \pm 4, \pm 8\}$ (well defined since $-1 \in Q$).
[Supplement] *No monochromatic* K_4. Translations $x \mapsto x + c$ and multiplications $x \mapsto qx$ with $q \in Q$ preserve the red graph. Suppose $\{0, 1, a, b\}$ after normalisation is a red K_4 (translate one vertex to 0, then multiply by the inverse of another, which lies in Q). Then $a, b \in Q \cap (1+Q) \setminus \{0, 1\} = \{2, 9, 16\}$, but the pairwise differences of $2, 9, 16$ are ± 7 and ± 14, which are not in Q. Hence there is no red K_4. Multiplication by the non-square 3 maps red edges to blue and blue to red, so there is no blue K_4 either.

$R(4,5) = 25$ and $43 \leq R(5,5) \leq 48$ (bounds as quoted in the book); $R(k,k)$ is unknown for $k \geq 5$.

Theorem 13.7

For $k, \ell \geq 2$,

$$R(k, \ell) \leq \binom{k + \ell - 2}{k - 1}. \tag{13.3}$$

Proof. Induction on $k+\ell$. For $k=2$ both sides equal ℓ, and symmetrically for $\ell=2$. Otherwise, by Corollary 13.4, the induction hypothesis and Pascal's rule,

$$R(k,\ell) \le \binom{k+\ell-3}{k-1} + \binom{k+\ell-3}{k-2} = \binom{k+\ell-2}{k-1}.$$

■

Corollary 13.8

$R(k,k) \le 4^{k-1}$ for $k \ge 2$.

Proof. By Theorem 13.7, $R(k,k) \le \binom{2k-2}{k-1} \le \sum_i \binom{2k-2}{i} = 2^{2k-2}$. ■

Campos, Griffiths, Morris and Sahasrabudhe (2023) showed $R(k,k) < (4-\epsilon)^k$ for some $\epsilon > 0$. Exponential lower bounds come from the probabilistic method (Chapter 15).

13.2 Generalizations of the Ramsey theorem

Motivation. The same induction works with any number of colours and even when we colour r-element subsets instead of edges. The hypergraph version has a striking geometric consequence: among enough points in general position, some n form a convex polygon.

Example 13.9

If every edge of K_{17} is coloured with one of three colours, there is a monochromatic triangle. *Solution.* A vertex V has 16 edges, so at least six have the same colour, say green, with endpoint set g. A green edge inside g closes a green triangle with V. Otherwise g is 2-coloured on at least six vertices and contains a red or blue triangle (Example 13.1).

Theorem 13.10

For positive integers $n_1,\dots,n_k$ there is a least $N = R(n_1,\dots,n_k)$ such that every colouring of the edges of K_n, $n \ge N$, with colours $1,\dots,k$ has, for some i, a K_{n_i} all of whose edges have colour i.

Proof (completed [Supplement]). Induction on $n_1+\cdots+n_k$. If some $n_i = 1$, then $N = 1$ works (a single vertex is a K_1 of every colour). Otherwise let $R_i = R(n_1,\dots,n_i-1,\dots,n_k)$, which exist by induction, and

$$N = \sum_{i=1}^{k} (R_i - 1) + 2.$$

A vertex V of K_N has $N-1 = \sum_i (R_i-1)+1$ edges, so by pigeon-hole some colour i occurs on at least R_i of them. Let S be the set of their other endpoints. By the definition of R_i, K_S contains a K_{n_j} of colour j for some $j \ne i$ (done), or a K_{n_i-1} of colour i, which together with V is a K_{n_i} of colour i. ■

Instead of edges we may colour the r-element vertex sets (K_r-subgraphs) of K_n; $r=2$ is the edge case.

Theorem 13.11 (Hypergraph Ramsey theorem)

Colour each r-element subset of an n-set with one of k colours, and let $n_1, \dots, n_k$ be positive integers. There is a least $N = R_r(n_1, \dots, n_k)$ such that for $n \geq N$ there are an index i and an n_i-set all of whose r-subsets have colour i.

Proof [Supplement]. The book omits the proof. We use induction on r, and for fixed r on $n_1 + \cdots + n_k$.

Step 1 ($r = 1$). Colouring points: $N = \sum_i (n_i - 1) + 1$ works by pigeon-hole.

Step 2 (trivial cases). If some $n_i < r$, any n_i-set has no r-subsets at all, so it qualifies vacuously; take $N = n_i$.

Step 3 (induction step). Let $r \geq 2$, all $n_i \geq r$, and let $R_i = R_r(n_1, \dots, n_i - 1, \dots, n_k)$ (smaller sum) and

$$N = 1 + R_{r-1}(R_1, \dots, R_k)$$

(exists since $r - 1 < r$). Fix a point V of an N-set X, and colour each $(r-1)$-subset S of $X - V$ by the colour of $S \cup \{V\}$. By the choice of N there are i and $T \subseteq X - V$ with $|T| = R_i$ such that $S \cup \{V\}$ has colour i for every $(r-1)$-set $S \subseteq T$. Apply the definition of R_i to the original colouring on T. Either some n_j-subset of T ($j \neq i$) is monochromatic of colour j, and we are done, or some $(n_i - 1)$-set $U \subseteq T$ has all r-subsets of colour i. Then $U \cup \{V\}$ has all its r-subsets of colour i: those avoiding V lie in U, and those containing V are $S \cup \{V\}$ with $S \subseteq T$. ∎

Theorem 13.12 (Erdős–Szekeres)

For every n there is a least $ES(n)$ such that among any $N \geq ES(n)$ points in the plane, no three collinear, some n form a convex n-gon.

Proof. We show that $R_3(n, n)$ points suffice. Number the points $1, \dots, N$. For a triangle, list its vertices in increasing order; colour it red if that path turns clockwise and blue if counterclockwise. By Theorem 13.11 there are n points all of whose triangles have the same colour.

Step 1 (reduction) [Supplement]. n points (no three collinear) fail to be in convex position iff one of them lies inside a triangle formed by three others. Indeed, if a point P is not a vertex of the convex hull of the others, triangulate that hull by diagonals from one vertex; P lies in one of these triangles, and not on its boundary since no three points are collinear.

Step 2 (no point inside a triangle). Let $\chi(x, y, z) = \pm 1$ be the orientation of (x, y, z). It is invariant under cyclic shifts and changes sign under transpositions, and the colour of a triangle with $x < y < z$ is $\chi(x, y, z)$. Suppose D lies inside triangle ABC. Then D is on the same side of each side line as the opposite vertex, so

$$\chi(A, B, D) = \chi(B, C, D) = \chi(C, A, D) = \chi(A, B, C) =: \sigma.$$

By a cyclic shift we may take A smallest among A, B, C, and by a reflection (which reverses all orientations and keeps the configuration monochromatic) we may take $A < B < C$, so all four triangles have colour σ. Then:

(a) If $D < A$: $\chi(D, A, C) = \chi(A, C, D) = -\chi(C, A, D) = -\sigma$.

(b) If $A < D < B$: $\chi(A, D, B) = -\chi(A, B, D) = -\sigma$.

(c) If $B < D < C$: $\chi(B, D, C) = -\chi(B, C, D) = -\sigma$.

(d) If $C < D$: $\chi(A, C, D) = -\sigma$.

Each case gives a triangle of colour $-\sigma$, a contradiction. So the n points are in convex position. ∎

The original Erdős–Szekeres argument gives $ES(n) \leq \binom{2n-4}{n-2} + 1$. They also showed $ES(n) \geq 2^{n-2} + 1$ and conjectured equality, known for $n \leq 6$. Suk (2017) proved $ES(n) \leq 2^{n+o(n)}$.

13.3 Ramsey theory in geometry

Motivation. Colourings of the whole plane obey Ramsey-type laws too: some configuration with prescribed distances must be monochromatic. Equivalently, the unit-distance graph of the plane has chromatic number at least 4 (in fact at least 5).

Example 13.13

If every point of the plane is red or blue, some two points at unit distance have the same colour.
Solution. Two vertices of an equilateral triangle of side 1 share a colour.

Example 13.14

The same holds with three colours.
Solution. Suppose not. In any unit equilateral triangle ABC all colours differ. Reflect A across BC to get D, so that BCD is also unit equilateral. Then D differs from B and C, so D has the colour of A, and $|AD| = \sqrt{3}$. As ABC was arbitrary, any two points at distance $\sqrt{3}$ have the same colour. Hence the circle k of radius $\sqrt{3}$ about a red point is entirely red. But k contains two points at distance 1 (any chord length up to $2\sqrt{3}$ occurs), a contradiction.

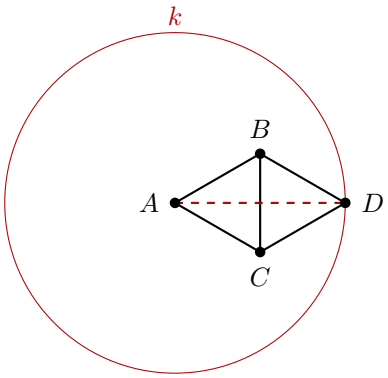


Figure. *Example 13.14: D is forced to have the colour of A, so every point of the circle k of radius $\sqrt{3}$ about A has that colour.*

Example 13.15

Colour the plane red/blue, and let T be a triangle with angles $30°, 60°, 90°$ and hypotenuse 1. Some triangle congruent to T has monochromatic vertices.
Solution. By Example 13.13 there is a unit segment AB with both ends red, say. On the circle with diameter AB, take $D_1, \dots, D_4$ dividing it, together with A, B, into six equal arcs. Any triangle ABD_i is congruent to T (Thales). If some D_i is red we are done. Otherwise all D_i are blue; two of them that are diametrically opposite form a diameter, and together with a third D_j they form a blue copy of T.

In graph language, the unit-distance graph of the plane has chromatic number (the **chromatic number of the plane**) greater than 3, by Example 13.14. It is at most 7, and de Grey (2018) proved it is at least 5; the exact value is open.

13.4 Selected exercises

Motivation. The book exercises below show the Ramsey mechanism (pigeon-hole at one vertex, then induction) in three settings: monotone subsequences, triangles in many colours, and monochromatic solutions of $a+b=c$.

Exercise 13.2 (Monotone subsequences)

Prove that every permutation of length $nm+1$ has an increasing subsequence of length $n+1$ or a decreasing subsequence of length $m+1$. (The entries of a subsequence need not be in consecutive positions.)

Solution.

Step 1 (labels). For each entry x let $\lambda(x)$ be the length of the longest increasing subsequence ending at x. If some $\lambda(x) \geq n+1$, we are done; so assume $\lambda(x) \in [n]$ for all x.

Step 2 (pigeon-hole). The $nm+1 > mn$ entries receive labels from $[n]$, so by Theorem 1.4 at least $m+1$ entries have the same label.

Step 3 (equal labels decrease). Let x precede y with $\lambda(x) = \lambda(y)$. If $x < y$, appending y to a longest increasing subsequence ending at x would give $\lambda(y) \geq \lambda(x)+1$. Hence $x > y$, and the entries with a common label, read from left to right, form a decreasing subsequence. By Step 2 it has length at least $m+1$.

The bound is sharp: for $n=2$, $m=2$ the permutation $2\,1\,4\,3$ of length 4 has neither an increasing nor a decreasing subsequence of length 3. ■

Exercise 13.6 (Monochromatic triangles in k colours)

Let $n_k = k!\left(1+\frac{1}{1!}+\frac{1}{2!}+\cdots+\frac{1}{k!}\right)+1$. Prove that every colouring of the edges of K_{n_k} with k colours contains a triangle with monochromatic edges.

Solution. Let $m_k = n_k - 1 = \sum_{j=0}^{k} \frac{k!}{j!}$, an integer. For $k \geq 2$,

$$m_k = \sum_{j=0}^{k-1} k \cdot \frac{(k-1)!}{j!} + \frac{k!}{k!} = k\, m_{k-1} + 1.$$

We prove by induction on $k \geq 1$ that every k-colouring of the edges of K_{m_k+1} has a monochromatic triangle.

Step 1 (initial step). $m_1 = 2$, and in a 1-colouring of K_3 the triangle itself is monochromatic.

Step 2 (pigeon-hole at a vertex). Let $k \geq 2$ and fix a vertex v of K_{m_k+1}. It has $m_k = k\, m_{k-1}+1$ edges, so by Theorem 1.4 some colour, say colour 1, appears on at least $m_{k-1}+1$ of them. Let S be the set of their other endpoints, $|S| \geq m_{k-1}+1$.

Step 3 (inside S). If some edge inside S has colour 1, it forms a triangle of colour 1 with v. Otherwise the edges inside S use only the $k-1$ colours $2,\ldots,k$, and by the induction hypothesis (applied to $m_{k-1}+1$ vertices of S) there is a monochromatic triangle in S.

For $k=2$ this gives $n_2 = 6$, i.e. $R(3,3) \leq 6$, and for $k=3$ it gives $n_3 = 17$, matching Example 13.9. ■

Exercise 13.16 (Schur's theorem)

Colour every positive integer with one of k colours. Prove that there is an integer $N = N(k)$ such that for every $n > N$ there are integers $a, b, c < n$ of the same colour with $a + b = c$ (with $a = b$ allowed).

Solution.

Step 1 (the constant). Let $N = R(3, 3, \ldots, 3)$ with k entries 3, which exists by Theorem 13.10. Let $n > N$.

Step 2 (an edge colouring). Take the complete graph on the vertices $0, 1, \ldots, N-1$, and give the edge $\{i, j\}$ the colour of the positive integer $|i - j| \in [N-1]$.

Step 3 (a monochromatic triangle). By the choice of N there are vertices $i < j < l$ whose three edges have the same colour. Put $a = j - i$, $b = l - j$ and $c = l - i$. These have the same colour, $a + b = c$, and $1 \le a, b, c \le N - 1 < n$. ■

Chapter 14. Subsequence Conditions on Permutations

14.1 Pattern avoidance

Motivation. Children of distinct heights line up so that each child sees every shorter child ahead of them: the lineup must contain no entries $a \dots b \dots c$ with $a < c < b$. Counting permutations that avoid such a "pattern" produces the Catalan numbers for every pattern of length 3. For length 4 the patterns behave differently, and only exponential bounds are within easy reach.

In 1423567, child 2 cannot see 1 (blocked by 4); 6723415 is a good lineup. A lineup is good iff there are no entries a, b, c, in this order (not necessarily adjacent), with $a < c < b$.

Definition 14.1 (132-pattern)

Entries a, b, c of a permutation, in this left-to-right order, with $a < c < b$ form a 132**-pattern** (they are ordered like $1, 3, 2$). Similarly for 123, 231, etc.

Definition 14.2 (132-avoiding)

A permutation with no 132-pattern is 132**-avoiding**.

Let $f(n)$ count 132-avoiding n-permutations ($f(0) = 1$). If n is in position i, every entry left of n exceeds every entry right of n (else $x < y$ with x left and y right gives the pattern $x\,n\,y$). So the right side is a 132-avoiding permutation of $\{1, \dots, n-i\}$ and the left side one of $\{n-i+1, \dots, n-1\}$, and conversely any such pair gives a 132-avoiding permutation. Hence

$$f(n) = \sum_{i=1}^{n} f(i-1) f(n-i) \qquad (n \geq 1). \tag{14.1}$$

Corollary 14.3

The number of 132-avoiding n-permutations is the Catalan number $c_n = \frac{1}{n+1}\binom{2n}{n}$.

Proof. (14.1) with $f(0) = 1$ is the recurrence of the Catalan numbers: comparing coefficients in (8.6) gives $c_n = \sum_{i=1}^{n} c_{i-1} c_{n-i}$ (§8.1.2.1). The two sequences agree by induction, and $c_n = \frac{1}{n+1}\binom{2n}{n}$. ■

Definition 14.4 (Pattern)

Let p be an n-permutation and $q = q_1 \cdots q_k$ a k-permutation, $k \leq n$. Entries $a_1, \dots, a_k$ of p (left to right) form a q**-pattern** if $a_i < a_j \iff q_i < q_j$.

Definition 14.5 (Avoidance)

p is q**-avoiding** if no k entries of p form a q-pattern. $S_n(q)$ (or $\mathrm{Av}_n(q)$) denotes the number of q-avoiding n-permutations.

Symmetries. The **reverse** $p_n \cdots p_1$ and the **complement** $\bar{p} = (n+1-p_1) \cdots (n+1-p_n)$ map q-avoiders bijectively onto avoiders of the reversed or complemented pattern. 231 is the reverse of 132, 312 its complement, and 213 the reverse of 312. Hence

$$S_n(132) = S_n(231) = S_n(312) = S_n(213), \qquad S_n(123) = S_n(321).$$

An entry smaller than all entries before it is a **left-to-right minimum**; these entries form a decreasing subsequence that starts with p_1 and contains 1.

Lemma 14.6

$S_n(123) = S_n(132)$ for all $n \geq 1$.

Proof. We construct a bijection from 123-avoiders to 132-avoiders that fixes the left-to-right minima (values and positions).

Step 1 (constraints). Let $m_1 > m_2 > \cdots > m_r$ be left-to-right minima at positions $i_1 = 1 < i_2 < \cdots < i_r$; call the other positions *slots*, a slot between i_j and i_{j+1} having *threshold* m_j. Filling the slots with the remaining values gives a permutation with exactly these left-to-right minima iff every slot receives a value larger than its threshold. Thresholds decrease from left to right.

Step 2 (the map f). Given a 123-avoider p, fill the slots from left to right, each time with the smallest unused value exceeding the threshold. This never gets stuck: at the s-th slot with threshold m, all of the first s slots have thresholds $\geq m$, so in p they hold s values $> m$. Only $s-1$ values have been used, so a value $> m$ remains. By Step 1, $f(p)$ has the same left-to-right minima.

Step 3 ($f(p)$ avoids 132). If non-minima x before y both exceed a left-to-right minimum m_j to their left, then when x was placed (threshold $\leq m_j < y$), y was available, so $x < y$. Now let a, b, c form a 132-pattern. Replacing a by the last left-to-right minimum at or before it keeps the pattern, so let $a = m_j$. Then $b, c > m_j$ lie right of m_j, so they are not left-to-right minima, and by the observation $b < c$, a contradiction.

Step 4 (the inverse g). Given a 132-avoider q, fill its slots with the non-minima in *decreasing* order. This respects the thresholds: in q the first s slots hold s values exceeding the s-th threshold, so the s-th largest value does too. $g(q)$ is a union of two decreasing sequences, hence avoids 123.

Step 5 (g and f are inverse). A 123-avoider has its non-minima in decreasing order (if $x < y$ were non-minima in this order, the left-to-right minimum before x, smaller than x, together with x, y would form a 123), so $g(f(p)) = p$. A 132-avoider q equals the greedy filling of Step 2. Indeed, if at some slot (threshold m_j) q holds x while a smaller eligible unused $y > m_j$ comes later, then m_j, x, y is a 132-pattern. Hence $f(g(q)) = q$. ∎

For example, $p = 465132$ has minima $4, 1$ and $f(p) = 456123$.

Theorem 14.7

For every pattern q of length 3 and all $n \geq 1$, $S_n(q) = c_n = \frac{1}{n+1}\binom{2n}{n}$.

Proof. By the symmetries and Lemma 14.6, all six values $S_n(q)$ coincide, and $S_n(132) = c_n$ by Corollary 14.3. ∎

For length 4, symmetries and further arguments leave three classes: 1342, 1234, 1324, with

$$\begin{aligned} S_n(1342):&\quad 1, 2, 6, 23, 103, 512, 2740, 15485 \\ S_n(1234):&\quad 1, 2, 6, 23, 103, 513, 2761, 15767 \\ S_n(1324):&\quad 1, 2, 6, 23, 103, 513, 2762, 15793 \end{aligned}$$

for $n \leq 8$. In fact $S_n(1342) < S_n(1234) < S_n(1324)$ for $n \geq 7$; surprisingly, the monotone pattern is in the middle. We prove the second inequality.

Theorem 14.8

$S_n(1234) < S_n(1324)$ for all $n \geq 7$.

A **right-to-left maximum** is an entry larger than all entries to its right. The proof is given after Lemma 14.11.

Definition 14.9 (Class)

Permutations x, y are in the same **class** if they have the same left-to-right minima at the same positions and the same right-to-left maxima at the same positions.

For instance 51234 and 51324 are in the same class; 24315 and 24135 are not.

Constraints of a class [Supplement]. Fix a class and call the remaining positions slots. A slot s has a lower bound $l(s)$, the nearest left-to-right minimum to its left, and an upper bound $u(s)$, the nearest right-to-left maximum to its right. A filling of the slots lies in the class iff $l(s) <$ value $< u(s)$ for every slot. (Then no slot entry becomes a new minimum or maximum, and the given minima and maxima keep their status.) Both l and u are non-increasing from left to right.

Lemma 14.10

Every nonempty class contains exactly one 1234-avoiding permutation.

Proof.

Step 1 (existence). Fill the slots with the remaining values in decreasing order. This respects the bounds [Supplement]. Take any member of the class. Its first s slots hold s values exceeding $l(s)$, so the s-th largest value exceeds $l(s)$. Its last $N - s + 1$ slots (of N) hold values below $u(s)$, so the $(N - s + 1)$-th smallest, which is the s-th largest, is below $u(s)$. The result is a union of three decreasing sequences (minima, maxima, slots), so a 1234-pattern would have two entries in one of them, impossible by pigeon-hole.

Step 2 (uniqueness). In a 1234-avoider of the class, if slot entries $a < b$ appeared in this order, then the left-to-right minimum before a (smaller than a), a, b, and the right-to-left maximum after b (larger than b) would form 1234. So the slot entries are decreasing, and the filling is the one of Step 1. ■

An **inversion** of p is a pair $i < j$ with $p_i > p_j$.

Lemma 14.11

Every nonempty class contains at least one 1324-avoiding permutation.

Proof.

Step 1 (good patterns). If p contains a 1324-pattern $a\,c\,b\,d$ ($a < b < c < d$), replace a by the nearest left-to-right minimum at or before it and d by the nearest right-to-left maximum at or after it. The result is still a 1324-pattern, now starting at a left-to-right minimum and ending at a right-to-left maximum (a **good pattern**). The middle entries c, b are slot entries: a left-to-right minimum can only play the role of "1" and a right-to-left maximum only that of "4".

Step 2 (swapping). Exchange c and b. The value b moves left, with $b > a \geq$ the new lower bound and $b < c < u$ there. The value c moves right, with $c < d \leq$ the new upper bound and $c > b > l$ there. So the bounds hold, and the new permutation is in the same class. Swapping an inverted pair reduces the number of inversions by at least one.

Step 3 (termination). Repeat while a good pattern exists. After at most $\binom{n}{2}$ steps, we reach a permutation of the same class with no good pattern, hence (Step 1) no 1324-pattern. ■

Proof of Theorem 14.8. By Lemmas 14.10 and 14.11, $S_n(1234) \leq S_n(1324)$; it remains to find a class with two 1324-avoiders. Write $a_1{*}a_2{**}b_1$ for the class of length-6 permutations with left-to-right minima a_1, a_2 at positions $1, 3$ and right-to-left maximum b_1 at position 6. For $n = 7$

the class $3{*}1{*}7{*}5$ contains the two 1324-avoiders 3612745 and 3416725. For $n > 7$ use the class $n(n-1)\cdots 8\,3{*}1{*}7{*}5$. ■

Stanley–Wilf. For every pattern q there is c_q with $S_n(q) \le c_q^n$ (Marcus–Tardos, 2004; cited). For monotone patterns this is easy:

Theorem 14.12

For $k \le n$, $S_n(12\cdots k) \le (k-1)^{2n}$.

Proof. Say x has **rank** i if the longest increasing subsequence ending at x has length i. Entries of equal rank form a decreasing subsequence (if $x < y$ had equal rank with x first, appending y would raise its rank). A $12\cdots k$-avoider has ranks in $[k-1]$, so it is a union of $k-1$ decreasing subsequences. The permutation is determined by assigning each value to one of the $k-1$ subsequences ($(k-1)^n$ ways) and each position to one of them ($(k-1)^n$ ways): each subsequence's values then fill its positions in decreasing order. ■

This agrees with $S_n(123) = c_n < 4^n$.

Definition 14.13 (Direct sum)

For $p \in S_a$, $q \in S_b$, the **direct sum** $p \oplus q \in S_{a+b}$ is p followed by q with every entry increased by a.

Example 14.14

$132 \oplus 2431 = 1325764$.

Theorem 14.15

If $S_n(q_1 \oplus 1) \le c_1^n$ and $S_n(1 \oplus q_2) \le c_2^n$ for all n, then

$$S_n(q_1 \oplus 1 \oplus q_2) \le \left(\sqrt{c_1} + \sqrt{c_2}\right)^{2n}.$$

Example 14.16

$S_n(2134) = S_n(1234) < 9^n$ (the equality is a known Wilf-equivalence, cited; the bound is Theorem 14.12), and $S_n(1243) < 9^n$ likewise. With $q_1 = 213$, $q_2 = 132$, Theorem 14.15 gives $S_n(213 \oplus 1 \oplus 132) \le (3+3)^{2n} = 36^n$.

Proof of Theorem 14.15. Let p avoid $q = q_1 \oplus 1 \oplus q_2$. Colour an entry red if it is the last (largest) entry of some $(q_1 \oplus 1)$-pattern, blue otherwise.

Step 1. The red entries avoid $1 \oplus q_2$. Given a red copy C, its first entry x ends a copy of $q_1 \oplus 1$ whose other entries lie left of x and are smaller than x, hence smaller than all of C. Together these give a copy of q.

Step 2. The blue entries avoid $q_1 \oplus 1$: the last entry of such a copy would be red.

Step 3 (counting). With k blue entries, p is determined by the set of blue values, the set of blue positions, the pattern of the blue entries and that of the red ones: at most $\binom{n}{k}^2 c_1^k c_2^{n-k}$ choices. Since a sum of squares of nonnegative numbers is at most the square of their sum,

$$S_n(q) \le \sum_{k=0}^{n} \left(\binom{n}{k} \sqrt{c_1}^{\,k} \sqrt{c_2}^{\,n-k}\right)^2 \le \left(\sum_{k=0}^{n} \binom{n}{k} \sqrt{c_1}^{\,k} \sqrt{c_2}^{\,n-k}\right)^2 = \left(\sqrt{c_1} + \sqrt{c_2}\right)^{2n}.$$

■

14.2 Stack sortable permutations

Motivation. Passing a permutation once through a stack sorts it exactly when it avoids 231. Sorting twice is no longer described by plain pattern avoidance, but the recursion $s(LnR) = s(L)s(R)n$, read on decreasing binary trees, reveals a hidden symmetry: t-stack sortable permutations with k descents are as numerous as those with $n-1-k$ descents.

Stack sorting. Read p from left to right. An incoming entry is pushed on the stack if it is smaller than the top; otherwise the top is popped to the output, repeatedly, until the entry can be pushed. At the end the stack is emptied. The output is $s(p)$; p is **stack sortable** if $s(p) = 12\cdots n$.

Example 14.17

$p = 2413$: push 2; 4 pops 2 and is pushed; push 1; 3 pops 1 and is pushed on 4; finally pop $3, 4$. So $s(2413) = 2134$, and 2413 is not stack sortable.

Proposition 14.18

Let $a < b$ be entries of p.
(1) If a precedes b in p, then a precedes b in $s(p)$.
(2) If b precedes a and no entry $c > b$ lies between them, then a precedes b in $s(p)$.
(3) If b precedes a and some $c > b$ lies between them (so b, c, a form a 231-pattern), then b precedes a in $s(p)$.

Proof.

(1) b cannot be pushed on top of the smaller a, so a leaves the stack before b enters.

(2) An entry leaves the stack only when a larger entry arrives or at the end. All entries between b and a are smaller than b, so b is still in the stack when a is pushed above it, and a leaves first. [Supplement] (The book's claim that the entries between are decreasing is not needed.)

(3) b must leave before c is pushed, and c is pushed before a arrives. ■

Theorem 14.19

p is stack sortable iff it avoids 231.

Proof. A 231-pattern bca puts b before a in $s(p)$ by part (3). Without such patterns every pair $a < b$ falls under (1) or (2) and appears in increasing order in $s(p)$. ■

So the stack sortable n-permutations are counted by c_n (Theorem 14.7). p is **two-stack sortable** if $s(s(p))$ is the identity. This property is not closed under taking patterns:

Example 14.20

$s(35241) = 32145$ and $s(32145) = 12345$, so 35241 is two-stack sortable. But its subsequence 3241 has $s(3241) = 2314$ and $s(2314) = 2134$.

Theorem 14.21 (West)

p is two-stack sortable iff it contains no 2341-pattern and no 3241-pattern except as part of a 35241-pattern.

Proof.

Step 1 (a 2341 obstructs). Let $b\,c\,d\,a$ form 2341 ($a<b<c<d$). By Proposition 14.18, b precedes c (part 1), and b and c precede a in $s(p)$ (part 3, via d). So b, c, a is a 231 in $s(p)$, and $s(p)$ is not stack sortable.

Step 2 (a bad 3241 obstructs). Let $y\,x\,z\,w$ form 3241 ($w<x<y<z$), not inside a 35241. Via z, both x and y precede w in $s(p)$. If no entry between y and x exceeds y, then x precedes y in $s(p)$ (part 2), and x, y, w is a 231 in $s(p)$. If some t between y and x exceeds y, then $t<z$ (otherwise $y\,t\,x\,z\,w$ is a 35241), so $y\,t\,z\,w$ is a 2341, and Step 1 applies.

Step 3 (sufficiency). Suppose $s(p)$ contains a 231-pattern $f\,g\,e$ ($e<f<g$). Since e comes after f and g in $s(p)$, Proposition 14.18 shows that f, g precede e in p and that some $h>g$ lies between g and e in p.

(a) If f precedes g in p, then $f\,g\,h\,e$ is a 2341.

(b) If g precedes f: as f precedes g in $s(p)$, no entry between g and f exceeds g (part 3). Hence h lies between f and e, and $g\,f\,h\,e$ is a 3241. It is not part of a 35241, which would need an entry larger than $h>g$ between g and f. ■

The number of two-stack sortable n-permutations is $\frac{2}{(n+1)(2n+1)}\binom{3n}{n}$ (cited). p is **t-stack sortable** if $s^t(p)$ is the identity.

Lemma 14.22

If $p = LnR$, then $s(p) = s(L)\,s(R)\,n$.

Proof. The entries of L all leave the stack before n enters, producing $s(L)$. Then n sits at the bottom while R passes over it, producing $s(R)$, and n leaves last. ■

Corollary 14.23

Every n-permutation is $(n-1)$-stack sortable.

Proof. Induction on n. By Lemma 14.22, $s(p) = s(L)s(R)\,n$ and $s(\sigma n) = s(\sigma)\,n$ for any σ. Hence $s^{n-1}(p) = s^{n-2}(s(L)s(R))\,n$, and $s(L)s(R)$ is a permutation of $[n-1]$, which is $(n-2)$-stack sortable by induction. ■

Corollary 14.24

$s^t(p)$ ends with $(n-t+1)(n-t+2)\cdots n$.

Proof. Induction on t [Supplement]. If $s^{t-1}(p) = \sigma\,(n-t+2)\cdots n$ with σ a permutation of $[n-t+1]$, then by Lemma 14.22 (applied to the trailing maxima one by one), $s^t(p) = s(\sigma)\,(n-t+2)\cdots n$, and $s(\sigma)$ ends with $\max\sigma = n-t+1$. ■

Decreasing binary trees. $T(p)$ has root n. The maximum of L is its left child and the maximum of R its right child (a missing side gives no child, but an only child is still a left or a right child). Recursively, $T(L)$ and $T(R)$ hang below. Reading $T(p)$ *in order* (left subtree, root, right subtree) recovers p.

Example 14.25

For $p = 263498175$, $T(p)$ is drawn below.

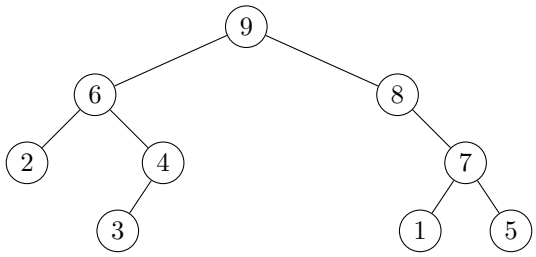


Figure. $T(263498175)$. *In order:* 263498175*; postorder:* $234615789 = s(p)$. *Here* 4 *has only a left child and* 8 *only a right child.*

Example 14.26

Reading the tree above in *postorder* (left subtree, right subtree, root) gives 234615789, which is $s(263498175)$.

Proposition 14.27

$T(p)$ read in postorder is $s(p)$.

Proof. Induction on n. For $p = LnR$, postorder reads $T(L)$, then $T(R)$, then n, giving $s(L)s(R)n$ by induction, which is $s(p)$ by Lemma 14.22. ■

i is a **descent** of p if $p_i > p_{i+1}$; $d(p)$ is the number of descents. Complementation shows that k and $n-1-k$ descents are equally frequent among all n-permutations. [Supplement] $p_i > p_{i+1}$ iff p_i has a right child in $T(p)$: the in-order successor of a vertex with a right child is the leftmost vertex of its right subtree (smaller), and otherwise it is an ancestor (larger). So $d(p)$ equals the number of right edges of $T(p)$.

Let $W_t(n,k)$ be the number of t-stack sortable n-permutations with k descents. Neither reverse nor complement preserves t-stack sortability (try 213 and 132), yet the data suggest $W_t(n,k) = W_t(n,n-1-k)$.

Definition 14.28 (The map f)

Define f on permutations (of any finite set of integers) by $f(1) = 1$ and, with n the maximum: $f(LnR) = f(L)\,n\,f(R)$ if L, R are nonempty; $f(Ln) = n\,f(L)$; $f(nR) = f(R)\,n$.

Example 14.29

$f(123) = 3\,f(12) = 3\,2\,f(1) = 321$, so $f(4123) = f(123)\,4 = 3214$.

Example 14.30

$f(1423) = f(1)\,4\,f(23) = 1432$.

Example 14.31

$f(412395867) = f(4123)\,9\,f(5867) = 3214\,9\,5876$.

Proposition 14.32

$d(p) + d(f(p)) = n - 1$ for every n-permutation p.

Proof. Induction on n; $n = 1$ is trivial.

Step 1 ($p = LnR$, $L, R \neq \emptyset$, n in position i). $d(p) = d(L) + d(R) + 1$ and $d(f(p)) = d(f(L)) + d(f(R)) + 1$, while by induction $d(L) + d(f(L)) = i - 2$ and $d(R) + d(f(R)) = n - i - 1$. The total is $n - 1$.

Step 2 ($p = Ln$). $d(p) = d(L)$ and $d(n\,f(L)) = d(f(L)) + 1$, total $(n-2)+1$.

Step 3 ($p = nR$). $d(p) = 1 + d(R)$ and $d(f(R)\,n) = d(f(R))$, total $1 + (n-2)$. ■

Lemma 14.33

$s(p) = s(f(p))$ for every permutation p.

Proof. Induction on n, using Lemma 14.22 and the induction hypothesis $s(L) = s(f(L))$, $s(R) = s(f(R))$ (f permutes the entries of L and R among themselves).

(1) $p = LnR$: $s(p) = s(L)s(R)n = s(f(L))s(f(R))n = s(f(L)nf(R)) = s(f(p))$.

(2) $p = nR$: $s(p) = s(R)n = s(f(R))n = s(f(R)n) = s(f(p))$.

(3) $p = Ln$: $s(p) = s(L)n = s(f(L))n = s(nf(L)) = s(f(p))$. ■

Corollary 14.34

p is t-stack sortable iff $f(p)$ is.

Proof. Both are equivalent to $s(p) = s(f(p))$ being $(t-1)$-stack sortable. ■

Theorem 14.35

$W_t(n,k) = W_t(n, n-1-k)$ for all n, k, t.

Proof. f is a bijection (it acts on $T(p)$ by flipping every only child to the other side, which is an involution on trees; see below). By Corollary 14.34 and Proposition 14.32 it maps t-stack sortable permutations with k descents onto those with $n-1-k$ descents. ■

Tree view. On $T(p)$, f turns every only left child into an only right child and vice versa, and keeps vertices with two children unchanged. This swaps the numbers of left and right edges, which is Proposition 14.32 again. For postorder it does not matter on which side an only child hangs, which is Lemma 14.33 again, and in fact holds for the subtree of every vertex.

Example 14.36

$p = 356124$ and $f(p) = 536421$ both give 351246 when their trees are read in postorder, and the same holds for the subtree of each entry.

Chapter 15. The Probabilistic Method

15.1 The notion of probability

Motivation. On a finite sample space with equally likely outcomes, probability is counting divided by counting. The union bound $P(\bigcup A_i) \le \sum P(A_i)$, trivial as it is, will drive the existence proofs of the next section.

Tossing a coin four times has $2^4 = 16$ outcomes, five of which ($HHHH$, $HHHT$, $HHTH$, $HTHH$, $THHH$) have at least three heads, so this happens with probability $5/16$.

Definition 15.1 (Probability)

Let Ω be a finite set of equally likely outcomes (the **sample space**) and $A \subseteq \Omega$ an **event**. The **probability** of A is $P(A) = \frac{|A|}{|\Omega|}$.

For disjoint A, B, $P(A \cup B) = P(A) + P(B)$; in general $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ by the sieve. The **complement** of A is $\bar{A} = \Omega - A$, and $P(\bar{A}) = 1 - P(A)$.

Proposition 15.2 (Union bound)

For events $A_1, \dots, A_n$, $P(A_1 \cup \cdots \cup A_n) \le P(A_1) + \cdots + P(A_n)$.

Proof. $|A_1 \cup \cdots \cup A_n| \le |A_1| + \cdots + |A_n|$: the left side counts every element of the union once, the right side at least once. ∎

Example 15.3

Six numbers are drawn from $[36]$. A ticket of six numbers contains at least one winning number with probability

$$1 - \binom{30}{6} \Big/ \binom{36}{6} \approx 1 - 0.3048 = 0.6952,$$

not $\frac{6}{36}$.

Example 15.4 (Birthday paradox)

Among 23 people (nobody born on February 29), two share a birthday with probability more than $\frac{1}{2}$. The probability that all birthdays differ is

$$\frac{365 \cdot 364 \cdots 343}{365^{23}} < \frac{1}{2},$$

and more people only increase the chance of a coincidence.

15.2 Non-constructive proofs

Motivation. If a random object has a bad property with probability less than 1, some object avoids it. This proves that colourings without large monochromatic cliques exist, giving $R(k,k) > 2^{k/2}$, although nobody knows how to construct such colourings explicitly.

Theorem 15.5 (Erdős)

$R(k,k) > 2^{k/2}$ for $k \geq 3$.

Proof. Let $n \leq 2^{k/2}$ and colour each edge of K_n red or blue by an independent fair coin; Ω is the set of all $2^{\binom{n}{2}}$ colourings, equally likely. A fixed K_k is monochromatic in 2 of the $2^{\binom{k}{2}}$ colourings of its edges, so with probability $2^{1-\binom{k}{2}}$. Let A_S be the event that the K_k-subgraph S is monochromatic. By Proposition 15.2,

$$P\Big(\bigcup_S A_S\Big) \leq \sum_S P(A_S) = \binom{n}{k} 2^{1-\binom{k}{2}}. \tag{15.1}$$

Using $\binom{n}{k} < \frac{n^k}{k!}$ and $n^k \leq 2^{k^2/2}$,

$$\binom{n}{k} 2^{1-\binom{k}{2}} < \frac{2^{k^2/2}}{k!} \cdot 2^{1-\frac{k^2}{2}+\frac{k}{2}} = \frac{2 \cdot 2^{k/2}}{k!} < 1$$

for $k \geq 3$: at $k = 3$ the value is $\frac{2 \cdot 2^{3/2}}{6} < 1$, and increasing k by one multiplies it by $\frac{\sqrt{2}}{k+1} < 1$. So with positive probability no K_k is monochromatic, and such a colouring of K_n exists. ∎

Together with Corollary 13.8, $(\sqrt{2})^k < R(k,k) \leq 4^{k-1}$; these are essentially the best known exponential bounds, apart from the recent $(4-\epsilon)^k$ upper bound.

Theorem 15.6

Let $n, m \geq 2$ with $m \geq 2\log_2 n$. Then the edges of $K_{n,n}$ can be 2-coloured with no monochromatic $K_{m,m}$.

Proof. Colour randomly. A fixed $K_{m,m}$ (choose m vertices on each side: $\binom{n}{m}^2$ choices) is monochromatic with probability $2 \cdot 2^{-m^2}$. By the union bound it suffices that $2\binom{n}{m}^2 < 2^{m^2}$. Indeed, $\binom{n}{m} \leq \frac{n^m}{m!}$ and $m! \geq 2$ give $2\binom{n}{m}^2 < n^{2m}$, and $n \leq 2^{m/2}$ gives $n^{2m} \leq 2^{m^2}$. ∎

Equivalently, there is an $n \times n$ 0–1 matrix with no constant $m \times m$ submatrix for $m \approx 2\log_2 n$; the best explicit constructions only achieve $m = c\sqrt{n}$.

15.3 Independent events

Motivation. Sometimes $P(A \cap B) = P(A)P(B)$ and sometimes not; independence means that learning B does not change the odds of A. Conditional probabilities defy intuition (Simpson's paradox, Monty Hall), and the reliable tool is the law of total probability, called Bayes' theorem in the book.

15.3.1 The notion of independence and Bayes' theorem

For a number chosen from [12], divisibility by 2 and by 3 satisfy the product rule, but divisibility by 2 and by 4 do not: knowing evenness halves the outcomes without losing any multiple of 4.

Definition 15.7 (Independence)

Events A, B are **independent** if $P(A \cap B) = P(A)P(B)$, and **dependent** otherwise.

Definition 15.8 (Conditional probability)

For $P(B) > 0$, $P(A \mid B) = \frac{P(A \cap B)}{P(B)}$ is the probability of A given B.

Proposition 15.9

For $P(B) > 0$, A and B are independent iff $P(A \mid B) = P(A)$.

Proof. Divide $P(A \cap B) = P(A)P(B)$ by $P(B)$, or multiply back. ■

Example 15.10

Four coin tosses, at least two heads (B). The probability of four heads is $P(A \mid B) = \frac{P(A)}{P(B)} = \frac{1/16}{11/16} = \frac{1}{11}$ (with $P(B) = \frac{1+4+6}{16}$).

Example 15.11

For a random n-permutation let $A = \{p_1 > p_2\}$ and $B = \{p_2 > p_3\}$. Then $P(A) = P(B) = \frac{1}{2}$ (reverse the pair) and $P(A \cap B) = P(p_1 > p_2 > p_3) = \frac{1}{6}$. So $P(A \mid B) = \frac{1}{3} \neq P(A)$, and the events are dependent.

Example 15.12 (Simpson's paradox)

Two colleges each admit domestic applicants at a higher rate than international ones. It does not follow that the whole university does:

	Liberal Arts	Engineering	University
Domestic	10/120 (8.3%)	10/10 (100%)	20/130 (15.4%)
International	1/15 (6.7%)	90/100 (90%)	91/115 (79.1%)

About 87% of the international applicants, but only 8% of the domestic ones, applied to Engineering, where admission is easy.

Theorem 15.13 (Bayes' theorem, law of total probability)

If A, B are mutually exclusive with $A \cup B = \Omega$ and C is any event, then

$$P(C) = P(C \mid A)P(A) + P(C \mid B)P(B). \tag{15.2}$$

Proof. C is the disjoint union of $C \cap A$ and $C \cap B$, so $P(C) = P(C \cap A) + P(C \cap B)$, and $P(C \cap A) = P(C \mid A)P(A)$, likewise for B. ■

So $P(C)$ is a *weighted* average of the conditional probabilities. In Example 15.12 the weights (the shares applying to each college) differ between the two groups, which explains the paradox. With equal weights the domestic applicants would win.

Example 15.14

Exactly one of Andrew (A) and Brandi (B) committed a crime; the culprit has a blood type

T shared by 5% of the population. If Andrew has type T (event X), then

$$P(A \mid X) = \frac{P(A \cap X)}{P(X)} = \frac{P(X \mid A)P(A)}{P(X)}. \tag{15.3}$$

Here $P(X \mid A) = 1$, $P(X \mid B) = \frac{1}{20}$, $P(A) = P(B) = \frac{1}{2}$, so $P(X) = \frac{1}{2} + \frac{1}{40} = \frac{21}{40}$ by (15.2), and $P(A \mid X) = \frac{20}{21}$.

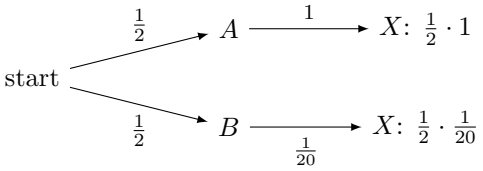


Figure. *Example 15.14 as a probability tree:* $P(A \mid X) = \frac{1/2}{1/2+1/40} = \frac{20}{21}$.

15.3.2 More than two events

Pairwise independence is not enough. Pick $x \in [10]$ and $y \in [20]$, and let $A = \{x \text{ odd}\}$, $B = \{y \text{ odd}\}$, $C = \{x - y \text{ odd}\}$. Each has probability $\frac{1}{2}$ and any two are independent, yet $P(A \cap B \cap C) = 0 \neq \frac{1}{8}$.

Definition 15.15 (Mutual independence)

$A_1, \ldots, A_n$ are **independent** if $P(A_{i_1} \cap \cdots \cap A_{i_k}) = P(A_{i_1}) \cdots P(A_{i_k})$ for every nonempty $\{i_1, \ldots, i_k\} \subseteq [n]$.

Theorem 15.16 (Bayes' theorem, general version)

If $A_1, \ldots, A_n$ are pairwise disjoint with union Ω and C is any event, then $P(C) = \sum_{i=1}^{n} P(C \mid A_i)P(A_i)$ (terms with $P(A_i) = 0$ omitted).

Proof. C is the disjoint union of the sets $C \cap A_i$, and $P(C \cap A_i) = P(C \mid A_i)P(A_i)$. ∎

Monty Hall. The contestant picks door 1; the host, who knows where the car is, opens door 2, showing a goat (event B). With $A_i =$ "car behind door i",

$$P(A_1 \mid B) = \frac{P(B \mid A_1)P(A_1)}{\sum_i P(B \mid A_i)P(A_i)} = \frac{\frac{1}{2} \cdot \frac{1}{3}}{\frac{1}{2} \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3}} = \frac{1}{3}, \tag{15.4}$$

so switching wins with probability $\frac{2}{3}$.

15.4 Expected values

Motivation. Expectation is linear even for heavily dependent variables. So averages of complicated counts (valleys, fixed points) split into sums of easy indicator expectations. And since some outcome is at least the average, expectations also prove existence.

A **random variable** is a function $X : \Omega \to \mathbb{R}$; sums and products are taken pointwise.

Definition 15.17 (Expectation)

If X takes finitely many values S, its **expected value** is $E(X) = \sum_{i \in S} i\, P(X = i)$, where $P(X = i) = \frac{|\{u : X(u) = i\}|}{|\Omega|}$. Equivalently,

$$E(X) = \frac{1}{|\Omega|} \sum_{u \in \Omega} X(u). \tag{15.5}$$

The value depends on Ω (write E_Ω if needed); countable ranges need convergent sums.

Definition 15.18 (Independent variables)

X, Y are **independent** if $P(X = s, Y = t) = P(X = s)P(Y = t)$ for all s, t.

15.4.1 Linearity of expectation

Theorem 15.19 (Linearity)

(1) $E(X + Y) = E(X) + E(Y)$ for random variables on the same space.
(2) $E(cX) = cE(X)$ for $c \in \mathbb{R}$.

Proof.

Step 1 (equally likely outcomes). Both follow from (15.5): $\sum_u (X + Y)(u) = \sum_u X(u) + \sum_u Y(u)$ and $\sum_u cX(u) = c \sum_u X(u)$.

Step 2 (without uniformity). Let X take values x_i and Y values y_j. Then

$$E(X + Y) = \sum_{i,j} (x_i + y_j) P(X = x_i, Y = y_j) = \sum_i x_i P(X = x_i) + \sum_j y_j P(Y = y_j),$$

summing out j and i respectively. For (2), $P(cX = cx_i) = P(X = x_i)$ if $c \neq 0$ (and $c = 0$ is trivial). ■

No independence is needed. An **indicator variable** is 1 if an event occurs and 0 otherwise, so its expectation is the probability of the event.

i is a **valley** of p if $2 \leq i \leq n - 1$ and $p_i < p_{i-1}$, $p_i < p_{i+1}$.

Theorem 15.20

For $n \geq 2$, a random n-permutation has on average $\frac{n-2}{3}$ valleys.

Proof. Let Y_i $(2 \leq i \leq n - 1)$ indicate that i is a valley. Each of p_{i-1}, p_i, p_{i+1} is equally likely to be the smallest of the three, so $E(Y_i) = \frac{1}{3}$. The number of valleys is $Y = \sum_{i=2}^{n-1} Y_i$, so $E(Y) = \frac{n-2}{3}$ by Theorem 15.19. ■

Theorem 15.21

A random n-permutation has on average one fixed point.

Proof. Let X_i indicate $p_i = i$; p_i is uniform on $[n]$, so $E(X_i) = \frac{1}{n}$. Then

$$E(X) = \sum_{i=1}^{n} E(X_i) = n \cdot \frac{1}{n} = 1. \tag{15.6}$$

■

15.4.2 Existence proofs using expectation

Theorem 15.22

If X takes finitely many values with maximum j, then $j \geq E(X)$. Hence some outcome u has $X(u) \geq E(X)$.

Proof. $E(X) = \sum_{i \in S} i\, P(X = i) \leq j \sum_{i \in S} P(X = i) = j$. ■

Theorem 15.23

A simple graph G on $[n]$ ($n \geq 2$) with $m \geq 1$ edges has a bipartite subgraph with more than $m/2$ edges.

Proof. Let Ω be the $2^{n-1} - 1$ splits of $[n]$ into two nonempty parts $\{A, B\}$; each split gives the bipartite subgraph of edges between A and B, with X edges. Let X_i indicate that edge i crosses. The splits separating its endpoints u, v are obtained by sending each of the other $n - 2$ vertices to u's or v's side, so

$$E(X_i) = P(X_i = 1) = \frac{2^{n-2}}{2^{n-1} - 1} > \frac{1}{2}.$$

Then $E(X) = \sum_{i=1}^{m} E(X_i) > \frac{m}{2}$, and Theorem 15.22 gives a split with more than $m/2$ crossing edges. ■

Example 15.24 (Betweenness)

Given triples $L_i = (a_i, b_i, c_i)$ of distinct elements of $[n]$, $i \leq k$, say that p satisfies L_i if b_i lies between a_i and c_i in p. Some n-permutation satisfies at least $k/3$ of the triples.
Solution. For a random p, each of a_i, b_i, c_i is equally likely to be in the middle, so the indicator Y_i of satisfying L_i has $E(Y_i) = \frac{1}{3}$. The number of satisfied triples has expectation $\frac{k}{3}$; apply Theorem 15.22.

15.4.3 Conditional expectation

$E(X \mid A) = \sum_i i\, P(X = i \mid A)$ is the expectation of X given A ($P(A) > 0$).

Theorem 15.25

If $A_1, \ldots, A_n$ are pairwise disjoint with union Ω (and positive probabilities), then $E(X) = \sum_{i=1}^{n} E(X \mid A_i) P(A_i)$.

Proof. By Theorem 15.16 with $C = \{X = j\}$, $P(X = j) = \sum_i P(X = j \mid A_i) P(A_i)$. Multiply by j and sum over j:

$$E(X) = \sum_j j \sum_i P(X = j \mid A_i) P(A_i) = \sum_i P(A_i) \sum_j j\, P(X = j \mid A_i) = \sum_i E(X \mid A_i) P(A_i).$$

■

Example 15.26

A die is thrown three times and the first throw is at least 4. The first throw is even (A_1) with probability $\frac{2}{3}$, giving on average $1 + 2 \cdot \frac{1}{2} = 2$ even throws, and odd (A_2) with probability $\frac{1}{3}$, giving on average 1. So the expected number of even throws is $\frac{2}{3} \cdot 2 + \frac{1}{3} \cdot 1 = \frac{5}{3}$.

Example 15.27

A team wins each game independently with probability $\frac{3}{4}$. Given at least three wins in the first four games (B), what is the expected number of wins in 12 games?
Solution. Let A_1, A_2 be "exactly 3" and "all 4" of the first four won. $P(A_1) = 4 \cdot \frac{1}{4}(\frac{3}{4})^3 = \frac{27}{64}$ and $P(A_2) = (\frac{3}{4})^4 = \frac{81}{256}$, so $P_B(A_1) = \frac{4}{7}$ and $P_B(A_2) = \frac{3}{7}$. The last eight games add $8 \cdot \frac{3}{4} = 6$ wins on average. By Theorem 15.25 in the space B,

$$E_B(X) = 9 \cdot \frac{4}{7} + 10 \cdot \frac{3}{7} = \frac{66}{7},$$

more than the unconditional 9.

15.5 Selected exercises

Motivation. The two book exercises below are the typical uses of indicator variables: an existence proof from an expectation, and a second-moment computation that needs the joint behaviour of pairs of indicators.

Exercise 15.17 (Tournaments with many Hamiltonian paths)

Prove that for every $n \geq 1$ there is a tournament on n vertices with at least $\frac{n!}{2^{n-1}}$ Hamiltonian paths. What can be said about Hamiltonian cycles?

Solution.

Step 1 (random tournament). Orient each edge of K_n on $[n]$ independently in either direction with probability $\frac{1}{2}$; all $2^{\binom{n}{2}}$ tournaments are equally likely.

Step 2 (indicators). For a permutation $\sigma = \sigma_1 \cdots \sigma_n$ of $[n]$ let X_σ be the indicator of the event that $\sigma_1 \to \sigma_2 \to \cdots \to \sigma_n$ is a directed path. This event prescribes the directions of $n-1$ distinct edges, so $E(X_\sigma) = 2^{-(n-1)}$.

Step 3 (expectation). The number of Hamiltonian paths is $X = \sum_\sigma X_\sigma$, so by Theorem 15.19, $E(X) = \frac{n!}{2^{n-1}}$. By Theorem 15.22, some tournament has $X \geq \frac{n!}{2^{n-1}}$.

For cycles ($n \geq 3$): a directed Hamiltonian cycle is determined by a cyclic order of $[n]$, and there are $(n-1)!$ of them (Example 6.10). Each prescribes the directions of n distinct edges, so it occurs with probability 2^{-n}. The same argument gives a tournament with at least $\frac{(n-1)!}{2^n}$ Hamiltonian cycles. ■

Exercise 15.18 (The variance of the number of fixed points)

For a random variable Y let $\mathrm{Var}(Y) = E((Y - E(Y))^2)$.
(a) Prove that $\mathrm{Var}(Y) = E(Y^2) - E(Y)^2$.
(b) Let $X(p)$ be the number of fixed points of a random n-permutation p, $n \geq 2$. Prove that $\mathrm{Var}(X) = 1$.

Solution.

(a) Let $\mu = E(Y)$, a constant. By Theorem 15.19,

$$E((Y-\mu)^2) = E(Y^2) - 2\mu E(Y) + \mu^2 = E(Y^2) - \mu^2.$$

(b)

Step 1 (indicators). Let X_i be the indicator of $p(i) = i$, so $X = \sum_i X_i$ and, as in Theorem 15.21, $E(X_i) = \frac{(n-1)!}{n!} = \frac{1}{n}$ and $E(X) = 1$.

Step 2 (second moment). Since $X_i^2 = X_i$,

$$X^2 = \sum_i X_i + \sum_{i \neq j} X_i X_j.$$

For $i \neq j$, $X_i X_j$ is the indicator of $p(i) = i$ and $p(j) = j$; exactly $(n-2)!$ permutations fix both, so $E(X_i X_j) = \frac{(n-2)!}{n!} = \frac{1}{n(n-1)}$. There are $n(n-1)$ ordered pairs $i \neq j$, hence

$$E(X^2) = n \cdot \frac{1}{n} + n(n-1) \cdot \frac{1}{n(n-1)} = 2.$$

Step 3 (conclusion). By (a), $\mathrm{Var}(X) = 2 - 1^2 = 1$. (For $n = 1$, $X \equiv 1$ and the variance is 0.)

So the number of fixed points has mean 1 and variance 1 for every $n \geq 2$, like a Poisson variable with parameter 1. ■